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PREVENTING AGE-RELATED ALPHA-MOTOR NEURON LOSS

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PREVENTING AGE-RELATED ALPHA-MOTOR NEURON LOSS

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LIST OF ABBREVIATIONS

AChR:	Acetylcholine receptor
AD:	Alzheimer's disease
ALS:	Amyotrophic lateral sclerosis
αMN:	Alpha-motor neuron
BBB:	Blood-brain barrier
BSCB:	Blood-spinal cord barrier
BTX:	Bungarotoxin
CUT&RUN-seq:	Cleavage Under Targets and Release Using Nuclease and Sequencing
FOXD3:	Forkhead Box D3
OKN-007:	Oklahoma Nitron 007; disodium 4-(tert-butyl-imino) methyl benzene-1,3-disulfonate N-oxide
MPTP:	1-Methyl-4-phenyl-1,2,3,6-tetrahydropyridine
NAC:	N-acetylcysteine
NMJ:	Neuromuscular junction
Nox1:	NADPH oxidase 1
PD:	Parkinson's disease
ROS:	Reactive oxygen species

LIST OF ABBREVIATIONS CONTINUED

SAM:	S-adenosyl methionine
SC:	Spinal cord
SOD1:	Copper Zinc Superoxide Dismutase
TBI:	Traumatic brain injury
TF:	Transcription factor

ABSTRACT

Aging is a universal problem concerning the whole society. Age-related loss of muscle mass and function leads to motor deficits and affects greatly the quality of life of an affected individual, but it also has a broader negative impact on economy, healthcare system, etc. Our laboratory demonstrated that the loss of innervation is a primary cause leading to sarcopenia, therefore it is important to address changes occurring in the aging spinal cord, especially those concerning alpha-motor neurons. Spinal cord aging, however, is an understudied subject, and the work presented in this dissertation aims to fill this gap and describe the changes occurring in the aging spinal cord that can contribute to the age-related α -MN loss and identify the interventions that can prevent the loss of α -MNs in aging.

In this study we uncovered a 41% decrease in α -MNs number in the lumbar spinal cord of old C57BL/6J wildtype mice. We also observe demyelination and mild inflammation, including astrocyte and microglia activation and an increase in levels of cytokines such as sICAM-1. We also identified changes in metabolites consistent with decreased neuronal viability, such as reduced levels of N-acetyl-aspartate. Cleaved caspase-3 is more abundant in spinal cord from old mice, suggesting that apoptosis contributes to neuronal loss. RNA-seq analysis revealed changes in the expression of a number of genes in spinal cord from old mice, in particular genes encoding ECM and a 172-fold increase in MMP-12 expression. BSCB integrity is also compromised in old mice.

In this work we also propose novel interventions aimed at preventing age-related α -MN loss. We demonstrate that knocking out MMP-9 and MMP-12 leads to a decrease in inflammation, microglia activation, as well as decreased BSCB permeability, which promotes α -MN survival in 25 months old wildtype mice. Furthermore, we treated

wildtype mice with OKN-007 in drinking water starting at 16 months of age, until mice reached 25 months of age. The treated mice exhibited a higher number of α -MNs, as well as lower levels of cleaved caspase-3, suggesting a decrease in apoptosis. OKN-007 decreased microglia proliferation and activation, without affecting astrocyte proliferation, as well as improved BSCB integrity. Moreover, the treatment decreased NMJ fragmentation and down-regulated markers of denervation, as well as significantly improved muscle contractile force, but did not affect hindlimb muscle mass. However, further studies are needed though to pinpoint the precise mechanism of action of OKN-007 in the context of aging.

OKN-007 was also successful in preventing neuronal loss in a mice model that exhibits an accelerated α -MN loss. We treated SOD1 G93A mutant mice and wildtype mice with OKN-007 in drinking water starting at P60, before the mice become symptomatic, until sacrifice around day P145, prior to end-stage. The treatment did not delay the disease onset, however, it significantly slowed down the disease progression. Immunostaining revealed that the treated mice has less fragmented NMJs. OKN-007 also significantly decreased microglia activation, without affecting though the astrocyte proliferation. Furthermore, perfusion MRI revealed that OKN-007 normalized the blood flow which is significantly decreased in untreated G93A mice comparing to age-matched wildtype controls. Our results suggest that, due to its neuroprotective and anti-inflammatory effect, OKN-007 may be a potential treatment aiming at slowing down ALS progression by increasing α -MN survival.

Lastly, we reported for the first time the presence of troponins, proteins canonically associated with the muscle tissue, in the α -MNs. The abundance of troponins I and C increases with age, while troponins I and T can translocate to the nucleus where they differentially bind DNA in an age-dependent manner and can contribute to dysregulation of gene expression in aging. Additionally, both troponins I

and T are also present in the nuclei from SOD1 KO and SOD1 G93A mice, which are an example of a model exhibiting accelerated α -MN loss (the former is considered a model of accelerated aging, while the latter is one of the mouse models of ALS), which suggests the link between the nuclear presence of troponins and increased α -MN loss. Further studies are needed to determine the precise effect of troponins' age-dependent differential binding on gene expression.

In conclusion, our data show that the spinal cord undergoes significant changes during aging, including progressive α -MNs loss that is associated with low-grade inflammation, apoptosis, changes in ECM, myelination, and vascular permeability. Knocking out MMP-9 and MMP-12 protects α -MNs, decreases inflammation and promotes the maintenance of BSCB integrity. The treatment with OKN-007 confers similar benefits in the aging spinal cord, as well as protects α -MNs and decreases inflammation in SOD1 G93A ALS mice model that exhibits an accelerated α -MN loss. Furthermore, we identify troponins as potential targets for therapeutic intervention.

CHAPTER I

INTRODUCTION AND LITERATURE REVIEW

1.1 SPINAL CORD AGING

In comparison to what is known regarding aging of the brain, surprisingly little is known about the spinal cord (SC) aging. Studies of central nervous system aging tend to focus on the brain, and the results are often extrapolated to the spinal cord, since it is a part of CNS. However, the processes taking place in the aging spinal cord might be distinct and spinal cord-specific. Furthermore, most spinal cord-related studies investigate spinal cord injury, and if aging is mentioned, it is considered merely a factor affecting the recovery after spinal cord injury (e.g., Geoffroy et al., 2017; Lundström et al., 2017). These studies focus mostly on the age-dependent decline in axonal growth after injury reported in various models, and interestingly, the implicated molecules, DAF-2 and DAF-16/FOXO, also happen to be involved in lifespan extension (Graciarena et al., 2014 – studies in zebrafish; Byrne et al., 2014 – studies in *C. elegans*).

The literature regarding CNS aging often reports age-related brain atrophy as a main aging phenotype. Atrophy has been reported in cerebrum in general (e.g., Madan and Kensinger, 2016; Fitzhugh et al., 2019; He et al., 2021), and cerebellum (Gellersen et al., 2021), as well as in the structures such as hippocampus (Erickson et al., 2012), motor cortex and corpus callosum (Seidler et al., 2010). However, one of the easiest assays used to measure age-related brain atrophy – simple brain weight measurement, is problematic in case of the spinal cord, because of the inconsistent amount of material obtained from the standard spinal cord isolation procedure. As a result, other strategies have to be implemented in order to assess spinal cord atrophy. One possible approach is alpha-motor neuron (α -MN) staining and counting. This method reveals **α -MN loss** in

anterior (ventral) horn of human lumbar spinal cord (Terao et al., 1996; Zhou et al., 1996; Cruz-Sánchez et al., 1998; Drey et al., 2014; Piasecki et al., 2016; Buchman et al., 2019). Besides α -MN loss, Kawamura et al. (1977a; 1977b) report also a decrease in axon number and diameter in ventral roots. Tomlinson and Irving (1977) observe up to 50% loss of α -MNs in individuals older than 60 years. α -MN loss was also reported in human cervical SC (Zhang et al., 1996). Surprisingly, Fu et al. (2015) report increased neuronal and non-neuronal cell number in aging mouse spinal cord (neuronal/non-neuronal cell number ratio remained unchanged), but the overall cell density decreased with age. Lee et al. (2017) also do not report age-related α -MN loss, as they did not observe α -MN loss and changes in morphology in 24 months old male Mongolian gerbils and neither does Maxwell et al. (2018), however most published reports, as well as our study (Piekarz et al., 2020) support age-related α -MN loss.

α -MN are not the only motor neuron population that deteriorates with age, as a similar phenomenon was described in phrenic MNs (a 22% loss in rats by 24 months of age comparing to 6 months old controls; Fogarty et al., 2018) and facial MNs (19-24% loss of MNs in facial nucleus in 24 months old rats compared to 3 months old controls; Katharesan et al., 2016), as well as in dorsal vagal nucleus (age-related loss in PD patients; Gai et al., 1992; in mice: Sturrock, 1990) and in enteric MNs in rats (Cowen et al., 2000). Age-related decline of motor functions can be further exacerbated by the death of dopaminergic neurons, as shown in wildtype C57BL/6 mice that exhibit locomotor impairments and age-dependent dopaminergic neuron loss (Noda et al., 2020). Furthermore, mouse models of progeria that exhibit an accelerated aging phenotype also show α -MN degeneration (for instance, progeroid *Ercc1* ^{Δ} mice in de Waard et al., 2010). Interestingly, sexual dimorphism was reported in human spinal cord aging (Yuan et al., 1999), as the area of gray matter decreases faster in males, while the area of white matter decreases faster in females, yet the implications of that fact are not

clear. In contrast, we do not observe sex-specific differences in neuronal loss in old mice (Piekarz et al., 2020).

In addition to neuronal loss, age-related **changes in neuronal morphology** were reported in the aging spinal cord. In 1977 Machado-Salas et al. described morphological changes in spinal cord and lower brain stem neurons in aging mouse (26 to 30 months old compared to 3 and 9 months old control mice). The researchers used Golgi staining technique to assess neuronal morphology and, while they did not observe significant changes in cortical neurons (even though they observed them in earlier studies on human brains, especially in prefrontal and temporal regions), progressive age-related changes were reported in the neurons located in spinal cord and lower brain stem. The changes consisted of irregular swelling of neuronal soma and proximal dendrites, nodularization of dendritic tree, loss of spines and distortion of axons. However, axons are difficult to visualize by silver deposition because of the presence of myelin sheath, so only initial or terminal segments, or unmyelinated axons can be stained properly. Formation of constrictions along the dendrites resulted in the loss of distal dendritic parts and a decrease in amount of functional neuropil, followed by neuronal loss. These changes were detected both in sensory and motor systems. The changes in the dendritic network in aging was also confirmed by altered (decreased) synaptophysin immunoreactivity in anterior horn of human SCs (Cruz-Sanchez et al., 1996). Staining with periodic acid-Schiff reagent and counterstain with hematoxylin revealed increased lipofuscin deposits in hypoglossal and dorsal vagal nuclei in old mice (3% vs 100%, no photo provided; Machado-Salas et al., 1977), which may contribute to neuronal swelling. However, there is no data from spinal cord in this study. The authors suspect that also tubular proteins might be of importance in this context. However, the authors do not show micrographs of control neurons, just affected, swollen neurons from old animals and drawings of the stages of degeneration. The paper also lacks statistical

analysis. Nevertheless, the group size was big enough to warrant confidence in the results. The increase in **lipofuscin deposits** was also confirmed by other investigators, both in human samples (Hirunagi 1983) and mouse ventral spinal cord and dorsal ganglia, as demonstrated by Sudan black B staining and electron microscopy (Samorajski et al., 1965). This was also shown in cell culture (Kim 1983) and in other models, such as aged dogs and pigs (Few, 1966).

Another pathology that can be observed in aging is **amyloid bodies accumulation** in human spinal cord (in posterior funiculus; Zhou et al., 1996). However, it is not known if donors were diagnosed with AD. In contrast, no “senile plaques” were detected in study conducted by Hirunagi (1983), even though staining of corpora amylacea is present to various degree in the spinal cord sections and the presence of neurofibrillary tangles was confirmed in three cases (donors AD status is unknown).

Another characteristic of brain aging is **demyelination** (Lintl and Braak, 1983; Sherin and Bartzokis, 2011); however, data concerning the spinal cord is sparse. Xie et al. (2013) report downregulation of myelin basic protein (MBP) and myelin oligodendrocyte glycoprotein (MOG) in spinal cord posterior funiculus and dorsal horn in 18 months old rat comparing to 2 months old control (also Parkinson et al., 2016 - decrease in myelin proteins shown by qPCR and immunoblotting). Myelin abnormalities were also reported in axons innervating soleus muscle in old rats (20-25 months old; Ansved and Larsson, 1990). There are also changes in myelin composition that include altered cholesterol metabolism. Cholesterol is an important component of myelin sheath and its homeostasis has been shown to be perturbed in aging, which may be a result of increased inflammation and can contribute to demyelination and age-related spinal cord dysfunction (Parkinson et al., 2016). Also Galbavy et al. (2017) report four cholesterol metabolism-related pathways to be differentially expressed in SC of 17-month old rats.

Increased neuroinflammation is another feature of aging, and **astrocytes and microglia activation** is often reported in aging spinal cord, as indicated by an increase in GFAP and Iba1 immunoreactivity in 18 months old rat dorsal horn and posterior funiculus of spinal cord (Xie et al., 2013; Duan et al., 2009). Besides increased GFAP, Parkinson et al. (2016) observe in dorsal and ventral funiculi also changes in Iba1 abundance, as well as an increase in CD68, a marker for microglia and macrophages. Kamiya et al. (2017) report GFAP as the most upregulated protein in a proteomic analysis of 2 years old and 8 weeks old rat SC. Lee et al. (2017) also observe an Iba1 upregulation in gerbil SC. Microglial activity was shown to contribute to age-related neuronal loss through TREM2 signaling (Linnartz-Gerlach et al., 2019), while depleting microglia prevents age-related decrease in motor unit number and NMJ disruption (Giorgetti et al., 2019). Interestingly, loss of TREM2 was shown to protect also from age-dependent cognitive decline (Qu and Li, 2020). Increased microglia activation was also reported in lumbar SC of aging dogs (Toedebusch et al., 2020). There is also significant age-related increase in astrocyte number (Xie et al., 2013). Moreover, the authors reported a significant negative correlation between both MBP and MOG downregulation and GFAP and Iba1 expression in aging rat brain, however they do not mention spinal cord in this analysis, even though the expression patterns of aforementioned molecules were similar both in brain and spinal cord. Activated astrocytes and microglia can contribute to demyelination by releasing toxic molecules, such as ROS, cytokines or complement system components. Proteomic analysis revealed also upregulation in peripherin and α B-crystallin proteins in aged rat SC (Kamiya et al., 2017). Peripherin can form aggregates that have been implicated as a toxic agent inducing neuronal death, therefore its accumulation during aging may contribute to neuronal loss. In contrast, α B-crystallin has a protective effect, as it is a heat shock protein and a chaperone with an anti-apoptotic activity (Robertson et al., 2001). The increase in inflammation-related

genes and glia activation further confirm the presence of age-related inflammation (Parkinson et al., 2016). Mild inflammatory state in aged rat SC was also reported by Nacka-Aleksić et al. (2017), as the levels of IL-1 β were increased (especially in males) and there was age and sex-dependent decrease in CX3CR1 (receptor for CX3CL1, microglia activation regulator). A para-inflammatory state was also reported in lumbar SC of middle-aged rats (17 months old) and pathway analysis (IPA) indicated complement cascade to be the most differentially expressed among all 13 canonical pathways that were identified (Galbavy et al., 2017). The complement pathway was shown to be activated in aging rhesus monkey brain, especially the early components of the pathway (Duce et al., 2006). The authors demonstrated that C3 covalently binds to myelin, suggesting that myelin and oligodendrocytes have potential to activate complement pathway. However, the presence of terminal complement components was not detected. The authors hypothesized that the activation of early complement components may be to some extent a normal biological process that is part of myelin metabolism, at least in the brain. Nevertheless, a clear complement pathway activation is observed with age. Another phenomenon that contributes to age-related α -MN loss, is **impaired autophagy** (Pareja-Cajiao et al., 2021). The authors report increased p62 levels, starting at 18 months of age and persisting at 24 months of age in cervical MNs in mice, which suggest a defect in degradation phase in the process of autophagy, as well as an increase in LC3 at 24 months of age, together with autophagosome accumulation.

Oxidative stress and decreased anti-oxidative defenses have also been implicated in aging (Sergiev et al., 2015). Antioxidant tolerance capacity was shown to be decreased in old mouse spinal cord (Duan et al., 2009): specifically, the expression of antioxidant enzymes such as heme oxygenase 1 (HO1) and NAD(P)H/quinone oxidoreductase 1 (NQO1) is decreased, as well as the expression of Nrf2, which is HO1 and NQO1 expression regulator. However, HO1 mRNA levels were shown to be

increased in aged rat SC (especially in males). At the same time the expression of iNOS was significantly increased, and there was a decrease in expression of the receptor for CX3CL1 which was shown to limit NO production by activated microglia. Together these findings suggest that oxidative balance is disrupted in aged rat SC, again especially in males (Nacka-Aleksić et al., 2017). However, McDonagh et al. (2016) suggest that defective redox signaling, rather than oxidative damage, contributes to the neuronal loss. The authors do not observe significant increase in oxidative damage in sciatic nerve of old mice, but report instead an increased superoxide activity and higher abundance of redox-sensitive proteins.

Another age-related change that was reported in the spinal cord concerns the **vasculature** (Qiu and Zhu, 2004). Capillary density in rat's spinal cord gray matter decreases significantly with age and both ventral spinal artery and central arteries often become tortuous in aged rats (above 22 months of age). Resulting decreased blood perfusion and lower accessibility to nutrients may contribute to neuronal stress and degeneration. However, we did not observe a change in blood vessel density in the lumbar spinal cord of 29 months old wildtype mice using anti-CD31 immunostaining (Piekarz et al., 2020).

SC aging consequences extend far beyond CNS, since they result in changes in the periphery, and can affect the muscular system. Age-related changes that take place in SC can affect individual's motor functions, as changes in lower motor neuron affect the whole motor unit (Lexell, 1997), which leads to impaired motor capabilities and contributes to muscle wasting as seen in age-related **sarcopenia** (Altun et al., 2012). Other peripheral changes with age concern both amplitude and duration of motor unit action potentials (Hayward, 1977) and slowed axonal conduction velocities (Doherty et al., 1993). Also, **neuromuscular junction is negatively affected**, as its morphology and function change with age. We can observe both pre- and post-synaptic

fragmentation and expansion as visualized by staining of pre-synaptic terminals (using for example neuronal ubiquitin decarboxylase PGP9.5) and motor endplates (staining with alpha-bungarotoxin which binds acetylcholine receptors). There is also less overlap of surface of presynaptic terminal and motor endplate (~70% for type IIb fibers, which are the fibers that are lost as first in aging; Mantilla, Sieck 2011). NMJ denervation and structural abnormalities, such as an enlargement of the postsynaptic area was also reported in a study on intercostal muscles in humans between 4-77 years of age (Wokke et al., 1990). The NMJ expansion also seen in aging might be a compensatory mechanism meant to counteract decreased physical activity that accompanies aging. Other changes within NMJ are decrease in synaptic vesicle density, which results in lower neurotransmitter release and synaptic transmission dysregulation (Jang and Van Remmen 2011). This phenomenon can also contribute to eventual further synaptic loss. Motor neurons can compensate for synaptic loss by axonal sprouting, but in aging this mechanism is also already impaired. Furthermore, aging-related NMJ degeneration disrupts signaling via neurotrophins, such as NT-3, NT-4, BDNF and GDNF, that are necessary for a proper activity-induced modification of synaptic communication, which further impairs neurotransmission (Jang and Van Remmen 2011). Besides all the above-mentioned structural changes occurring with age in the α -MNs, also neuronal function is affected, as well as synaptic plasticity, as shown in the studies on aging MNs in *Aplysia californica* (Kempsell and Fieber, 2015).

In summary, our knowledge about spinal cord aging is very limited compared to the amount of research dedicated to brain aging. Some processes can be universal for the whole CNS: both brain and spinal cord, but the scarcity of data concerning spinal cord does not allow yet for full comparison. Recently there has been increasing interest in investigating the effects of aging on the spinal cord, and the landscape emerging from the results includes α -MN loss and morphological changes, mild or 'para'-inflammatory

state combined with complement pathway activation, glial activation, increased oxidative stress and demyelination. Further studies are needed to elucidate underlying mechanisms and to connect all the separate pieces of the spinal cord aging puzzle.

1.2 INTERVENTIONS AIMING AT PREVENTING AGE-RELATED α -MOTOR NEURON LOSS

1.2.1 MATRIX METALLOPROTEINASES

Matrix metalloproteinases are a family of 23 zinc proteases that are highly expressed in development, however later in life their expression is very low, except for the process of wound healing or tissue remodeling after an injury. Their expression increases again in aging and is associated with neuroinflammation and pathological processes (Webster and Crowe, 2006; Jackson et al., 2010). MMPs degrade extracellular matrix components, but their role is not limited strictly to modifying the structure of ECM, as MMPs were also shown to influence numerous growth factors, such as EGF, basic FGF, VEGF, TNF- α , TGF- β , etc., by activating them through cleavage or releasing them from binding proteins (Sternlicht and Werb, 2001; Mott and Werb, 2004), which influences cellular growth. Furthermore, by remodeling ECM, MMPs affect cell migration, cell-to-cell communication and even apoptosis by interacting with Fas/FasL (McCawley and Matrisian, 2001; Musiał and Zwolińska, 2011). MMPs are strongly upregulated in injuries, such as spinal cord injury (MMP-12: Wells et al., 2003; MMP-1, 2, 9 and 12: Buss et al., 2007) and promote neuroinflammation (Könnecke and Bechmann, 2013). Although MMPs were linked to neurodegeneration, they can also have a beneficial effect in the central nervous system, in terms of regulating neurogenesis, myelination or axonal growth (Yong 2005). MMPs play also an important role in cancer progression and metastasis (e.g., Shiomi and Okada, 2003; Wang et al.,

2019; Zhang et al., 2019b; Rahmati Nezhad et al., 2021; Carey et al., 2021; Tellman et al., 2021), as well as in cell invasion in gastric epithelial cells infected with *Helicobacter pylori* (Bae et al., 2021).

The role of **MMPs in aging** is most often connected with skin photoaging, where UVB-induced skin damage leads to increased oxidative stress and upregulation of numerous MMPs, such as MMP-1, MMP-2 or MMP-9 (Quan and Fisher, 2015; Lee et al., 2021; Hwang et al., 2021; Jang and Kim, 2021). Another **mechanism** by which MMPs contribute to the aging process is by interfering with stem cell niche maintenance (Freitas-Rodríguez et al., 2017) and MMPs can be also found among SASP (senescence-associated secretory phenotype; Coppé et al., 2010). Besides playing an important role in skin aging, the role of MMPs in aging is also being recognized in brain aging and disorders (Romero et al., 2010; Morawski et al., 2014; Mukherjee and Swarnakar, 2015). However, MMPs do not only have a deleterious effect in aging, as they were linked to increased longevity in a study on Sicilian population that showed that long-living individuals exhibited increased MMP-2 activity in plasma (Cancemi et al., 2020).

Based on the literature and our previous study (Piekarz et al., 2020), two MMPs seem to be of a particular interest in the context of age-related α -MN loss: MMP-9 and MMP-12. **MMP-9** is expressed in numerous cell types, such as monocytes (Watanabe et al., 2018), macrophages and fibroblasts (Yabluchanskiy et al., 2013), and neutrophils where it is stored in granules and released upon degranulation (Yong 2005), as well as α -MNs (Fang et al., 2010; Kaplan et al., 2014; Piekarz et al., 2020). MMP-9 is connected to cardiovascular diseases (Yabluchanskiy et al., 2013), cancer malignancy and metastasis (Gautam et al., 2018; Dong et al., 2019), as well as to neuroinflammation and BBB breakdown (Song et al., 2015; Rempe et al., 2016; Rempe et al., 2018; Zhu et al., 2018; Hannocks et al., 2019; Qin et al., 2019; Nascimiento et al., 2021). A possible

mechanism regulating MMP-9's effect on BBB is modulating hedgehog pathway, as it was demonstrated in rat model of traumatic brain injury (Wu et al., 2020). It has also been implicated in cardiac and tenocytes aging (Yu et al., 2013; Yabluchanskiy et al., 2014; Padmanabhan Iyer et al., 2016). Moreover, MMP-9 was shown to cause pathological neovascularization in age-related macular degeneration (Kim et al., 2021). Suppressing MMP-9 (and MMP-2) and oxidative stress has also a protective effect on brain microvasculature in glucose-induced brain endothelium injury (Lv et al., 2021). MMP-9 as well as MMP-3 upregulation was also reported in brains of rat model of collagen-induced arthritis which exhibited lower abundance of tight junction proteins, such as zonula occludens-1 and occludin and increased transport of amyloid-beta through BBB (Lai et al., 2021). MMP-9 was also linked to neurodegeneration, as it is expressed in the spinal cord and the skin in SOD1 G93A ALS mouse model (Fang et al., 2010). Also, Kaplan et al. (2014) demonstrated that MMP-9 is behind the selective MN vulnerability observed in ALS, as it is upregulated selectively in fast motor neurons, which are the neuronal subpopulation that degenerates during the disease progression. MMP-9 expression in these neurons triggers axonal degeneration through increased ER stress and MMP-9 knockdown has a protective effect on MNs also in rNLS8 ALS mouse model (Spiller et al., 2019; see Łukaszewicz-Zajac et al., 2014 for review). Furthermore, population studies show that polymorphism in MMP-9-encoding gene was associated with predisposition for developing both ALS and PD (He et al., 2013). MMP-9 is also important in neuronal circuit development and excitability, as well as in synaptic plasticity (Yin et al., 2020), as MMP-9 KO was shown to influence apoptosis, dendrite formation and synaptogenesis in CA1 pyramidal neurons (Murase et al., 2016).

MMP-12, also known as macrophage elastase, is produced mostly by microglia and macrophages (Guan et al., 2019). MMP-12 expression was also reported in neurons and oligodendrocytes (Chelluboina et al., 2015b). MMP-12 has also links to apoptosis,

as its C-terminal domain induces apoptosis in tumors by translocating to the nucleus and initiates transcription of TRAIL, which triggers TRAIL-mediated apoptosis (Dandachi et al., 2017). MMP-12 affects gene expression and down-regulates anti-apoptotic genes, such as Bcl-xL, while up-regulating pro-apoptotic genes, such as Bad and Bcl-xS (Dandachi et al., 2017). Furthermore, MMP-12 was linked to increased brain damage in focal cerebral ischemia in rat model of middle cerebral artery occlusion and knocking it down decreases apoptosis, MMP-9 activation and TNF- α levels, but increases MBP levels suggesting a protective effect on myelin (Chelluboina et al., 2015b). MMP-12 was linked to increased inflammation (Razavian et al., 2016; Guan et al., 2019; Aristorena et al., 2019), however, similarly to other MMPs, its effect is not exclusively deleterious, since there have been reports suggesting that MMP-12 can have an anti-inflammatory effect in cornea injury mouse model by inactivating CCL2 (Wolf et al., 2019). However, MMP-12 is upregulated in aging brain and knocking it out decreases age-related neuroinflammation (Liu et al., 2013). MMP-12 was also linked to BBB disruption in focal cerebral ischemia (Chelluboina et al., 2015a), and contributes to progression of multiple sclerosis (see Rempe et al., 2016 and Chelluboina et al., 2018 for review).

Other MMPs have also been linked to neuroinflammation and neurodegeneration. For instance, MMP-3 was shown to induce apoptosis and death of dopaminergic neurons (N27 cell line culture derived from mesencephalon of rat and CATH.a cell line) through stimulating the production of mitochondrial ROS which in turn activates Nox1 (Choi et al., 2008; Cho et al., 2009; Kim et al., 2010; Choi et al., 2011a; Choi et al., 2014). In an MPTP mouse model of Parkinson's disease (PD), MMP-3 also induces neuroinflammation and BBB disruption (Shin et al., 2012; Chung et al., 2013; Leem et al., 2020). A recent study reported that lower expression of MMP-3, as well as MMP-9 lead to decreased dopaminergic neuron loss and decreased inflammation in rat model of PD (Jayaraj et al., 2019). In the same model, pharmacological blocking of

peroxisome proliferation-activated receptor gamma prevented the loss of dopaminergic neurons by decreasing microglia activation and downregulating the expression of MMP-3 in substantia nigra (Sadeghian et al, 2012). Treatments with PTIQ and ghrelin were also shown to rescue dopaminergic neurons in MPTP-induced mouse PD model by the same mechanism (blocking microglia activation and downregulating MMP-3 expression; Kim et al., 2007; Moon et al., 2009; Son et al., 2012). MMP-3 also cleaves alpha-synuclein and contributes to PD progression, and it's present in Lewy bodies in brains of PD patients (Choi et al., 2011b). MMP-7 is upregulated in brains of patients suffering from Alzheimer's disease (Sorrentino et al., 2021).

Given all the above-mentioned reports linking increased MMPs expression with neuroinflammation, apoptosis, neuronal loss and BBB disruption, as well as our previous study showing an increase in MMPs expression and abundance in aging spinal cord (Piekarz et al., 2020), MMPs, especially MMP-12 and MMP-9, seem to be an interesting target of interventions aiming at preventing age-related loss of α -MNs.

1.2.2 OKN-007 AS A THERAPEUTIC IN AGING

Aging is associated with neuronal loss and decline of muscle mass and function, known as sarcopenia, which has profound consequences both on the levels of individuals, such as lower quality of life, and the whole society and health care system (Beaudart et al., 2014; Steffl et al., 2017; Waltz et al., 2018). Our laboratory has previously shown that the denervation is a critical factor initiating sarcopenia (Muller et al., 2007) and in the previous study we observed a 41% loss of α -MNs in old wildtype C57Bl/6 mice (Piekarz et al., 2020). Currently there is no cure or treatment for the age-related α -MN loss.

OKN-007 (OKlahoma Nitron 007; disodium 4-(tert-butyl-imino) methyl benzene-1,3-disulfonate N-oxide, also known as NXY-059 and HPN-07) is a nitron-based

molecule that is being investigated as a treatment for glioblastoma (GBM; Towner et al., 2019a; Floyd et al., 2013). It is a disulfonyl-phenyl derivative of PBN (N-tert-Butyl- α -phenylnitrone; Altinoz and Elmaci, 2018), which was first investigated in 1985-1988 in brain ischemia/reperfusion injury in gerbils, however the nitrones were used in free radical-related research since the 1960s (Floyd et al., 2013). OKN-007 is a promising treatment in cancer, since it was reported to increase survival and decrease tumor volume in GBM xenografts (de Sousa et al., 2015a; Towner et al., 2019) and in diffuse intrinsic pontine glioma (Thomas et al., 2020), as well as in liver cancer (Yoon et al., 2018). OKN-007 also increases tumor sensitivity to temozolomide through decreasing angiogenesis and cell proliferation, while favoring apoptosis in models such as human G55 cell line or F98 gliomas in rats (Towner et al., 2013; Towner et al., 2019a; Smith et al., 2019). OKN-007 is currently in multiple clinical trials for the treatment of GBM (e.g., NCT03649464, NCT03587038, NCT01672463 - clinicaltrials.gov).

Although the **mechanism of action** is not fully elucidated, OKN-007 is believed to work at least in part, by inhibiting TGF- β 1, which also effects ECM and downregulates MMPs, especially MMP-12, as indicated by an RNA microarray analysis (Towner et al., 2019a). OKN-007 is also SULF2 inhibitor, which is a therapeutic target in liver cancer treatment (Yoon et al., 2018; Lanzi et al., 2017; Floyd et al., 2011). Furthermore, Hedgehog/GLI1 signaling has been reported to play role in OKN-007 therapeutic effect in hepatocellular carcinoma (Zheng et al., 2013). Moreover, OKN-007 suppresses free radical production in model of cerebral ischemia (Altinoz and Elmaci, 2018), and in F98 rat glioma model (de Souza et al., 2015b) through spin trapping (Floyd et al., 2011), as well as suppresses the expression of VEGFR-2 (de Souza et al., 2015c) and basic fibroblast growth factor (Floyd et al., 2013). However, OKN-007's mechanism of action in aging may differ from those reported in cancer.

In addition to **scavenging free radicals**, nitron compounds have been shown to have an **anti-inflammatory** effect. The OKN-007 parent compound, PBN, prevented microglia activation and loss of NMDA receptors in kainic acid-induced rat epilepsy model (Floyd et al., 2000). In IL-1 β -induced acute brain injury (Fan et al., 2010) PBN was shown to inhibit COX-2, decrease iNOS expression, and affect NF κ B signaling. In contrast, OKN-007 was reported to downregulate iNOS in C6 glioma in rats; Floyd et al., 2011. Treatment with OKN-007 also decreases inflammation and in LPS-induced encephalopathy in rats (Towner et al., 2019b). Another reported **neuroprotective** mechanism of action is inhibition of the release of cytochrome C from mitochondria by maintaining Akt phosphorylation in rat cerebral ischemia model (NXY-059; Yoshimoto et al., 2002). PBN was also reported to alleviate neuritic abnormalities in APP/PS1 mice, suggesting that nitrones can influence neuronal plasticity, however without affecting amyloid-beta plaques (Garcia-Alloza et al., 2010). Additionally, OKN-007 was shown to have a neuroprotective effect in blood-brain barrier (BBB) in vitro model, since it normalized increased BBB permeability after oxygen-glucose deprivation, as well as lowered tissue plasminogen activator levels, which restores proper function of endothelial cells, however it failed in clinical trials for acute ischemic stroke (Culot et al., 2009). However, the compound did not increase cell survival in human stem cell-derived neurons subjected to oxidative stress or ischemia-like injury by oxygen and glucose deprivation (Antonic et al., 2018), which may suggest that the neuroprotective effect of OKN-007 is mediated through other cells and processes absent in purely neuronal culture, such as an interactions with activated microglia, BBB/BSCB integrity, etc., and that the neuronal environment changes that occur with age may be an important factor driving the age-related neuronal loss.

Because oxidative stress and neuroinflammation were shown to contribute to cognitive decline in **neurodegenerative diseases** and in **aging**, and since nitrones have

an anti-inflammatory effect and have free radical scavenging ability, it was proposed that PBN-nitrones can have a therapeutic effect in those ailments (Floyd et al., 2013), however the potential therapeutic effect of OKN-007 in neurodegenerative diseases and aging has not been yet investigated. So far, OKN-007 (denominated in the study HPN-07) in combination with N-acetylcysteine (NAC) was shown to decrease tau levels in rat model of blast-induced TBI (Du et al., 2016), and has a protective effect on the inner ear and cochlear nucleus in rats exposed to noise (Lu et al., 2014; Ewert et al., 2017), and in blast-induce tinnitus (Lu et al., 2021), as well as in the rat blast injury model (Ewert et al., 2012). Also, monotherapy with OKN-007 (denominated in the study NXY-059) was shown to be neuroprotective in rat model of fluid percussion TBI (Clausen et al., 2008).

All the afore-mentioned properties of OKN-007 make it an ideal candidate for a neuroprotective therapeutic in aging and, to our knowledge, our study is the first to show its neuroprotective effect on α -MNs in aging.

1.2.3 OKN-007 IN ALS AS A MODEL OF ACCELERATED α -MOTOR NEURON LOSS

Amyotrophic lateral sclerosis (ALS), known as Lou Gehrig's disease, is a progressive neurodegenerative disease that leads to death of motor neurons in the brain and the spinal cord, which causes paralysis and eventually death due to the respiratory failure. The cause of the disease is still unknown, however there are several hypotheses concerning the **mechanisms** behind the disease development. The immunologic hypothesis suggests the autoimmune mechanism behind the disease (Ralli et al., 2019), based on the evidence of T lymphocyte and macrophage infiltration, increase in cytokine abundance, presence of immune complexes in the bloodstream, etc. IgGs isolated from ALS patients were shown to induce neuronal death in the primary motor neuron cultures (Ralli et al., 2019). Also, microglia activation has been demonstrated in ALS that was hypothesized to be caused by mutant SOD1 aggregates (via TLRs binding). Moreover,

microglia expressing mutant SOD1 were shown to have impaired phagocytosis (Ralli et al., 2019; Mezzini et al., 2019). Also, environmental factors, such as smoking, head trauma, heavy metal exposure, have been implicated in ALS pathogenesis (Ralli et al., 2019). Only up to 10% of ALS cases are familial ALS (fALS) caused by mutations in genes such as SOD1, C9ORF72, FUS, and TARDBP (Mezzini et al., 2019). The proposed mechanism behind the role of SOD1 in the disease development is a toxic gain of function, which may lead to excitotoxicity, oxidative stress, ER stress and mitochondrial disruption (Mezzini et al., 2019). TARDBP gene encodes a DNA/RNA binding protein, TDP-43, and the mutations can lead to protein aggregation in the cytoplasm, which depletes it in the nucleus; toxic gain of function was also suggested (Prasad et al., 2019. Ragagnin et al., 2019; Nakamura-Shindo et al., 2020). A proposed mechanism behind the deleterious effects of a mutation of another RNA-binding protein, FUS, are similar to those suggested for TDP-43. The mutations in non-coding part of C9ORF72 gene were shown to lead to the decreased gene transcription, however the last two gene mutations are not sufficient to cause disease in mouse model (Mezzini et al., 2019). Interestingly motor neuron size seems to play a role in susceptibility to ALS, since larger FF-MNs (fast fatigable) are more vulnerable and cell soma of SOD1 G93A mice is larger than wildtype control and it increases as the disease progresses (Ragagnin et al., 2019). Also, MMP-9 was suggested to be one of the factors involved in selective MN vulnerability (Kaplan et al., 2014). All in all, suggested mechanisms include impairment of nucleocytoplasmic transport, altered protein turnover, mitochondrial dysfunction, impaired RNA metabolism, neuroinflammation, excitotoxicity, defects in axonal transport, and oligodendrocyte dysfunction (Mezzini et al., 2019).

There are currently four **FDA-approved treatments** for ALS: Riluzole (Rilutek) has an ant glutamatergic effect and increases patients' survival (by ~90 days); Edaravone (Radicava) is a free radical scavenger which slows down decline of motor

function (a 33% decrease after 6 months of treatment; Jaiswal 2019); Nuedexta aims at treating pseudobulbar affect that accompanies the disease (Smith et al., 2017), and Tiglutik is a liquid formulation of Riluzole for patients with difficulties swallowing, and there is still no cure for the disease (Dyer and Smith, 2016).

New treatment options are being explored but have not made their way to the clinic yet. Among novel, not yet FDA-approved therapies, there is for instance dietary supplementation with SAM (S-adenosyl methionine), which delayed the disease onset in SOD1 G93A ALS mouse model by 2-3 weeks (Suchy et al., 2010). The supplementation also increased α -MN survival, decreased neuroinflammation, and increased anti-oxidant activity, while reducing protein carbonylation. In a study from 2019 Sawada et al. showed that overexpression of synapsin-driven caveolin-1 (Cav-1) rescues α -MNs and mitochondrial morphology, also in SOD1 G93A mice. In the same year another study proposed cromolyn sodium as a potential treatment, since it increases α -MN survival, protects NMJs and decreases neuroinflammation (Granucci et al., 2019). The treatment also delays the disease onset and increases lifespan, however only in female mice. Verapamil is another therapeutic investigated as a potential treatment for ALS. This L-type channel Ca^{2+} blocker was shown to increase α -MN survival and lifespan through promoting autophagy (Zhang et al., 2019a). The drug has been also successfully used in combination with Riluzole (Yang et al., 2018). Anti-inflammatory drugs are also being investigated as a potential treatment modulating ALS progression. For instance, all-d-peptide RD2 (RD2RD2), an anti-inflammatory peptide administered intraperitoneally for 4 weeks decreased inflammation and slowed down the disease progression, as well as protected MNs in motor cortex of G93A mice (Post et al., 2021). Another approach involves stem cell therapy. Intravenous administration of human bone marrow stem cells into SOD1 G93A ALS mouse model improves BSCB integrity and repair, as well as decreases microhemorrhaging (Eve et al., 2018; Garbuzova-Davis et al., 2018). Also

human bone marrow endothelial progenitor cells were used successfully in symptomatic ALS mice (Garbuzova-Davis et al., 2019), as well as human mesenchymal stem cells that after combined intrathecal and intramuscular administration decrease necroptosis in SC of SOD1 G93A rats (Řehořová et al., 2019). Gene therapy has also been explored in the context of treating ALS (see Amador et al., 2021 for review), such as AAV-mediated mutated SOD1 silencing in both G93A mice and marmosets (e.g., Foust et al., 2013; Borel et al., 2016; Bravo-Hernandez et al., 2020).

SOD1 G93A ALS mouse model is a frequently used model in ALS research aimed at discovering new treatments, as we can see in the above-mentioned studies. This model was developed by introducing the human SOD1 gene with a mutation common in familial ALS, namely a point mutation at aminoacid at position 93 causing a substitution of glycine by alanine (Dal Canto and Gurney, 1994; Ripps et al., 1995). Mice become symptomatic around P90-P100 and exhibit gradual loss of α -MNs, hindlimb tremor, NMJ degeneration and progressive muscle weakness leading to paralysis. End-stage occurs around P150. The model recapitulates as well intense neuroinflammation, in terms of for example astrocyte activation (Duval et al., 2018) as well as microglia activation, and the latter leads to α -MN death through activating NF- κ B pathway (Frakes et al., 2014). Impaired cytoplasm-nuclear transport leading to the disruption of DNA repair processes also contributes to neuronal death (Li et al., 2019). G93A mice also exhibit dysregulated lipid metabolism (Chaves-Filho et al., 2019), as well as mitochondrial dysfunction, as glycolysis, β -oxidation, and mitochondrial metabolism are affected in this mouse model, and the mitochondrial dysregulation precedes the disease onset (Pharaoh et al., 2019). Vascular changes also accompany the disease progression and start before neuronal degeneration and go hand in hand with BSCB disruption due to a decrease in abundance of tight junction proteins, such as claudin-5, occluding and ZO-1 (Zhong et al., 2008).

In our study we propose that **OKN-007** can be a promising potential treatment for ALS. Since one of the ALS treatments, Radicava, is a free radical scavenger, one of the reasons OKN-007 could be a potential treatment for ALS is that it also acts as free radical scavenger. Also, ALS has a strong inflammatory component, and nitrones were shown to decrease microglia activation (Floyd et al., 2000). Increased BBB permeability reported in ALS is another potential target, since OKN-007 normalizes increased BBB permeability in vitro (Culot et al., 2009) and BSCB permeability in aging wildtype mice, as we demonstrate in Chapter IV. In addition, since MMP-9 has been implicated in susceptibility to ALS (Kaplan et al., 2014), OKN-007 seems to be a promising candidate for ALS treatment, since it was shown to decrease MMP expression in glioblastomas (Towner et al., 2019a). Lastly, another OKN-007's property which makes it an ideal drug candidate for ALS is its beneficial effect on neuronal survival.

1.2.4 TROPONINS IN THE NERVOUS SYSTEM AND LINKS TO THE AGING PROCESS

Troponins are a group of three proteins forming a complex that is crucial for muscle contraction and can be found in the skeletal and cardiac muscles (Gomes et al., 2002; Takeda et al., 2003). The three troponins are **troponin I, T and C** and each of them serves a distinct purpose in the process of contraction. The troponins are encoded by multiple genes and have numerous isoforms specific for fast- and slow-twitching fibers, as well as for cardiac muscle. Troponin I is an inhibitory subunit which binds actin and inhibits actin-induced myosin-ATPase activity, troponin T anchors the complex to tropomyosin, while troponin C has calcium binding capability. Calcium binding induces conformational change of troponin I that influences complex's interaction with the thick filament in the muscle. A disruption of the complex has profound consequences for the

muscle physiology and leads to the development of myopathies (Chang et al., 2008; Dai et al., 2020).

Besides forming the complex necessary for muscle contraction, troponins were shown to also have so-called **non-canonical localization**, namely to be expressed in non-muscle cells. The non-canonical expression of troponins was demonstrated both in healthy (e.g., in liver: Berezowsky and Bag, 1992; Grewal and Bag, 1996) and diseased tissue (in numerous tumors: Casas-Tintó et al., 2016, as well as cancer cell lines: Johnston et al., 2017). In this context troponins have often cytotoxic effect, as for instance troponin T fragments induce apoptosis in C2C12 and HEK cell lines (Jeong et al., 2009) and the presence of troponin T in the nucleus was shown to promote apoptosis in aging skeletal muscle (Zhang et al., 2013a and b). Non-canonical troponin expression often involves also nuclear translocation and association with DNA, which has effect on chromosomal stability, cell division and gene expression (Sahota et al., 2009; Johnston et al., 2017, Nunez Lopez et al., 2018). Moreover, troponin T was shown to translocate to the nucleus and act as a transcription factor that influences the expression of voltage sensing calcium channel subunit $\alpha 1$ in the skeletal muscle by binding to the promoter of *Cacna1* gene (Zhang et al., 2016).

Non-canonical presence of troponins includes also **neuronal tissue**, as troponin C was reported to be present in the brain (Fine et al., 1975), however its role in the neurons remains unknown. It was speculated though that the troponins may play a role in axonal transport. A subsequent publication demonstrated the presence of troponin C in dorsal root ganglion neurons of chicken and rat (Roisen et al., 1983), while Berezowsky and Bag (1992) further confirmed the presence of troponin C in the brain at the level of mRNA. In the following years the presence of troponin T was reported in various regions of mouse brain, such as hippocampus, hypothalamus, cortex, amygdala

and cerebellum (Lowe et al., 2007) and troponin C was detected in the layer 4 pyramidal neurons of visual cortex (Lyckman et al., 2008).

The non-canonical presence of troponins in the neurons, as well as their links to apoptosis and aging process make troponins an interesting target in the investigation of age-related α -MNs loss. The fact that troponin T was shown to act as a transcription factor suggests that troponins can indeed influence gene expression and similar process may take place in the nuclei of aging α -MNs.

DISSERTATION OVERVIEW

The aim of the work presented in this dissertation is to understand the changes occurring in the aging spinal cord that can contribute to the age-related α -MN loss and identify the interventions that can prevent the loss of α -MNs in aging.

Chapter II: Molecular changes associated with spinal cord aging

Age-related muscle weakness and loss of muscle mass (sarcopenia) is a universal problem in the elderly and our previous studies indicate α -MNs play a critical role in this process. The **goal** of the current study is to describe changes occurring in the aging spinal cord that contribute to loss of innervation and the downstream degenerative processes that occur in skeletal muscle. We **hypothesized** that similarly to neuronal loss in the aging brain, there is an age-related α -MN loss which is exacerbated by increased inflammation and BSCB permeability. To test our predictions, we aged in our colony wildtype mice and assessed α -MN number, neuronal morphology, demyelination, as well as inflammation levels and changes in the transcriptome.

Our results demonstrate that the number of α -MNs is decreased in the spinal cord of wildtype mice during aging, beginning in middle age (18 months of age) and reaching a 41% loss by 27 months of age. We also observe demyelination and mild inflammation, including astrocyte and microglia activation and an increase in levels of

cytokines such as sICAM-1. We identified changes in metabolites consistent with decreased neuronal viability, such as reduced levels of N-acetyl-aspartate. Cleaved caspase-3 is more abundant in spinal cord from old mice, suggesting that apoptosis contributes to neuronal loss. RNA-seq analysis revealed changes in the expression of a number of genes in spinal cord from old mice, in particular genes encoding ECM and a 172-fold increase in MMP-12 expression. Furthermore, BSCB permeability is increased in old mice, which may contribute to alterations in spinal cord homeostasis and exacerbate neuronal distress. Together these data show for the first time that the spinal cord undergoes significant changes during aging, including progressive α -MNs loss that is associated with low-grade inflammation, apoptosis, changes in ECM, myelination, and vascular permeability.

Chapter III: Knocking out MMP-9 and MMP-12 has a differential and sex-specific effect on age-related α -motor neuron loss and decline in muscle mass

Matrix metalloproteinases recently receive more attention in the context of aging and neuronal loss (Choi et al., 2008; Cho et al., 2009; Kim et al., 2010; Romero et al., 2010; Choi et al., 2011a; Choi et al., 2014; Morawski et al., 2014; Mukherjee and Swarnakar, 2015). Our previous study (Piekarz et al., 2020) showed that MMPs are upregulated in the spinal cord of old wildtype mice. MMP-12 is the most upregulated transcript in aging spinal cord, and MMP-2, MMP-9 and MMP-14 expression is also affected. Based on those results and existing literature that shows MMPs' role in neuroinflammation, microglia activation and BBB disruption (Shin et al., 2012; Chung et al., 2013; Razavian et al., 2016; Guan et al., 2019; Aristorena et al., 2019; Leem et al., 2020) we selected MMP-12 and MMP-9 as targets for manipulation in spinal cord aging. Our **goal** was to investigate the link between the expression of MMP-9 and MMP-12 and increased inflammation and α -MN loss in aging. We **hypothesized** that knocking out MMP-9 and MMP-12 leads to a decrease in inflammation, microglia activation, as well as

decreased BSCB permeability, which would promote α -MN survival. To answer that question, we obtained MMP-9 KO and MMP-12 KO mice, bred and aged them in our colony, until the mice reached 25 months of age and were euthanized. The neuroinflammation, BSCB integrity and α -MN number were then assessed to see whether decreased MMP expression is protective in aging. Both MMP-9 KO and MMP-12 KO old mice exhibited higher α -MN number comparing to age-matched wildtype control. Interestingly, when male and female mice were analyzed separately, only the male mice showed a significant increase in α -MN number. The sex-specific response was also visible if it comes to neuroinflammation, as only male mice (both MMP-9 and MMP-12 KO) showed decreased microglial density and activation. Neither knockout had any effect on astrocyte activation in neither male nor female mice. Moreover, MMP-12 KO mice exhibited lower BSCB permeability, which was also reflected in a higher abundance of claudin-5, one of the tight junction proteins crucial for maintaining tight junction integrity.

Chapter IV: OKN-007 decreases age-related alpha-motor neuron loss and improves muscle function in wildtype mice

Neuronal loss and decline of muscle mass and function, also known as sarcopenia (Waltz et al., 2018) are common features of aging. Other common characteristics of the aging process are increased oxidative stress and inflammation, and a compound known as OKN-007 was shown to have both anti-oxidative and anti-inflammatory properties (Floyd et al., 2011; de Souza et al., 2015b; Altinoz and Elmaci, 2018). OKN-007 is being investigated extensively in cancer research as a treatment for glioblastoma (Floyd et al., 2013; Towner et al., 2019), however the compound's properties make it an ideal candidate for a therapeutic in neurodegeneration and aging.

Our **goal** was to determine, if treatment with OKN-007 decreases age-related inflammation and has a neuroprotective effect on α -MNs. We **hypothesized** that, since

PBN, OKN-007 parent compound, has anti-inflammatory properties and decreases microglia inflammation (Fan et al., 2010), the treatment with OKN-007 would decrease age-related inflammation in the aging spinal cord. Moreover, since MMPs were linked to increased BBB permeability (Yang et al., 2007; Yang and Rosenberg, 2011; Rempe et al., 2016; Rempe et al., 2018) and OKN-007 was shown to normalize MMPs expression which is upregulated in aging (Towner et al., 2019; Piekarz et al., 2020), we predicted that OKN-007 can decrease BSCB permeability which is observed in spinal cord aging (Piekarz et al., 2020). We further hypothesized that both decreased inflammation and improved BSCB integrity promote α -MN survival, especially that OKN-007 was shown to have neuroprotective properties (Clausen et al., 2008).

To test that hypothesis, we treated wildtype C57BL/6J mice with OKN-007 in drinking water starting at 16 months of age, until mice reached 25 months of age and were sacrificed and tissues were collected for further analyses. Indeed, the treated mice exhibited a higher number of α -MNs (on average 15.45 α -MNs per ventral horn comparing to 11.44 in untreated controls), as well as lower levels of cleaved caspase-3, suggesting a decrease in apoptosis. OKN-007 decreased microglia proliferation and activation, without affecting astrocyte proliferation, as well as improved BSCB integrity. Moreover, the treatment decreased NMJ fragmentation and down-regulated markers of denervation, as well as significantly improved muscle contractile force, but did not affect hindlimb muscle mass. However, the treatment improved the age-related lean mass loss. The RNA-seq analysis revealed that the transcriptome of treated mice is altered and the changes involve the transcripts linked to pathways and diseases such as fibrotic pathway, organismal death, cell death, movement disorders, and quantity of neurons. Our findings suggest that OKN-007 can be a potential therapeutic aiming at preventing α -MN loss and improving muscle function in aging. Further studies are needed though to pinpoint the precise mechanism of action of OKN-007 in the context of aging.

Chapter V: OKN-007 treatment slows down disease progression in SOD1 G93A ALS mouse model by decreasing alpha-motor neuron loss and microglia activation, without affecting muscle mass at P145

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease characterized by an accelerated loss of motor neurons in both brain and spinal cord. In a previous study, we demonstrated that OKN-007, an antioxidant with anti-inflammatory and neuroprotective properties decreases age-related α -MN loss in wildtype mice, therefore our **goal** in this study was to determine, whether the compound can protect α -MNs also in mouse models of accelerated α -MN loss, such as SOD1 G93A mice. We **hypothesized** that, since ALS has a strong inflammatory component, and OKN-007 was shown to have anti-inflammatory effect and to decrease microglia activation (Fan et al., 2010; Floyd et al., 2011; de Souza et al., 2015b; Altinoz and Elmaci, 2018), as well as to promote neuronal survival (Clausen et al., 2008; Lu et al., 2014; Ewert et al., 2017), the treatment with OKN-007 of SOD1 G93A mice will lead to increased α -MN survival.

To test our hypothesis, we treated SOD1 G93A mutant mice and wildtype mice with OKN-007 in drinking water starting at P60 (before the mice become symptomatic, which occurs around P90-100) until sacrifice around day P145, just prior to end-stage. We monitored the disease progression and changes in body weight twice a week. The treatment did not delay the disease onset, however, it significantly slowed down the disease progression. At sacrifice, the treated group did not progress beyond the score of 3, and majority of the mice reached only the score of 2, whereas the highest score in the untreated group was 4 and the most frequent score was the score of 3. The treatment prevented α -MN loss, however it did not affect body weight or the hindlimb muscle mass. We also looked at the NMJ structure and the immunostaining revealed that the treated mice has less fragmented NMJs. OKN-007 also significantly decreased microglia activation, without affecting though the astrocyte proliferation. Furthermore, perfusion

MRI revealed that OKN-007 normalized the blood flow which is significantly decreased in untreated G93A mice comparing to age-matched wildtype controls. Our results suggest that, due to its neuroprotective and anti-inflammatory effect, OKN-007 may be a potential treatment aiming at slowing down ALS progression by increasing α -MN survival.

Chapter VI: Troponins are expressed in alpha-motor neurons of wildtype mouse and can translocate to the nucleus to potentially act as transcription factors

Troponins were shown to have a so-called non-canonical localization which denotes the presence of troponins in non-skeletal muscle cells, such as cancer cells (Johnston et al., 2017). The literature also mentions the expression of troponin C in the neurons (Fine et al., 1975; Roisen et al., 1983), however this surprising troponin's localization wasn't further investigated and its role in the neuron remains unknown. Our previous study (Piekarz et al., 2020) reported the expression of troponin I fast-skeletal type in the spinal cord of old wildtype mouse (C57BL/6) and showed that its mRNA levels increase with age ~40-fold.

The **goal** of our study was to confirm the presence of troponins in the spinal cord and pinpoint its precise localization, as well as investigate their role in the aging process. Given the troponins' capability of translocating to the nucleus and binding DNA (Sahota et al. 2009; Johnston et al., 2017), we **hypothesized** that troponins can translocate with age to the nucleus of α -MNs and influence gene expression in a way that would contribute to the aging process. Firstly, to confirm the presence of troponins in α -MNs on the level of the protein, we performed immunoblotting, which show the presence of all three troponins in the spinal cord of young and old wildtype mice. Additionally, the results show that the abundance of troponin I and C, but not T, increases with age. Furthermore, the immunostaining revealed that the troponins localize to α -MNs. Secondly, nuclear isolation and subsequent immunostaining demonstrated the presence of troponins T and I, but not C in the nuclei of α -MNs of old wildtype mice. Moreover,

both troponins were also present in the nuclei from SOD1 KO and SOD1 G93A mice, which are an example of a model exhibiting accelerated α -MN loss (the former is considered a model of accelerated aging, while the latter is one of the mouse models of ALS), which suggests the link between the nuclear presence of troponins and increased α -MN loss. Lastly, we confirmed the presence of troponins I and T in the nuclei of α -MNs of wildtype old mice, as well as young mice and determined their binding sites by performing CUT&RUN-seq. We attribute the discrepancy between the results of nuclear isolation and CUT&RUN-seq regarding the nuclear presence of the troponins in young mice to the difference in sensitivity of both methods, as the CUT&RUN-seq is far more sensitive than western blotting. The results show an age-dependent differential binding of the troponins I and T in the mouse genome. Together, the data strongly support the presence of troponins in the α -MNs and their involvement in the aging process. Further studies are needed to determine the precise effect of troponins' age-dependent differential binding on gene expression, however we hypothesize that there are three potential mechanisms through which the troponins can influence gene expression: (1) by affecting the expression of the genes located nearby the binding sites, (2) by affecting the expression of the genes located on different chromosomes but that are in a physical proximity of troponin binding sites due to the complex 3D chromatin structure, and (3) by competing for binding sites with other TFs with common DNA motifs, such as FOXD3 which shares the binding sequence with the old mice-specific binding sites of troponin T.

Summary

Spinal cord aging is characterized by α -MN loss, as well as by changes in neuronal morphology (swelling of the soma) and axonal loss. Other changes that accompany SC aging are increased inflammation due to increased microglia and astrocyte activation, demyelination, increased BSCB permeability and upregulation of fibrotic pathway resulting in a higher transcription of MMPs, especially MMP-12, and

other ECM components (Chapter II). We demonstrate that knocking out MMP-9 and MMP-12 has a beneficial effect on the aging SC, as it decreases age-related α -MN loss, as well as decreases neuroinflammation and improves BSCB integrity (Chapter III). Furthermore, a treatment with OKN-007, an anti-inflammatory and neuroprotective compound, also promotes α -MN survival in aging, decreases microglia activation and protects BSCB (Chapter IV). The treatment also proves beneficial in SOD1 G93A ALS mouse model which exhibits accelerated α -MN loss. Administering OKN-007 to the mice before they become symptomatic (from P60 to ~P145) prevents α -MN loss, decreases microglia activation and slows down the disease progression without affecting disease onset (Chapter V). We also discovered and described for the first time that troponins, proteins canonically associated with the contractile apparatus present in the cardiac and skeletal muscles, are also present in α -MNs and the abundance of troponins I and C increases with age. Troponins I and T can translocate to the nucleus where they differentially bind DNA in an age-dependent manner and can contribute to dysregulation of gene expression in aging (Chapter VI). Together, these studies describe the changes occurring in the aging spinal cord and identify the interventions that decrease age-related α -MN loss, as well as promote α -MN survival in a mouse model of accelerated α -MN loss, which can also lead to a new treatment for ALS.

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CHAPTER II

MOLECULAR CHANGES ASSOCIATED WITH SPINAL CORD AGING

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KMP created the experimental design together with HVR; KMP performed the experiments, analyzed the data, plotted the results in figures 2.1; 2.2 b-d; 2.4 a-b, e-f; 2.5; 2.6; 2.9; Supplemental Figures A.1-2; plotted the results in figure 2.3 b,d; 2.4 d; 2.7; Supplemental Figure A.3; created Supplemental Data Tables 1-4. KMP drafted the manuscript.

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Abstract

Age-related muscle weakness and loss of muscle mass (sarcopenia) is a universal problem in the elderly. Our previous studies indicate that alpha-motor neurons (α -MNs) play a critical role in this process. The goal of the current study is to uncover changes in the aging spinal cord that contribute to loss of innervation and the downstream

degenerative processes that occur in skeletal muscle. The number of α -MNs is decreased in the spinal cord of wildtype mice during aging, beginning in middle age and reaching a 41% loss by 27 months of age. There is evidence for age-related loss of myelin and mild inflammation, including astrocyte and microglia activation and an increase in levels of sICAM-1. We identified changes in metabolites consistent with compromised neuronal viability, such as reduced levels of N-acetyl-aspartate. Cleaved caspase-3 is more abundant in spinal cord from old mice, suggesting that apoptosis contributes to neuronal loss. RNA-seq analysis revealed changes in the expression of a number of genes in spinal cord from old mice, in particular genes encoding extracellular matrix components (ECM) and a 172-fold increase in MMP-12 expression. Furthermore, blood-spinal cord barrier (BSCB) permeability is increased in old mice, which may contribute to alterations in spinal cord homeostasis and exacerbate neuronal distress. Together these data show for the first time that the spinal cord undergoes significant changes during aging, including progressive α -MNs loss that is associated with low-grade inflammation, apoptosis, changes in ECM, myelination, and vascular permeability.

Introduction

Our previous studies have demonstrated that changes in motor neurons and preservation of intact muscle innervation are important contributing factors in age-related muscle atrophy and loss of muscle contractile function. We have reported an age-related loss of neuromuscular junction (NMJ) innervation (Jang and Van Remmen, 2011), reduced synaptic transmission (Ivannikov and Van Remmen, 2015) and decreased nerve conductance velocity (Walsh et al., 2015). Here we extend our studies to define age-related changes in spinal cord that could have a direct impact on alterations in the neuromuscular junction and loss of innervation during aging. While there are numerous studies on age-related changes in brain function, aging of the spinal cord remains an

understudied topic. Loss of motor neurons (MNs) and an increase in neuronal cell body size with age have previously been reported in aging spinal cord (Zhou et al., 1996; Kawamura et al., 1977a; Kawamura et al., 1977b; Tomlinson and Irving, 1977; Machado-Salas et al., 1977); however, the underlying mechanisms and impact of these changes remain unknown.

The goal of this study was to investigate the landscape of changes occurring in aging spinal cord that may contribute to alpha motor neuron loss. We conducted an analysis of gene expression changes using RNA-seq and measured age-related changes in a number of potential contributing factors including inflammation, apoptosis, metabolite changes, demyelination, vascular permeability, and protein synthesis. The RNA-seq results reveal a number of changes in the transcriptome of old mouse spinal cord comparing to young, in particular, induction of MMPs and extracellular matrix (ECM) components. Our comprehensive analysis indicates a number of other changes consistent with alpha motor neuron death including activation of apoptosis, changes in metabolites consistent with neuronal dysfunction, decreased myelination and an increase in BSCB permeability.

Results

α -Motor neuron number and morphology changes with age

We observed an age-dependent loss of α -MNs in the ventral horn of the spinal cord from older mice (Fig. 2.1 a-c). The α -MN number decreases progressively with age, and middle-aged mice (17 months old) exhibit a 17% α -MN loss comparing to 3-6 months old young mice that does not reach statistical significance, while the old mice (24-27 months) have 41% less MNs per ventral horn than young mice. The rate of α -MNs loss increases between 17 and 24-27 months and we observe an average loss of four α -MNs during this time frame. α -MN morphology also changes with age, with the most

pronounced change being increased neuronal cell body area (Fig. 2.1 d-e). The timing of the cell body

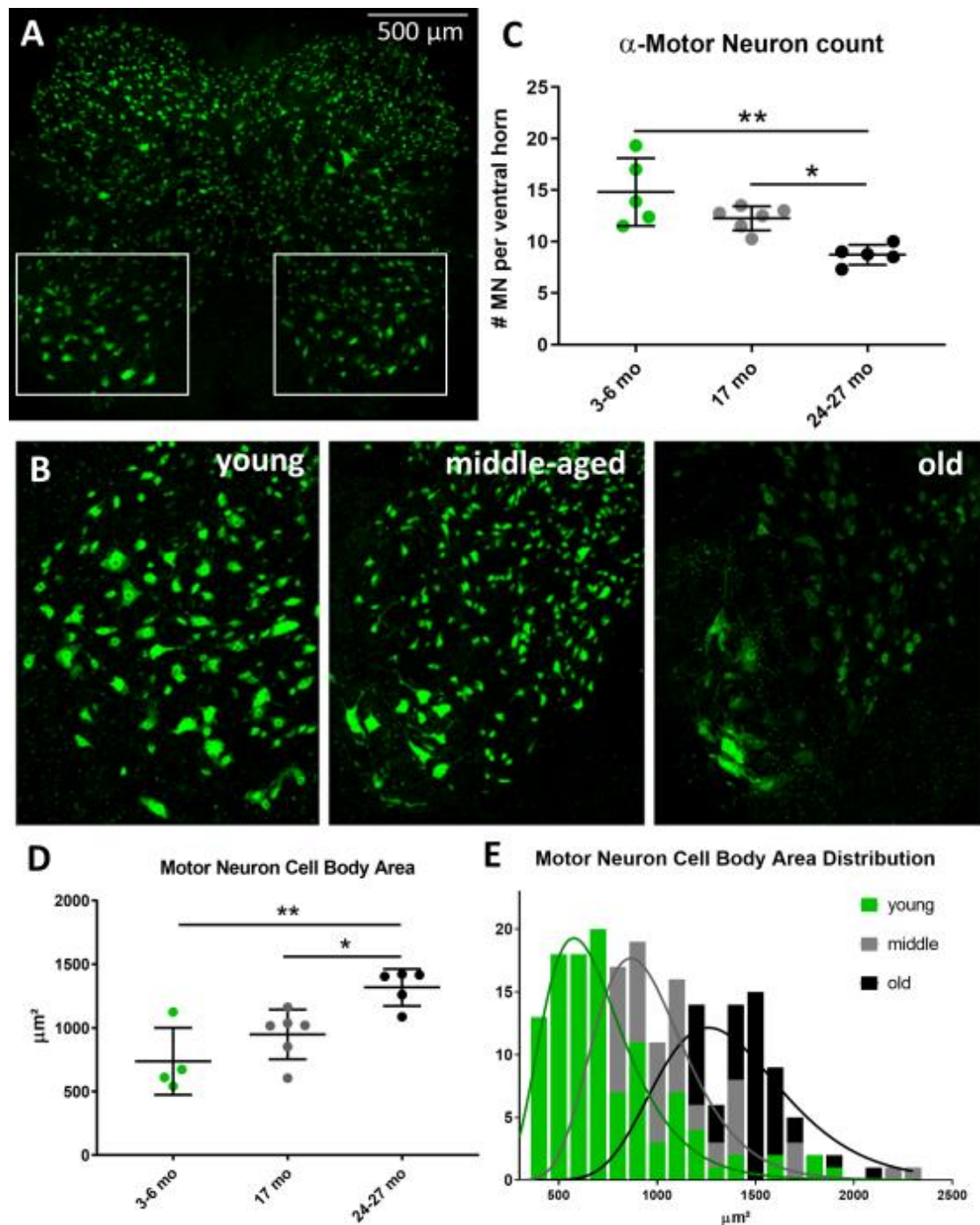


Figure 2.1. Age-related α -MN loss and morphological changes. **A** – Anti-NeuN staining of a spinal cord section (4x) with marked location of the region of interest,

ventral horns, where α -MNs reside; **B** – a representative image of a ventral horn (10x) of a young (3-6 mo), middle-aged (17 mo) and old (24-27 mo) mice, used for α -MN count and area measurement; **C** – alpha-MN count. There is a significant α -MN loss (one-way ANOVA, $F = 11.38$, $p = 0.0014$) in old mice ($n=5$, mean $\pm$ SD = 8.713 ± 0.97) comparing to young ($n = 5$, mean $\pm$ SD = 14.82 ± 3.28 , Tukey's multiple comparisons post hoc test, $p = 0.0010$), and middle-aged group ($n = 6$, mean $\pm$ SD = 12.25 ± 1.18 , $p = 0.0327$); scorer was blinded; **D** – There is an age-related increase in α -MN cell body area (one-way ANOVA, $F = 9.844$, $p = 0.0029$) in old mice ($n=5$, mean $\pm$ SD = 1318 ± 144.4) comparing to young ($n = 4$, mean $\pm$ SD = 737.9 ± 263.6 , Tukey's multiple comparisons post hoc test, $p = 0.0027$), and the middle-aged mice ($n = 6$, mean $\pm$ SD = 949.2 ± 195 , $p = 0.0260$); scorer was blinded; **E** – the distribution of α -MN cell body area is right-shifting with age, indicating age-dependent increase in cell body size.

swelling seems to follow a similar pattern as the α -MN loss. As shown in Figure 2.1 e, there is an age-related right shift in neuronal cell body area distribution, as well as an increased variance in α -MN cell body area in old mice. In addition to measuring α -MN count on spinal cord sections immunostained for NeuN, we also performed nuclear fast red staining (Appendix A Supplemental Figure A.1) to make sure that the effect we see is not NeuN staining-dependent, and the results of all methods consistently support α -MN loss in aged spinal cord.

Axonal loss and a slight demyelination accompany spinal cord aging

Axons present in aged spinal cord demonstrate similar changes as those occurring in aging α -MN cell bodies, namely a decrease in number and increase in size (Fig. 2.2 a-c). Analysis by electron microscopy reveals axonal loss (a 40% decrease) and swelling, as well as a decreased myelin content in spinal cord from old mice compared to young mice. The ratio of the axon diameter to myelin thickness increases with age, due to a decrease in myelin thickness and an increase in axonal diameter. The slopes of the regression equation describing the relationship between the myelin thickness and the axon diameter do not differ (Fig. 2.2 b), suggesting that the general property that bigger axons tend to have a thicker myelin sheath, remains true in spinal cord from old mice. However, the intercepts of the regression equation are significantly different ($p <$

0.0001), demonstrating that axons of a similar diameter have a thinner myelin sheath in old mice comparing to young mice. Furthermore, we observe an age-related decrease in

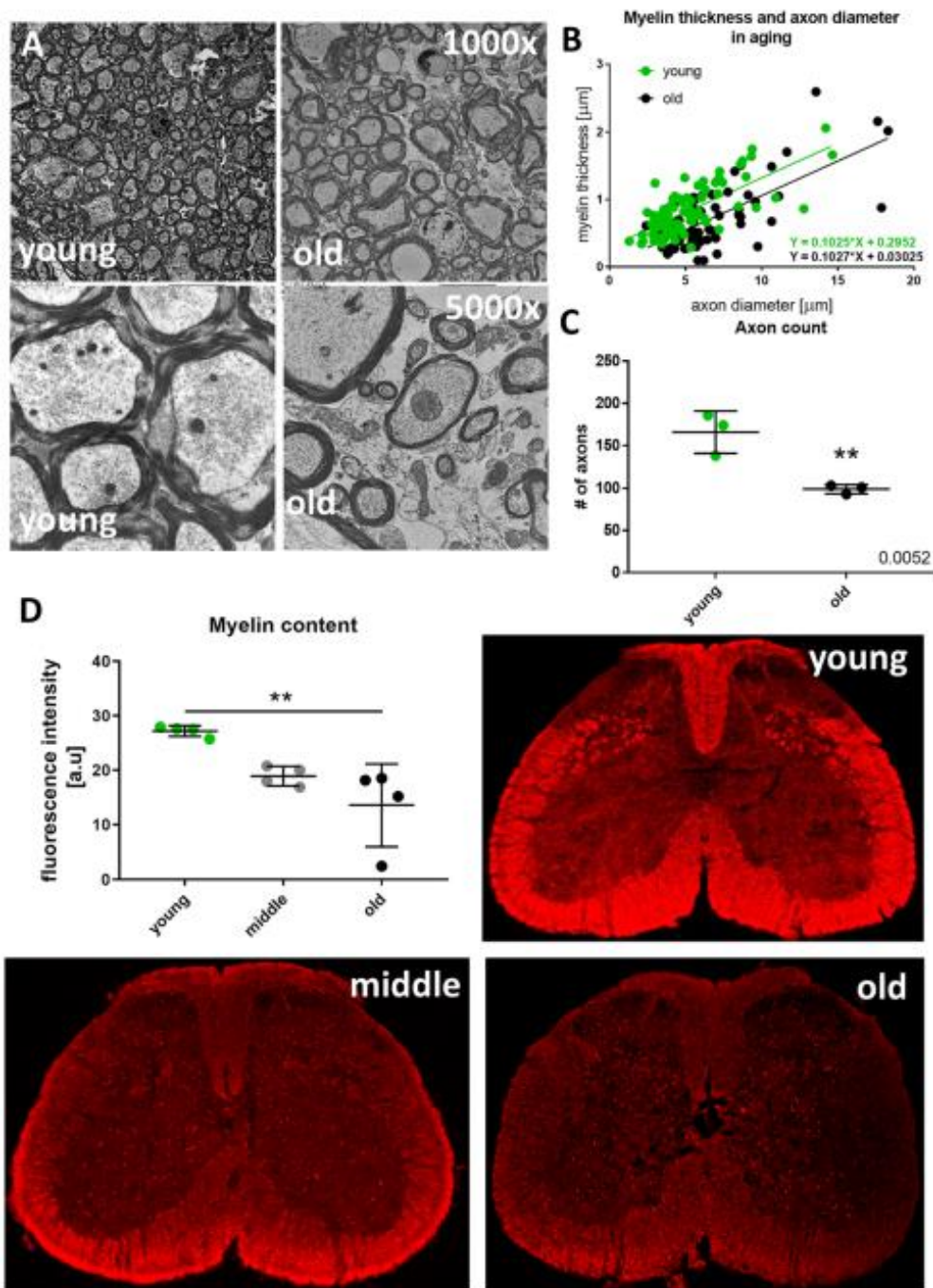


Figure 2.2. Age-related axonal loss and demyelination. **A** – A representative electron micrography (1000x – upper row, 5000x – lower row) of the white matter adjacent to the ventral horn of young (4 mo) and old (24-25 mo) mice; **B** – myelin thickness and axon

diameter relationship in young and old mice. There is a significant difference between intercepts in linear regression equations ($p < 0.0001$), while the slope remains the same ($p = 0.9935$); **C** – Aging is associated with axonal loss, since there is a significant decrease in the number of axons (one-tailed unpaired t-test, $t = 4.545$, $p = 0.0052$) between young ($n = 3$, mean $\pm$ SD = 166 ± 24.98) and old mice ($n = 3$, mean $\pm$ SD = 99 ± 5.29); **D** – a representative image of spinal cord section (4x, lumbar) stained with fluoromyelin and quantification of fluorescence intensity; we observe a decrease in fluorescence intensity between young (3-7 mo, $n = 4$, mean $\pm$ SD = 27.20 ± 0.98) and old mice (27-29 mo, $n = 4$, mean $\pm$ SD = 13.57 ± 7.57 ; ANOVA, $F = 9.196$, $p = 0.0067$ with Tukey's post hoc, $p = 0.0054$). The difference between middle-aged (17-18.5 mo, $n = 4$, mean $\pm$ SD = 18.90 ± 1.797) and young ($p = 0.0685$), as well as old mice ($p = 0.2697$) was not significant.

fluorescence intensity in fluoromyelin staining, which indicates a decrease in myelin abundance in aging spinal cord (Fig. 2.2 d).

Aging spinal cord exhibits increased apoptosis and decreased neuronal health and integrity

Because we observed an age-related MN loss, we measured caspase 3 activation to determine whether an increase in apoptosis plays a role in this process. As shown in Figure 2.3 a-b, we found a significant increase in cleaved caspase 3 in spinal cord from old mice. In agreement with the loss of motor neurons observed in the old spinal cord, we found changes in metabolites levels revealed by MR spectroscopy, including a decrease in NAA abundance (a 14% decrease in old mice comparing to young) that indicates decrease in neuronal health (Fig. 2.3 c-d). Moreover, we also found an increase in glutamate levels (a 52% increase) that can also contribute to an increased MN vulnerability with age through an increased excitotoxicity, a potential mechanism of neuronal loss in neurodegenerative diseases. Levels of myo-inositol are elevated (a 37% increase) in spinal cord from old mice, which can influence the levels of secondary messengers, since myo-inositol is a precursor for inositol triphosphate. Both myo-inositol and taurine (which also increased 73% in old spinal cord) are osmotically active molecules, and changes in their concentrations could potentially disrupt ionic balance in the cell and favor cell death, as well as contribute to the swelling of neuronal soma

observed in aged MNs. However, we did not observe decrease in blood vessel density with age (Appendix A Supplemental Figure A.2), after immunostaining for CD31, therefore the decreased neuronal health is not due to a decrease in availability of nutrients or oxygen.

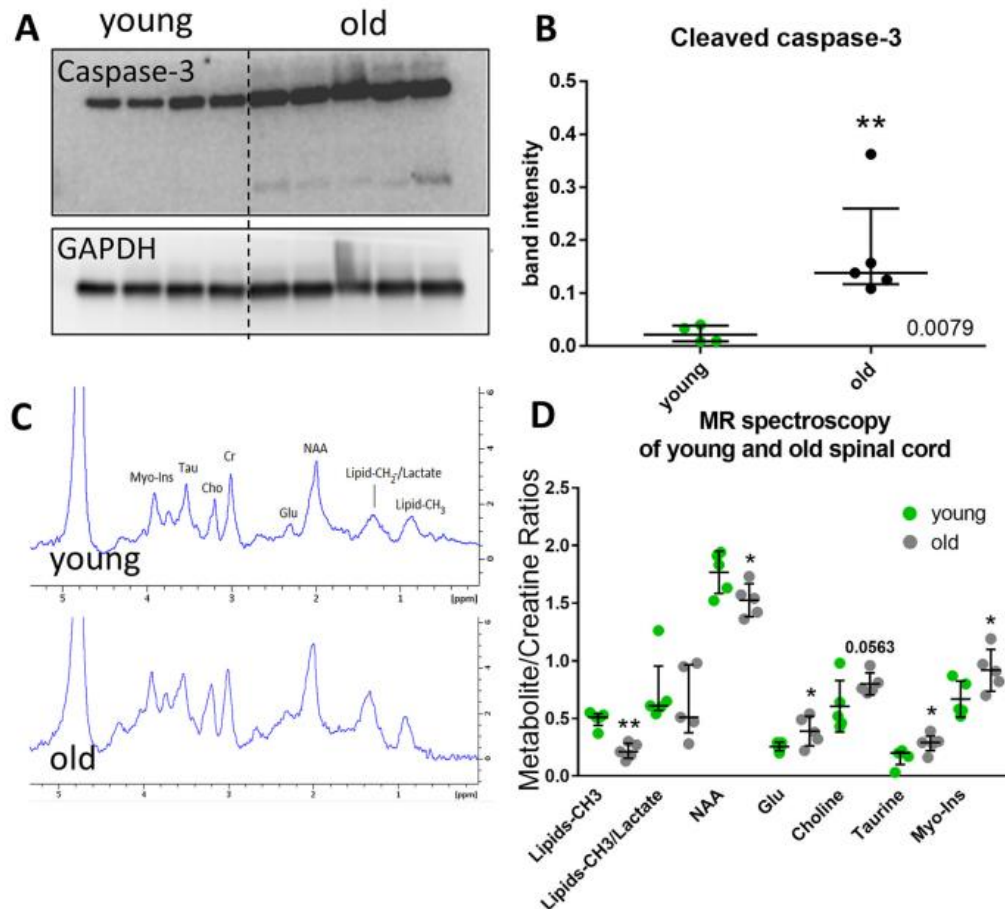


Figure 2.3. SC aging is associated with increased apoptosis and decreased neuronal health and integrity. **A** – western blot of spinal cord lysates (20 µg of total protein) of young (4 mo) and old (24-25 mo) mice probed for caspase-3; **B** – quantification of cleaved caspase-3 band from the blot showed in panel A after normalizing to GAPDH. There is a significant increase in cleaved caspase-3 abundance (Mann Whitney test, $p = 0.0079$) in old spinal cord ($n = 5$, median = 0.1383, IQR = 0.14) comparing with young ($n = 4$, median = 0.02182, IQR = 0.029); **C** – a representative MR spectrum of spinal cord of young (3-4 mo) and old mice (26-27); **D** – changes in individual metabolites measured by MR spectroscopy. There is a decrease in lipids and NAA in old spinal cord comparing to young, while the levels of glutamate, taurine and myo-inositol increase with age (one-tailed t-test with Welch's correction when necessary, or Mann-Whitney test, $n = 5$ per group); lipids: $p = 0.0040$ (young: median = 0.51, IQR = 0.10, old: median = 0.21, IQR = 0.13); NAA: $t = 2.325$, $p = 0.0243$ (young: mean $\pm$ SD =

1.766 $\pm$ 0.18, old: mean $\pm$ SD = 1.524 $\pm$ 0.14); glutamate: $t = 2.235$, $p = 0.0398$ (young: mean $\pm$ SD = 0.256 $\pm$ 0.037, old: mean $\pm$ SD = 0.39 $\pm$ 0.13); taurine: $p = 0.0437$ (young: mean $\pm$ SD = 0.164 $\pm$ 0.077, old: mean $\pm$ SD = 0.284 $\pm$ 0.082); myo-inositol: $t = 2.324$, $p = 0.0243$ (young: mean $\pm$ SD = 0.668 $\pm$ 0.016, old: mean $\pm$ SD = 0.918 $\pm$ 0.18).

Aging spinal cord demonstrates low grade inflammation which is not accompanied by cytokine release

Aging is associated with increased inflammation (“inflammaging”) and elevated inflammation in brain is a common manifestation of a number of neurodegenerative diseases (Franceschi et al., 2000; Akiyama et al., 2000). To determine whether there is an increase in inflammation with age in the spinal cord, we measured two common inflammatory markers, Iba1, a marker of microglia, and GFAP, a marker of astrocyte activation, in spinal cord sections (Fig. 2.4 a-d). Surprisingly, the increase in glial proliferation and activation was not associated with an increased cytokine release measured using a cytokine protein array (Fig. 2.4 e-f). Only 3 out of 40 cytokines (sICAM-1, eotaxin, and IL-12p70) changed between young and old spinal cord and reached statistical significance after adjusting for multiple comparisons. In contrast, our RNA-seq analysis did reveal an enrichment in transcripts connected to the immune system-related pathways, such as dendritic cell maturation, complement system, T helper cell differentiation, granulocyte adhesion and diapedesis (Fig. 2.4 g; results of IPA analysis on RNA-seq data can be found in the Appendix A Supplemental Data Tables A.1-4). However, the most enriched pathway in old spinal cord was the fibrotic pathway, which pointed towards changes in extracellular matrix.

Extracellular matrix components and metalloproteinases levels increase in old spinal cord

The transcripts that showed the largest change with age in the RNA-seq analysis were ECM components such as collagens 28A1 (this collagen was previously associated with

microglia cells. Microglia proliferation occurs in spinal cord from old mice (Mann Whitney test, $p = 0.0143$; $n = 4$, median = 17.58, IQR = 12.15) comparing to young ($n = 4$, median = 4.167, IQR = 2.086); **C** – a representative image of GFAP immunofluorescence in ventral horn of young (6-7 mo) and old (27-28 mo) mice; **D** – quantification of GFAP signal. There is an increase in GFAP fluorescence intensity in old spinal cord ($n = 3$, mean $\pm$ SD = 3.64 ± 1.114 ; unpaired t-test, $t = 2.225$, $p = 0.0450$) comparing to young ($n = 3$, mean $\pm$ SD = 1.235 ± 1.5); **E** – a representative image of membranes from cytokine array of young (3-4 mo) and old (25 mo) mice spinal cord lysates, 1h exposure; **F** – a heatmap of cytokine abundance in young ($n = 5$) and old ($n = 6$) spinal cord lysates measured with the cytokine array; **G** – RNA-seq analysis - top ten most enriched pathways that are differentially expressed with age (9 mo vs 26-27 mo) identified with IPA software and ordered by p-value (with B-H correction).

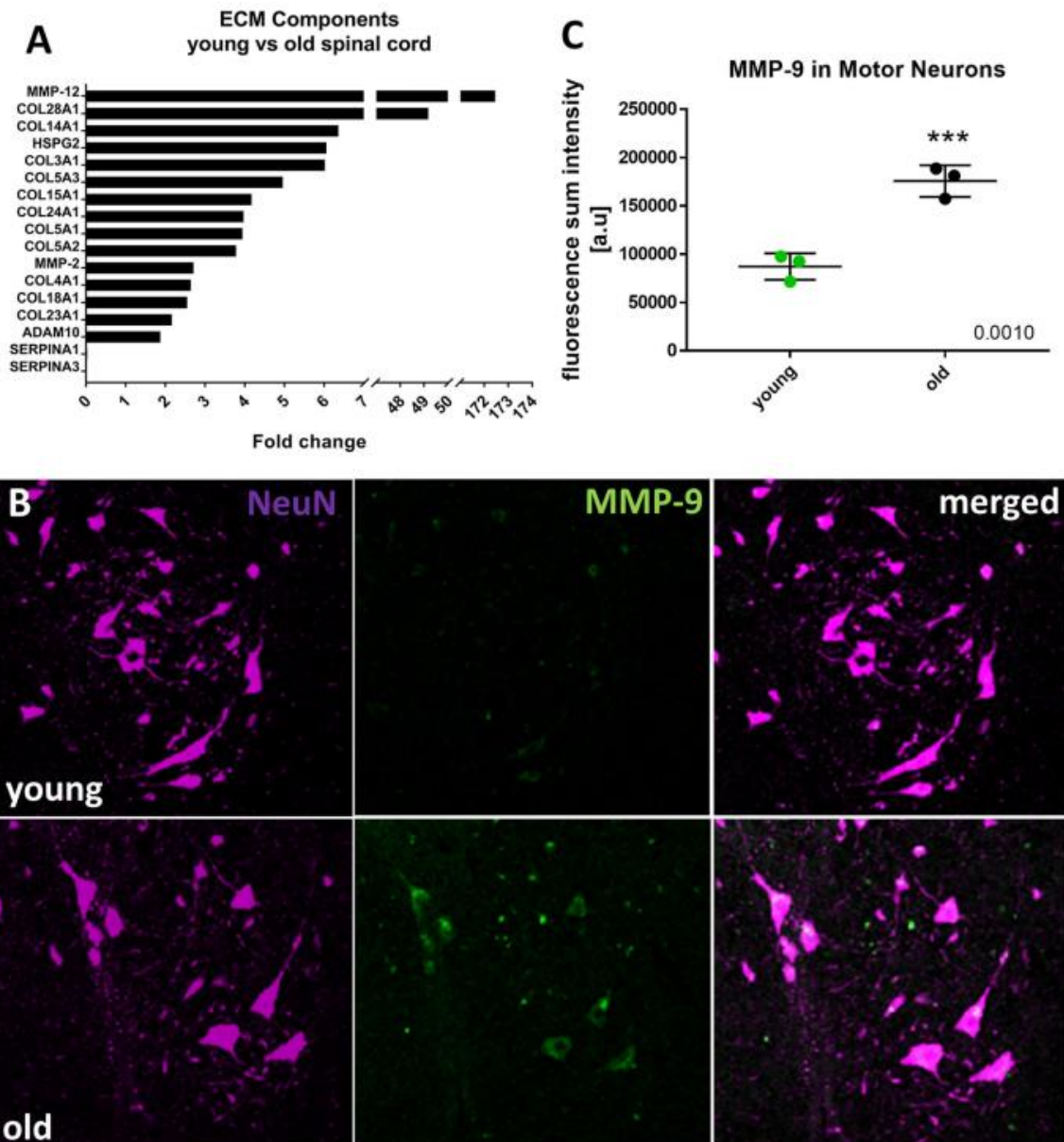


Figure 2.5. Age-related changes in ECM and the expression of MMPs. **A** – Example of ECM components in fibrotic pathway that are differentially expressed in aging (IPA analysis on RNA-seq data); **B** – MMP-9 immunofluorescence and co-localization with neuronal marker NeuN in young (6-7 mo) and old (27 mo) spinal cord sections; **C** – quantification of MMP-9 fluorescence sum intensity in MNs. There is an increase in MMP-9 in MNs in old spinal cord (one-tailed unpaired t-test, $t = 7.187$, $p = 0.0010$; $n = 3$, mean $\pm$ SD = 175678 ± 16322) comparing to young ($n = 3$, mean $\pm$ SD = 87058 ± 13773).

among other MMPs that change with age. While we were unable to use immunostaining to measure MMP-12 protein levels, we confirmed the age-dependent increase of MMP-9 abundance at the protein level (Fig. 2.5 b-c). The co-immunostaining demonstrated that most of the MMP-9 signal colocalizes with MNs, however MMP-9 exhibits also some non-neuronal localization.

Blood-spinal cord barrier permeability increases with age

Elevated MMPs (such as MMP-9 and MMP-12) have been associated with increased blood-brain barrier permeability in aging, as well as in the basal lamina disruption in cancer metastasis (Merdad et al., 2014; Shay et al., 2015). Thus, we asked whether the blood-spinal cord barrier (BSCB) is also affected in aging mice. Using injection of horseradish peroxidase (HRP) to measure blood vessel permeability, we found an increase in HRP extravasation in spinal cord from old versus young mice (Fig. 2.6 a-b). Since the animals were not perfused, we can visualize blood vessels as well as the regions of HRP extravasation, however these can be clearly distinguished. Blood vessels are clearly delineated in 2 months old mice and differ from the large, irregular regions that correspond to HRP extravasation at 18 and 27 months of age. Old mice (27 months) have significantly more regions of HRP extravasation compared to young mice (2 months). Interestingly, the HRP extravasation regions are not uniformly distributed across the spinal cord cross section. The greatest increase in the number of extravasation regions with age is in the ventral horn, where the α -MNs reside (Fig. 2.6

c). However, BSCB permeability to Gd-DTPA assay using MRI revealed increased permeability only in cervical spinal cord (Appendix A Supplemental Figure A.3). This may be due to technical limitations of the method, since the lumbar part of the spinal cord is smaller than the cervical part, which presents a challenge while trying to detect such a change in this region of the spinal cord. Also, HRP visualization using DAB is a method that has much higher sensitivity than Gd-DTPA assay.

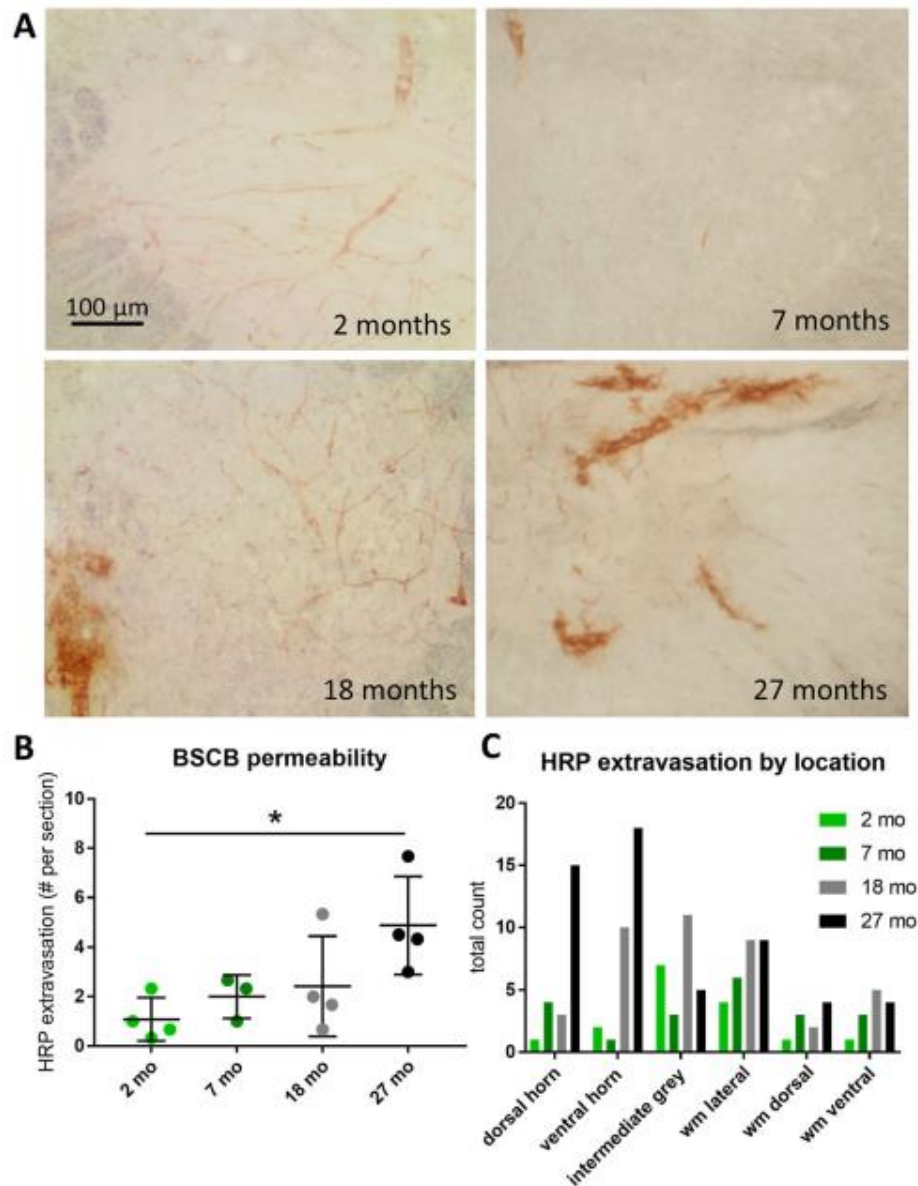


Figure 2.6. BSCB permeability increases with age. **A** – a representative image of a ventral horn (10x) across different ages after HRP injection; **B** – There is an increase in the number of HRP extravasation regions in old spinal cord (27 mo, $n = 4$, mean $\pm$ SD = 4.875 ± 1.98 comparing to young (2 mo, $n = 4$, mean $\pm$ SD = 1.08 ± 0.88 , one-way ANOVA, $F = 4.086$, $p = 0.0355$, with Tukey's multiple comparisons post hoc test, $p = 0.0275$). The HRP extravasation at 7 mo ($n = 3$, mean $\pm$ SD = 2 ± 0.88) and 18 mo ($n = 4$, mean $\pm$ SD = 2.42 ± 2.025) is not statistically different from other ages; **C** – plotting HRP extravasation by location reveals that the SC region with the highest age-related increase in BSCB permeability is the ventral horn.

Eicosanoids and isoprostanes abundance does not change in aging spinal cord

Eicosanoids are bioactive lipids that play a crucial role in inflammation and homeostasis (Dennis and Norris, 2015). An increase in eicosanoid levels was previously reported in neurodegenerative diseases (Tassoni et al., 2008), MS and demyelination (Palumbo and Bosetti, 2013), as well as in aging (see Das, 2018, for review). However, we did not observe a significant change in eicosanoid abundance in aging spinal cord (Fig. 2.7 a-c). Similarly, the level of F2-isoprostanes, a marker of lipid peroxidation, remained unchanged in the spinal cord between 7 and 27 months of age (Fig. 2.7 d).

Aging affects spinal cord protein synthesis rate of mitochondrial but not cytosolic protein fractions

As far as we are aware, there are no previous studies that have examined the effect of aging on protein synthesis in the spinal cord. By using mitochondrial protein synthesis as a direct measure of mitochondrial biogenesis (Miller and Hamilton, 2012), we determined that rates of mitochondrial biogenesis were lower in old animals (23 months) compared to young (6 months; Fig. 2.8 a, c). In addition, the fraction of the mitochondrial protein pool that was actively turning over was lower in young mice compared to old mice (Fig. 2.8 e). This finding indicates that for part of the mitochondrial protein pool of older mice became resistant to turnover. However, neither of these parameters were significantly different in the cytosolic fraction (Fig. 2.8 b, d, f), indicating that these age-related differences were only in mitochondria.

Neuromuscular junction structure is affected in aging

Alterations in the interaction between motor neurons and skeletal muscle at the neuromuscular junction is a key candidate culprit behind loss of muscle mass and function during aging. Because our data showed clear loss of MNs during aging, we

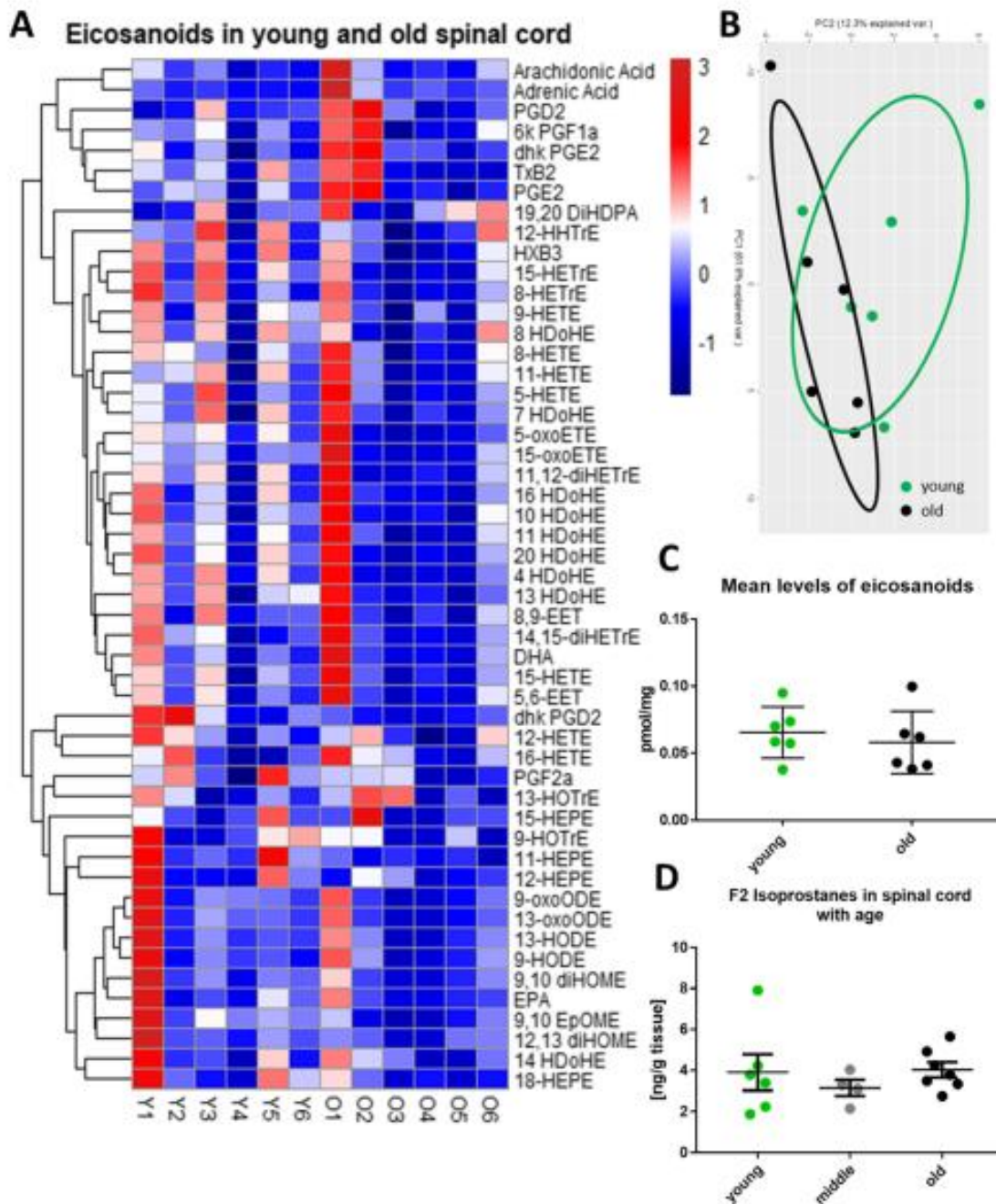


Figure 2.7. The abundance of eicosanoids and isoprostanes levels are not changing with age in the whole spinal cord lysates. **A** – a heatmap of normalized abundance of eicosanoids in young (n = 6) and old mice (n = 6), measured by lipidomics panel; **B** – PCA analysis of eicosanoid abundance in young and old spinal cord; **C** – average levels (geometric mean) of eicosanoids in young and old spinal cord; **D** – F2 isoprostanes abundance and young (7 mo, n = 7), middle-aged (16-17 mo, n = 4) and old (27 mo, n = 7) mice spinal cord lysates.

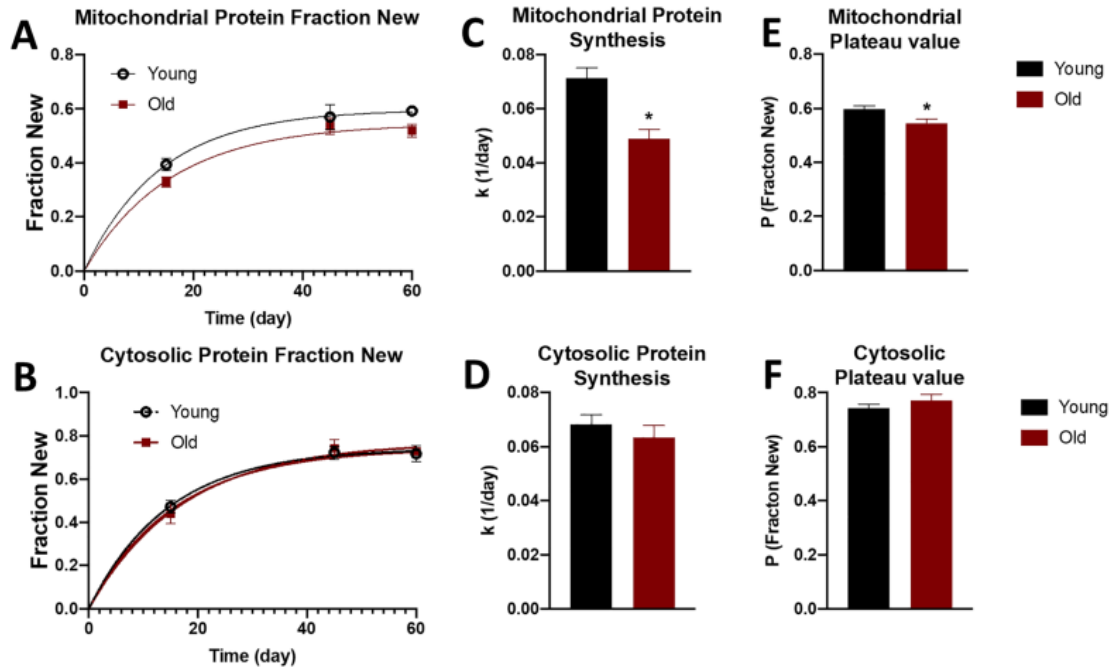


Figure 2.8. Protein synthesis rates of mitochondrial and cytosolic protein fractions of spinal cord tissue. Mitochondrial (**A**) and cytosolic (**B**) rise to plateau of protein fraction new, which was used to calculate synthesis (**C** and **D**) and plateau value (**E** and **F**); n = 3-5/timepoint, or 11-15 per age in total; * p<0.05 compared to young.

asked whether the timeline of changes in motor neurons and the NMJs was similar. To determine how early NMJ changes occur, we compared NMJs structure of young (4.5 months), middle-aged (17 months) and old (27 months old) mice by performing an immunostaining on gastrocnemius muscle (Fig. 2.9). NMJ area increases in both middle-aged and old animals compared to young mice, and this structural change appears to precede NMJ fragmentation that is evident only in old animals (NMJs with 5 or more

fragments were considered fragmented). The changes at NMJ (the significant increase in NMJ area in middle-aged mice) precede then the MN loss (we observe some MN loss in the middle-aged mice, but it is not yet statistically significant at this point). We did not observe denervation in the middle-aged mice. The percentage of fully innervated NMJs is decreased compared to young mice, although the difference was not significant. As expected, the percentage of denervated NMJs increases in old mice, while the percentage of fully innervated NMJs decreases significantly.

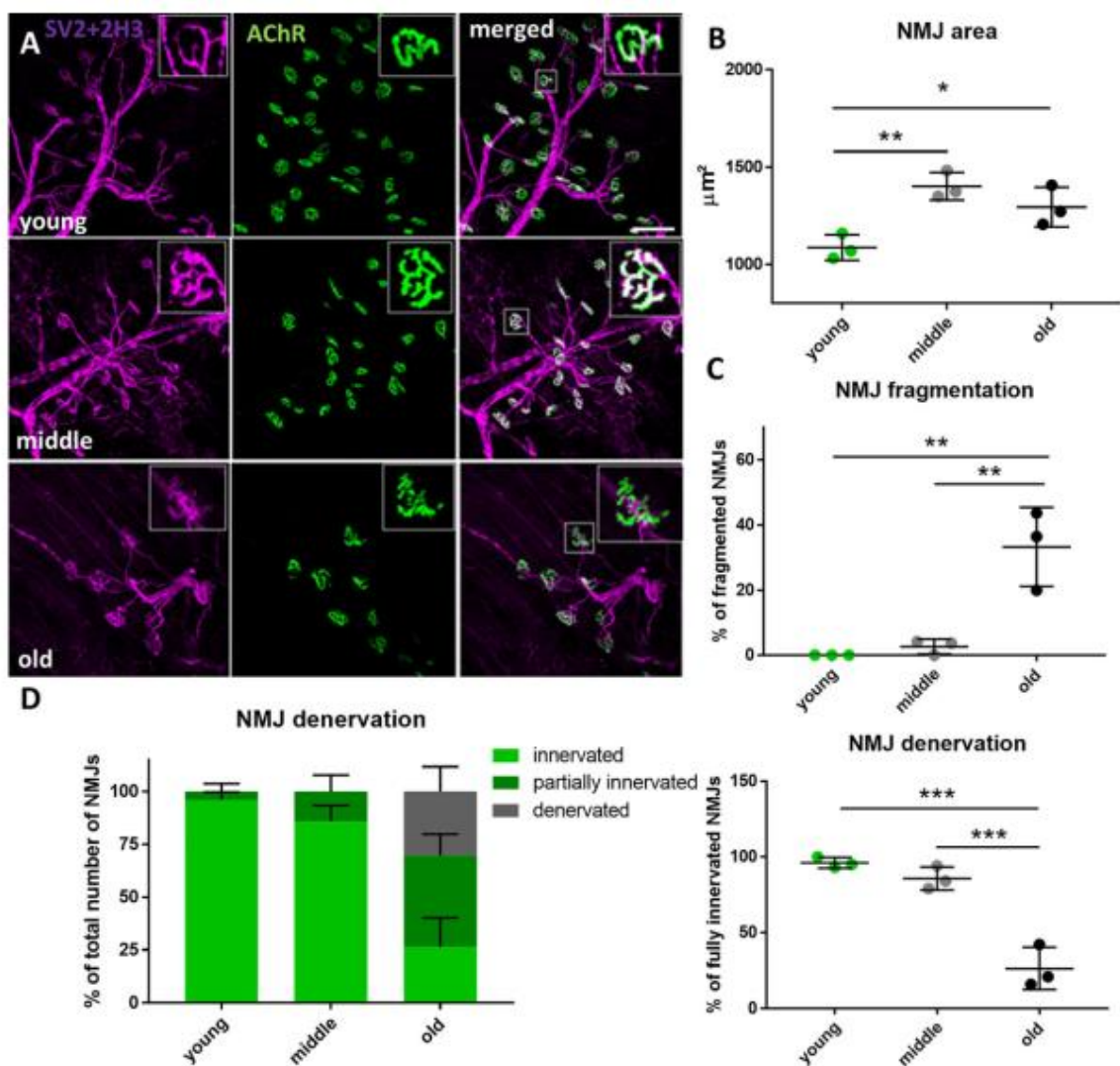


Figure 2.9. Age-related changes in NMJ structure. **A** – A representative image (20x) of NMJ of young (4.5 mo), middle-aged (17 mo) and old (27 mo) mice, immunostained

for SV2 (nerve terminals) and 2H3 (axons), as well as with bungarotoxin conjugated to a fluorophore (to visualize AChR). Insets show zoom of a single NMJ; **B** – There is an increase in NMJ (AChR) area in middle-aged ($n = 3$, mean $\pm$ SD = 1402 ± 70.68 , one-way ANOVA, $F = 11.75$, $p = 0.0084$, with Tukey's multiple comparisons post hoc test, $p = 0.0074$) and old mice ($n = 3$, mean $\pm$ SD = 1295 ± 101.9 , $p = 0.045$) comparing to young mice ($n = 3$, mean $\pm$ SD = 1086 ± 65.74); **C** – There is a significant age-related difference in NMJ fragmentation (NMJs with 5 or more fragments were considered fragmented) in middle-aged ($n = 3$, mean $\pm$ SD = 2.623 ± 2.284 ; one-way ANOVA, $F = 20.42$, $p = 0.0021$, with Tukey's multiple comparisons post hoc test, $p = 0.0044$) and young mice ($n = 3$, mean $\pm$ SD = 0 ± 0 , $p = 0.0029$) comparing to old mice ($n = 3$, mean $\pm$ SD = 33.32 ± 12.09); **D** – percentage of fully innervated, partially innervated and denervated NMJs in aging and comparison of the percentage of fully innervated NMJs across different age groups. There is a decrease in the number of fully innervated NMJs in old mice ($n = 3$, mean $\pm$ SD = 26.33 ± 13.8 ; one-way ANOVA with Tukey's multiple comparisons post hoc test, $F = 48.62$, $p = 0.0002$) comparing to middle-aged ($n = 3$, mean $\pm$ SD = 85.67 ± 7.638 , $p = 0.0006$) and young mice ($n = 3$, mean $\pm$ SD = 96 ± 3.606 , $p = 0.0002$).

Discussion

The goal of this study was to define differences between young and old mice spinal cord that may be related to loss of neuromuscular junction structure and function and increased denervation of skeletal muscles in sarcopenia.

We measured an age-related loss of α -MNs in the lumbar part of the spinal cord in wildtype mice, characterized by a 41% decrease in α -MN cell body number and axonal loss of 40% by 27 months of age. Our results are in agreement with other studies supporting age-related α -MN loss, e.g., there are reports of α -MN loss (up to 50%) in human lumbar spinal cord from individuals older than 60 years (Zhou et al., 1996; Kawamura et al., 1977a; Kawamura et al., 1977b; Tomlinson and Irving, 1977). In contrast, a recent study by Maxwell et al. (2018) reported no age-related α -MN loss based on changes of gene expression and abundance of NeuN protein measured by immunoblotting, and transcript levels rather than using α -MN counts. One possible reason for the contrasting results between this and the other studies may be because NeuN is expressed not only by α -MNs, but by other neuronal populations as well. Thus, changes in NeuN protein or transcript levels in spinal cord lysates would lead to

inaccurate conclusions about any specific neuronal population size. Also, the gene expression and protein levels across lifetime are variable and can change in various diseases (e.g. decrease in NeuN protein level and loss of antigenicity was reported in stroke etc.; Duan et al., 2016), and thus may lead to an over- or underestimation of cell number. Overall, the majority of studies support a decrease in α -MN number as a function of age.

The significance of the increase in cell body size, reported by Machado-Salas et al. (1977), and also seen in our study, remains uncertain. It is possible that the increase in cell body size is the first step in neuronal degeneration process. The α -MN with larger cell bodies might be more vulnerable, since it is known that larger MNs are more affected in ALS patients (Conradi and Ronnevi, 1993) and are more susceptible to polio virus infection (Hodes, 1949; Hodes et al., 1949). Age-related changes in the concentration of osmolytes that we observe in MRS (N-acetyl-aspartate (NAA), taurine and myo-inositol) might be a factor contributing to increased cell body area (Erschbamer et al., 2011). It is also interesting to note that studies reporting no age-related α -MN loss do not report increased cell body size. For example, Liu et al., (1996) found that the number of 'large' neurons (α -MNs) did not change in aged cats, while the cell body area did not increase, suggesting that smaller cell body size could somehow be protective.

Our results also suggest that demyelination occurs during aging. This is consistent with our previous report that nerve conduction velocity is decreased in aging mice (Walsh et al., 2015). Demyelination has been reported in aging brain (Lintl and Braak, 1983; Sherin and Bartzokis, 2011), but studies of demyelination in aging spinal cord are sparse.

In addition to α -MN loss, we also found biochemical evidence of changes in metabolite levels using MR spectroscopy that confirms a decline in neuronal health and integrity. One commonly studied marker of neuronal health is NAA. The decrease in

NAA in aged spinal cord we report here corresponds well with the age-related loss of α -MNs that we observe. NAA was previously shown to decrease in the brain with age (Brooks et al., 1999), as well as in MS (Li et al., 2013) and Alzheimer's disease (AD) patients, and correlates with cognitive decline (Jessen et al., 2001), it has also been used as a marker of neuronal loss (Singh et al., 2002). NAA is used in myelin lipid synthesis as a source of acetyl groups, thus the decrease in NAA is also consistent with the slight demyelination that we observe. Furthermore, the decreased levels of lipids (a 56% decrease) observed in MR spectroscopy (Fig. 2.3 c-d) support a chronic demyelination in old spinal cord. Myo-inositol, besides being an osmolyte, is often used as glial cell marker (Best et al., 2014), and the changes in its levels that we report are consistent with glial cell proliferation and activation in aged spinal cord. Moreover, myo-inositol increases in AD patients and in demyelinating diseases (Singh et al., 2002). An increase in myo-inositol, a precursor for various secondary messengers, and in glutamate levels could contribute to neuronal distress via excitotoxicity. Thus, our MR spectroscopy results are consistent with neuronal loss, increased cell body size, increased apoptosis, demyelination and para-inflammatory state that we observe in our immunofluorescence and immunoblotting assays.

Since previous studies have reported an increase in pro-inflammatory cytokines and glial activation in aging brain, as well as in rat spinal cord (Sparkman and Johnson, 2008; Xie et al., 2013; Galbavy et al., 2017; Nacka-Aleksić et al., 2017), we were interested in determining, whether there was an increase in inflammation in spinal cord from aging mice. Surprisingly, the glial cell activation in aged spinal cord that we measured using immunofluorescence was not associated with a pronounced increase in cytokine levels measured using cytokine panel, as the level of only 3 out of 40 cytokines was increased. The most increased cytokine was sICAM, which is a marker of age-related inflammation that decreases in fasting in humans (Stekovic et al., 2019), and its

serum level was connected to mild cognitive impairment in diabetic patients (Gorska-Ciebiada et al., 2015). It is possible to observe the low cytokine release in presence of glial activation, since there are studies reporting glial activation without increased cytokine production (Bell-Temin et al., 2015), as M2a and M2c-activated microglia weren't associated with increased cytokine release. The existing literature, as well as our study, support the presence of both para-inflammatory state and demyelination in the aging spinal cord.

One of the most striking changes we observed is the upregulation of ECM components in aged spinal cord revealed by our RNA-seq analysis. This corresponds to changes in spinal cord structure that are visible in electron micrographs. Furthermore, our RNA-seq and IPA analyses demonstrate that the most upregulated pathway in the aging spinal cord is the fibrotic pathway. There is a relevant link between ECM, MMPs and neuronal viability, as summarized by McCawley and Matrisian (2001). The importance of MMPs in CNS homeostasis and pathology has been noted previously (see Yong, 2005 for a review) and recently MMPs are also getting more recognition in the field of aging (Freitas-Rodriguez et al., 2017). MMPs alter ECM properties, which changes the α -MN environment. Moreover, MMPs and ECM can regulate apoptosis by activating apoptosis-inducing ligands, such as Fas ligand (Musiał and Zwolińska, 2011) and by upregulating pro-apoptotic genes (Dandachi et al., 2017). It is possible that the dramatic upregulation of MMP-12 (172-fold increase) that we observe in aged spinal cord can contribute to increased apoptosis in α -MN. Also, MMP-12 knock down mice exhibit decreased apoptosis and increased myelin basic protein (MBP) levels in ischemic brain (Chelluboina et al., 2015b; Chelluboina et al., 2018). Interestingly, MBP is one of MMP-12 substrates, and the main source of MMP-12 is microglia, which undergo proliferation and activation in aged spinal cord, as we demonstrated in our study. MMP-12 was also reported to contribute to increased inflammation in aging brain (Liu et al.,

2013). Moreover, MMP-12 was shown to play role in increasing BSCB permeability after spinal cord injury (Chelluboina et al., 2015a). Upregulation of MMP-9 also contributes to barrier disruption (Noble et al., 2002), which corresponds well with changes in BSCB permeability and MMP-9 abundance that we see in aged mice. Increased BSCB permeability means that toxic metabolites could enter into the spinal cord parenchyma and further favor α -MN death. Therefore, age-related changes in the abundance of MMPs and ECM components may favor α -MN loss, and are an interesting target for future studies.

The current study makes the first direct measures of protein synthesis in spinal cord of young and old mice. We found that mitochondrial protein synthesis, a direct measure of mitochondrial biogenesis (Miller and Hamilton, 2012), was lower in older mice than young mice. This outcome is different from our previous studies in brain and muscle where mitochondrial protein synthesis rates are maintained or increased with age (Reid et al., 2019; Miller et al., 2019). Not only was the rate of mitochondrial protein synthesis lower in old than young, but the fraction of the protein pool that was actively turning in the older was also significantly lower in old than young. It is interesting that neither of these changes were present when examining cytosolic proteins. Therefore, it appears that there is some deterioration of proteostatic maintenance in mitochondrial proteins that was not present in non-mitochondrial proteins, which likely negatively impacts mitochondrial function and is consistent with at least one previous study that showed decreased mitochondrial respiration in spinal cord, but not brain, of older rodents (Yonutas et al., 2015). Future studies should determine if the age-related deterioration of mitochondrial proteostasis is a cause or consequence of the aging process.

In conclusion, in this study we attempted to give a comprehensive survey of changes occurring in aging spinal cord and α -MNs. The present study represents a

starting point to elucidate mechanisms behind age-related α -MN loss. We report for the first time the potential role of MMP-12 in α -MN aging and increased apoptosis. Further studies are needed to determine the meaning of the increased α -MN cell body size, as well as changes in ECM composition.

Materials and Methods

1. Animals

All experiments were approved by the Institutional Animal Care and Use Committee at the Oklahoma Medical Research Foundation (OMRF). Young (2-8 months), middle-age (16-18 months) and old (24-28 months) wildtype mice (C57BL/6J) mice from the Jackson Laboratory and NIA were used for the experiments (males and females). The mice were maintained in specific pathogen-free barrier environment with unlimited access to food and drinking water.

2. Immunofluorescence

The lumbar part of the spinal cord was fixed in 4% PFA overnight, then washed for 15 min in PBS twice, and incubated overnight in 30% sucrose in PBS. The excess of sucrose was removed by blotting the tissue with a moisten kimwipe and the tissue was incubated in OCT (Sakura Finetek, Torrance, CA) for 1h, then embedded in a mixture of OCT and TFM (EMS, Hatfield, PA) 1:1 and frozen on dry ice. The following day, the blocks were sectioned on cryotome (Leica; 15-20 μ m thick) and mounted on Superfrost Plus Slides (VWR, Radnor, PA) and stored at -80°C. For immunostaining, the frozen sections were rehydrated in ddH₂O for 2 min, washed three times for 5 min in 0.3% Triton X-100 in PBS then blocked for 2h at RT in blocking buffer (5% normal goat serum or donkey serum, 1% BSA and 0.3% Triton X-100 in PBS), then incubated overnight at

4°C with primary antibodies in the blocking solution. The next day the slides were washed in 0.3% Triton X-100 in PBS 3x 5 min and incubated with secondary antibodies in blocking buffer for 2h at RT. Then, the slides were washed again in 0.3% Triton X-100 in PBS 3x 5 min, air dried, mounted using EMS Glycerol Fluoromount with PPD anti-fading agent mounting medium (EMS, Hatfield, PA) and protected with a coverslip and sealed with nail polish. The cells were then visualized using Nikon C2 confocal microscope.

3. Motor neuron count and cell body area measurements

Images of every fourth section (4 sections in total per animal) were taken using (10x objective) confocal microscope, and NeuN positive cells that were identified as motor neurons were counted for each ventral horn. Motor neurons were distinguished by size, morphology and location (lamina IX), and their number was averaged per animal. MN area was measured using Nikon NIS-Elements Advanced Research software by tracing around the cell body. Only MNs with a visible nucleus were included in the analysis to avoid underestimating the cell body size. The person that analyzed the images was blinded to sample identity.

4. Electron microscopy

Electron microscopy was performed by OMRF Imaging Core to visualize white matter adjacent to the ventral horn. All axons visible in the field of view were counted and the results were averaged. Axon diameter and myelin thickness were measured for each axon using ImageJ software.

5. Immunoblotting

Tissues were homogenized in RIPA buffer with protease inhibitor and the samples were resolved on a 12.5% polyacrylamide gel, then transferred overnight onto a previously

methanol-activated PVDF membrane. The membrane was then stained with Ponceau stain to visualize the lanes, and cut, rinsed in TBS-T, blocked for 1h in 5% milk in TBS-T at RT and incubated overnight with primary antibodies in blocking buffer at 4°C. The following day the membrane was washed three times for 10 min in TBS-T, incubated for 1h at RT with secondary antibodies in blocking buffer, washed again three times for 10 min in TBS-T and developed with ECL. The band intensity was then measured using ImageJ software.

6. Fluoromyelin staining

Fluoromyelin staining was performed as follows: 20 µm-thick frozen sections of lumbar spinal cord (the blocks were prepared as described above) were incubated for 20 min in 0.3% Triton in PBS and then with Fluoromyelin Red fluorescent myelin stain (Invitrogen, #F34652) 1:300 in PBS at RT for 10 min. Next, the slides were washed in PBS 3x 10 min, mounted and visualized under confocal microscope. The images taken with 4x objective were used for the analysis in ImageJ software. Using free-hand selection tool, the white matter area was selected and fluorescence intensity was measured, then the fluorescence intensity in the gray matter was measured and subtracted from the fluorescence intensity in white matter.

7. Nuclear Fast Red staining

Nuclear Fast Red was performed by Imaging Core Facility at Oklahoma Medical Research Foundation. MNs were distinguished based on their size, shape and location. The scorer was blinded to sample identity.

8. Neuromuscular junction (NMJ) staining

Neuromuscular junctions were stained following a previously described method (Falk et al., 2015), with modifications. Fresh gastrocnemius muscle was dissected, cleaned from

connective tissue and cut in small, flat pieces in cold PBS. The tissue was then fixed in 10% STUmol (Poly Scientific R&D, Bay Shore, NY) in ddH₂O for 1h on a rocker, washed three times for 5 min and permeabilized in 2% Triton X-100 in PBS for 30 min. After blocking overnight at 4°C in 5% NGS, 4% BSA, 1% Triton X-100 in PBS the samples were incubated with primary antibodies for 24-48h. Then, the samples were washed six times for 30 min in PBS and incubated overnight with a secondary antibody and bungarotoxin conjugated to a fluorophore. Next, the samples were washed again in PBS (six times for 30 min), blotted with a moistened kimwipe and mounted onto Superfrost Plus Slides (VWR) with EMS Glycerol Fluoromount with PPD anti-fading agent mounting medium (EMS, Hatfield, PA), and coverslipped.

9. Neuromuscular junction analysis

Images of NMJs (20x) were used for measuring NMJ area, fragmentation and denervation. NMJ area was measured with Nikon NIS-Elements Advanced Research software by tracing around the acetylcholine receptors (30-50 per animal, n=3 per group). NMJ fragmentation was assessed by counting AChR fragments. NMJs composed of five or more fragments were considered fragmented. NMJ denervation was determined by assessing the overlap between the nerve terminals (SV2 staining) and AChR and expressed as fully innervated, partially innervated (partial overlapping) or denervated (little or no overlapping).

10. Cytokine panel

Cytokine abundance in the spinal cord lysates (200 µg of total protein) was measured using Proteome Profiler Array (Mouse Cytokine Array Panel A #ARY006, R&D Systems, Minneapolis, MN), accordingly to the manufacturer's protocol and visualized by autoradiography. The film exposed for 1h was scanned and the resulting image was

further analyzed. Dot intensity was measured in ImageJ and after subtracting averaged background, the signal was averaged among the duplicates and compared between the groups. Heatmap was generated in R Studio using pheatmap library.

11. RNA-seq

Total RNA was extracted from 30 mg of the spinal cord of the young and old mice using TRIzol reagent (Invitrogen, CA, USA) according to the manufacturer's protocol. After the RNA isolation, the samples were processed by OMRF Clinical Genomics Center (<https://omrf.org/research-faculty/core-facilities/next-generation-sequencing/>). To assure proper quality standards, the RNA integrity and concentration were estimated with the use of Agilent Tapestation. Libraries were prepared using the TruSeq Stranded mRNA Library Kit (Illumina) and the samples were subsequently sequenced on Illumina NextSeq 500. The resulting RNA-seq data were then processed by OMRF Core facility and analyzed by OMRF Genomics and Data Science group.

12. Blood-Spinal Cord Barrier permeability to HRP assay

Anaesthetized mice were injected ip with 75 mg/kg HRP (type II, #P8250, Sigma, St. Louis, MO) in 0.9% saline and kept on a heating pad for 15 min, then sacrificed and spinal cord was collected and processed the same way as in the immunofluorescence protocol. The blocks were then sectioned (50 µm thick) and HRP was visualized with the use of DAB. 250 µl of 1% DAB and 250 µl of 0.3% H₂O₂ were added to 5 ml of 0.01M PBS of pH 7.2. Slides were incubated for 3 min with 200 µl of DAB solution at RT, then rinsed twice in PBS-Tween 20 for 2 min, dried and coverslipped. The images taken with 1x were used to count HRP extravasation regions. Since the animals weren't perfused, the blood vessels remained highlighted in the sections and weren't considered during the analysis.

13. MRI

For MRI assessment mice were anesthetized with isoflurane (1.0-2.0% with 0.6-0.8 L/min O₂). Mice were placed into the MRI instrument, a 7 Tesla 30 cm-bore Bruker Biospec MRI system (Bruker Biospin Corporation, Billerica, MA) with a standard Bruker volume coil. 20 µL of the contrast agent (0.5 mmol Gadolinium diethylenetriamine penta-acetic acid/mL (Gado-DTPA™, BioPAL, Worcester, MA)) was diluted to 100 µL in saline, and was administered intravenously via tail-vein catheter. The following scans were done: 1H Magnetic Resonance Spectroscopy (1H-MRS), and pre and post Contrast, and Enhanced T1-weighted rapid acquisition with relaxation enhancement imaging (CE T1 RARE). A T1-RARE pulse sequence was used with the following parameters: repetition time (TR) of 1500 ms, echo time (TE) of 7 ms, 4 averages, 25 slices with a thickness of 0.8 mm/slice, a matrix of 256 × 256, a field-of-view (FOV) of 4.5 × 4.5 cm², and motion and fat suppression.

For Contrast Enhanced Blood-Spinal Cord Barrier Permeability assessment (CE-MRI) sagittal T1-weighted RARE images were used. MRI signal intensities were measured from circular regions of interests (ROIs) within the spinal cord, and 3-4 ROIs were taken in cervical, thoracic and lumbar regions of the spinal cord in pre and post contrast injection images and recorded as average value for each spinal region of each animal. The percent differences between signal intensities before and after contrast administration were calculated. Percent differences of Signal Intensity of each spinal region were recorded as mean ± standard deviation within each age group.

1H-MRS was acquired using a PRESS (Point REsolved SpectroScopy) sequence with a TE of 24.0 ms, a TR of 2500.0 ms, 256 averages, and a spectral width of 4006 Hz. A non-suppressed MR spectrum was acquired beforehand by applying eddy-current correction to maximize signal intensity and decrease the peak linewidths. Water was

suppressed with a VAPOR (variable power radio frequency pulses and optimized relaxation delays) suppression scheme. In all cases, the peak width (full width at half maximum) of the water peak was less than 30 Hz following localized shimming, which was conducted by using first and second order adjustments with Fastmap. A cubic voxel of $1.5 \times 2.0 \times 1.5 \text{ mm}^3$ was positioned in the mouse spinal cord. To analyze the MRS data, an in-house Mathematica program was used (version 8.0.4.0, Wolfram Research, Champaign, IL, USA). The spectra were scaled in ppm by calibrating against the water peak (4.78 ppm). The major metabolic peaks were identified as Lipids-CH₃ at 0.9 ppm, Lipids/Lactate at 1.3 ppm, N-acetylaspartate (NAA) at 2.02 ppm, glutamate (Glu) at 2.3 ppm, creatine (Cr) at 3.02 ppm, choline (Cho) at 3.22 ppm, taurine (Tau) at 3.43 ppm, and myo-inositol at 3.57 ppm. The peak area measurements of each metabolite were used to calculate ratios against Cr (metabolite/Cr).

14. Eicosanoids

The spinal cords of young (10 months) and old (24 months) mice were snap frozen and sent to the lipidomics core at UCSD (<http://www.ucsd-lipidmaps.org/>) for analysis. The heatmap of the results was generated using pheatmap library in RStudio. RStudio was also used to perform PCA analysis.

15. Isoprostanes

The abundance of F₂-isoprostanes in spinal cord lysates was established following a method described by Roberts and Morrow (2002) with some modifications. In summary, 45-60 mg of previously frozen spinal cord tissue was homogenized in 10 ml of ice-cold Folch solution (CHCl₃: MeOH, 2:1) with butylated hydroxytoluene (BHT). The samples were then incubated for 30 min at room temperature. Next, 2 ml of 0.9% NaCl were added, mixed vigorously, and the samples were centrifuged at 3,000g for 5 min at 4 °C.

The aqueous layer was discarded, while the organic layer was dried using N₂ at 37 °C. Solid phase was extracted and F2-isoprostanes were quantified using GC-MS, with [2H₄]8-Iso-PGF₂α as the internal standard.

16. Protein turnover

Young (6 mo) and old (23 mo) male C57Bl/6 mice were used for the labelling of newly synthesized proteins with deuterium oxide (D₂O) using our previously published procedures (Drake et al., 2013; Drake et al., 2015; Reid et al., 2019). After an intraperitoneal (ip) bolus injection of 99% D₂O, mice received 8% D₂O enriched drinking water 15, 30, or 45 of days of labeling (n = 3-5 per timepoint). The initiation of labeling was staggered so that all mice were sacrificed at the same age. Mice were euthanized by isoflurane, immediately followed by cardiac puncture. The spinal cord was dissected, immediately frozen in liquid nitrogen and stored at -80 C°. Approximately 30 mg of the spinal cord was pulverized using liquid nitrogen-cooled mortar and pestle. In brief, tissue was fractionated into subcellular compartments by differential centrifugation as previously described (Drake et al., 2013; Drake et al., 2015; Reid et al., 2019) to obtain mitochondrial (MITO) and cytosolic (CYTO) protein fractions. In addition, plasma was prepared using distillation for the determination of precursor pool enrichment. Protein fractions were derivatized for analysis of deuterium-enrichment by gas chromatography-mass spectrometry (Agilent 7890B GC and 5977B MS), while distilled water samples were analyzed by enhanced performance liquid water isotope analyzer (Los Gatos Research LWIA-912). Deuterium enrichment of protein was used to calculate protein fraction new using a MIDA correction for the equilibration of the precursor enrichment. From this data, one-phase associations were generated and were left unconstrained to calculate the plateau value (p, fraction new), or by constraining the plateau to the young value to calculate the rate (k, 1/day). The plateau of the fraction new is representative of

the proportion of the protein pool (with 1.0 equal to 100% of the protein pool) that is turning over. For the statistics, synthesis and plateau values used an unpaired two-tailed t-test.

17. Antibodies

The following antibodies were used for the spinal cord immunofluorescence: anti-NeuN (Cell Signaling #D3S3I, rabbit mAB, 1:500), anti-Iba1 (Wako #019-19741, rabbit, 1:500), anti-GFAP (Abcam, ab7260, rabbit, 1:1000), donkey anti-rabbit Alexa 594 (1:200), goat anti-rabbit Alexa-488 (1:200). For neuromuscular junction staining: anti-SV2 (DSHB, #2315387, mouse monoclonal, 1:50), anti-2H3 (DSHB, #2314897, mouse monoclonal, 1:50), α -BTX-Alexa 488 (Invitrogen, #B13422, 1:1000), goat anti-mouse Cy3 (Jackson ImmunoResearch, #115-165-146, 1:250); immunoblotting: anti-caspase-3 (Santa Cruz, #sc7272, mouse, 1:1000), goat anti-mouse IgG-HRP (Santa Cruz, #sc-2005, 1:10000). Anti-CD31 antibody was kindly provided by Lijun Xia from OMRF.

18. Statistical analysis

After confirming normality, averages (for normally distributed data) or medians (for data that doesn't follow a normal distribution) were analyzed using Graphpad Prism 8 software. For a comparison between two groups, unpaired student t-test with or without Welch's correction (depending on unequal or equal variance) was employed for normally distributed data, or Mann-Whitney test for non-normally distributed datasets. Similarly, for comparisons between more than two groups, one-way ANOVA with Tukey's post hoc test or Kruskal-Wallis with Dunn's multiple comparisons post hoc test were performed.

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CHAPTER III

KNOCKING OUT MMP-9 AND MMP-12 HAS A DIFFERENTIAL AND SEX-SPECIFIC EFFECT ON AGE-RELATED A-MOTOR NEURON LOSS AND DECLINE IN MUSCLE MASS

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CONTRIBUTIONS BY DOCTOR OF PHILOSOPHY CANDIDATE:

KMP created the experimental design together with HVR; KMP performed all the experiments, analyzed the data, created the figures and tables, and drafted the manuscript.

Abstract

Age-related loss of alpha-motor neurons (α -MNs), increased inflammation, and increased blood-spinal cord barrier (BSCB) permeability are all common characteristics of spinal cord aging that contribute to decline in muscle mass with age. In the present study we manipulated matrix metalloproteinases (MMPs) to see, if it improves spinal cord health in aging. Both MMP-9 KO and MMP-12 KO mice exhibited increased α -MN survival at 25 months of age, however when the sexes were analyzed separately, only male MMP-9 KO mice showed higher α -MN number comparing to wildtype controls. Both MMP-9 KO and MMP-12 KO male, but not female mice, showed a decrease in microglia density and activation with age, but there was no effect on astrocyte proliferation in any of the groups. MMP-12 KO mice had decreased BSCB permeability and higher abundance of claudin-5, one of the important tight junction proteins. Only MMP-12 KO female mice showed an increase in gastrocnemius muscle weights, while

male MMP-12 KO mice had higher body weight at 25 months of age comparing to age-matched wildtype control mice. Our study shows that manipulating MMPs has an impact of aging spinal cord and that there are sex-specific effects in both knockouts, as α -MNs are more protected in males (MMP-9 KO), which also exhibit lower inflammation than females, in terms of decreased microglia density and activation (both MMP-9 KO and MMP-12 KO), as well as higher body weight (MMP-12 KO). However, only females show increase in hindlimb muscle mass (gastrocnemius in MMP-12 KO).

Introduction

We previously reported that spinal cord aging is associated with α -MN loss as well as with low-grade inflammation (Piekarz et al., 2020). Loss of innervation contributes to muscle wasting and promotes sarcopenia (Muller et al, 2007), which has a deleterious effect on mobility and quality of life, and also has economic implications (Beudart et al., 2014; Steffl et al., 2017; Waltz et al., 2018). Our previous study revealed that MMP-12 is the most upregulated transcript in aging spinal cord in wildtype mice, and MMP-2, MMP-9 and MMP-14 transcripts were also affected (Piekarz et al., 2020). Based on those results and existing literature we selected MMP-12 and MMP-9 as targets for manipulation in spinal cord aging.

MMPs are a family of 23 zinc proteases that degrade extracellular matrix components and are crucial in development, as well as in wound healing later in life (Webster and Crowe, 2006; Jackson et al., 2010). MMPs can also influence growth factors, such as EGF, basic FGF, VEGF, TNF- α , TGF- β , etc., by activating them through cleavage or releasing them from binding proteins or extracellular matrix (Sternlicht and Werb, 2001; Mott and Werb, 2004), therefore they can alter cellular growth. By altering ECM, MMPs can affect cell migration, as well as cell-to-cell communication and apoptosis by influencing Fas/FasL (McCawley and Matrisian, 2001; Musiał and

Zwolińska, 2011). MMPs are strongly upregulated after spinal cord injury both in humans (MMP-1, 2, 9 and 12; Buss et al., 2007) and mouse model (MMP-12; Wells et al., 2003) and promote neuroinflammation (see Könnecke and Bechmann, 2013 for review). MMPs interfere with stem cell niche maintenance (Freitas-Rodríguez et al., 2017). Although MMPs were linked to neurodegeneration, they can also have a beneficial effect in the central nervous system, in terms of regulating neurogenesis, myelination or axonal growth (see Yong 2005 for review).

Besides affecting tissue remodeling with age and skin aging (Quan and Fisher, 2015), the role of MMPs in aging is also being recognized in brain aging and related neuronal disorders (Romero et al., 2010; Morawski et al., 2014; Mukherjee and Swarnakar, 2015). MMPs were also shown to be part of SASP (senescence-associated secretory phenotype; Coppé et al., 2010). On the other hand, MMPs can be related to increased longevity, as it was shown in a study on Sicilian population that long-living individuals exhibited increased MMP-2 activity in plasma (Cancemi et al., 2020). Our previous study showed that MMP-12 is upregulated in aged spinal cord (Piekarz et al., 2020), it was also shown to be upregulated in aging brain (Liu et al., 2013).

MMP-12, also known as macrophage elastase, is produced mostly by microglia/macrophages (Guan et al., 2019). MMP-12 expression was also reported in neurons and oligodendrocytes (Chelluboina et al., 2015b). It was shown that C-terminal domain of MMP-12 induces apoptosis in tumors by translocating to the nucleus and initiates transcription of TRAIL, which triggers TRAIL-mediated apoptosis (Dandachi et al., 2017). It also down-regulated anti-apoptotic genes, such as Bcl-xL, while up-regulating pro-apoptotic genes, such as Bad and Bcl-xS (Dandachi et al., 2017). MMP-12 was linked to increased brain damage in focal cerebral ischemia in rat model of middle cerebral artery occlusion, knocking it down decreased apoptosis, MMP-9

activation, TNF- α levels, while increasing MBP levels (Chelluboina et al., 2015b). MMP-12 was linked to increased inflammation (Razavian et al., 2016; Guan et al., 2019; Aristorena et al., 2019), however, there have been reports suggesting that MMP-12 itself can have an anti-inflammatory effect in cornea injury mouse model by inactivating CCL2 (Wolf et al., 2019). Knocking out MMP-12 decreased age-related inflammation in brain (Liu et al., 2013). MMP-12 was also linked to BBB disruption in focal cerebral ischemia (Chelluboina et al., 2015a), and to progression of multiple sclerosis (see Chelluboina et al., 2018 and Rempe et al., 2016 for review).

MMP-9 is produced by multiple cell types, including monocytes (Watanabe et al., 2018), macrophages and fibroblasts (Yabluchanskiy et al., 2003), but it can be also stored in granules in neutrophils and released upon degranulation (Yong 2005). We previously demonstrated that it also colocalizes to α -MNs (Piekarz et al., 2020). MMP-9 is connected to cardiovascular diseases (Yabluchanskiy et al., 2003), cancer malignancy (Dong et al., 2019) and metastasis (Gautam et al., 2018), as well as weakens BBB integrity in neuroinflammation (Qin et al., 2019). It has also been implicated in cardiac aging (Yabluchanskiy et al., 2014; Padmanabhan Iyer et al., 2016), and increased MMP-9 (as well as MMP-2) activity was reported in aging tenocytes (Yu et al., 2013).

Here we aimed to determine, whether manipulating MMPs can alter spinal cord aging process and therefore have a beneficial effect on aging skeletal muscles. To answer this question, we aged MMP-9 KO and MMP-12 KO mice until they reached 25 months of age, and assessed α -MN number, inflammation levels, BSCB integrity, as well as muscle weight. We saw α -MN protection in aging in both MMP-12 KO and MMP-9 KO mice, however analyzing the results divided by sex revealed sex-dependent differences in response to MMP manipulation. MMP-9 KO males showed higher α -MN protection in aging, as well as lower inflammation. However only MMP-12 KO females exhibited

higher gastrocnemius muscle mass, while MMP-12 KO males had higher body weight comparing to sex-matched wildtype controls. Also, knocking out MMP-12 revealed to have a protective effect on BSCB integrity and the abundance of tight junction protein claudin-5.

Results

Knocking out MMP-9 and MMP-12 influences age-related α -motor neuron loss

Knocking out MMP-9 and MMP-12 decreased age-related α -motor neuron loss at 25 months of age in male and female mice combined (fig. 3.1 A, D). MMP-9 KO mice had on average 19.42 α -MNs (Dunn's post hoc $p = 0.0182$), MMP-12 KO mice – 19.60 (Dunn's post hoc $p = 0.0332$), while wildtype controls had only 14.78 α -MNs. However, when we analyzed both sexes separately, we saw a sex-specific effect of knocking out MMP-9. Male MMP-9 KO mice exhibited a highly significant increase in α -MN number (on average 21.31 α -MNs per ventral horn compared to 14.44 in male wildtypes, Dunn's post hoc $p = 0.0046$, fig. 3.1 B), while the number of α -MNs in females was not different from the WT controls (on average 16.59 α -MNs vs 15.44 in female wildtype mice, Dunnett's post hoc $p = 0.8924$, fig. 3.1 C). When looked at separately, neither male nor female MMP-12 KO mice demonstrated a difference in α -MN counts comparing to sex-matched wildtype control mice (fig. 3.1 B and C). Male MMP-12 KO mice had on average 16.67 α -MNs (vs 14.44 in male wildtypes, Dunn's post hoc $p = 0.7931$), while females had 21.07 α -MNs, however due to a high SD (5.153), it was not significantly different from female wildtype mice (15.44 α -MNs, Dunnett's post hoc $p = 0.1161$).

MMP-9 and MMP-12 KO decrease microglia density and activation, but do not affect astrocyte proliferation.

Since inflammation is one of the components of aging, and we reported before that there is an increase in microglia and astrocyte proliferation in the spinal cord of old mice

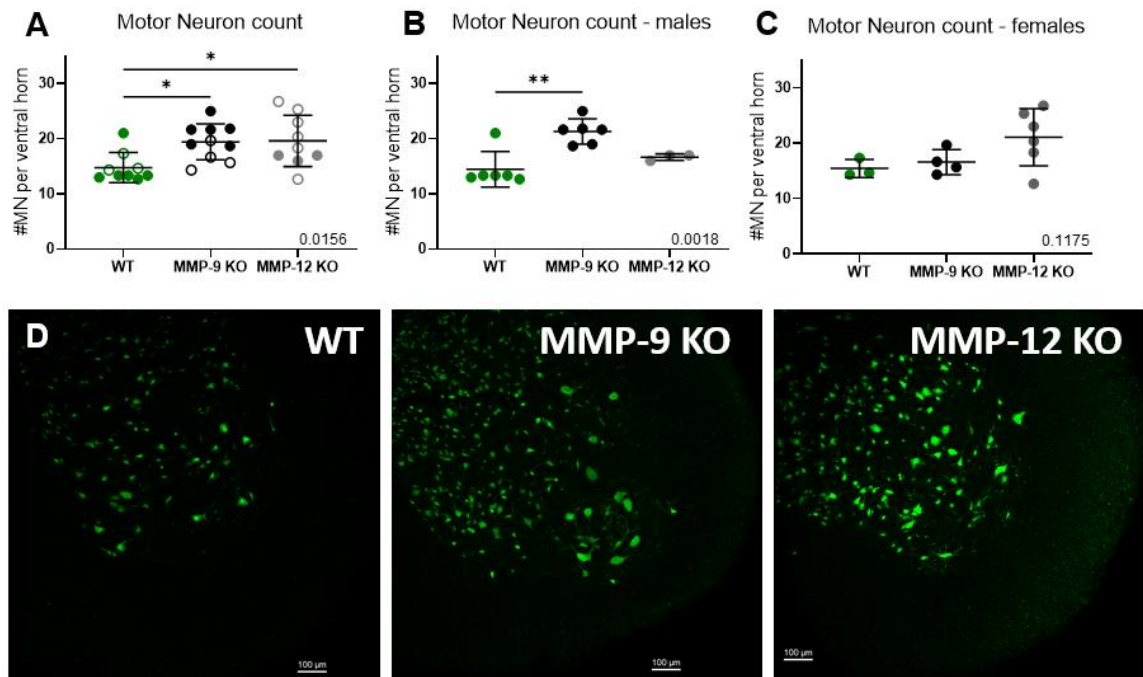


Figure 3.1 Knocking out MMP-9 and MMP-12 influences age-related α-motor neuron loss. **A** – at 25 months of age the number of motor neurons is higher in both knockouts (Kruskal-Wallis $p = 0.0156$; MMP-9 KO: $n = 10$, mean $\pm$ SD = 19.42 ± 3.256 , Dunn's post hoc $p = 0.0182$; MMP-12 KO: $n = 9$, mean $\pm$ SD = 19.60 ± 4.639 , Dunn's post hoc $p = 0.0332$) comparing to wildtype controls ($n = 9$, mean $\pm$ SD = 14.78 ± 2.723 ; open circles represent females); **B** – the increased MN survival is highly pronounced in MMP-9 KO males (Kruskal-Wallis $p = 0.0018$; MMP-9 KO: $n = 6$, mean $\pm$ SD = 21.31 ± 2.3 , Dunn's post hoc $p = 0.0046$; MMP-12 KO: $n = 3$, mean $\pm$ SD = 16.67 ± 0.58 , Dunn's post hoc $p = 0.7931$) comparing to wildtype controls ($n = 6$, mean $\pm$ SD = 14.44 ± 3.223); **C** – MMP-9 and MMP-12 KO female mice do not show a decrease in age-related MN loss (ordinary one-way ANOVA $p = 0.1175$; MMP-9 KO: $n = 4$, mean $\pm$ SD = 16.59 ± 2.269 , Dunnett's post hoc $p = 0.8924$; MMP-12 KO: $n = 6$, mean $\pm$ SD = 21.07 ± 5.153 , Dunnett's post hoc $p = 0.1161$) comparing to wildtype controls ($n = 3$, mean $\pm$ SD = 15.44 ± 1.643); **D** – a representative image of a ventral horn of lumbar spinal cord of an old WT, MMP-9 KO, and MMP-12 KO mouse, immunostained for NeuN.

(Piekarz et al., 2020), we were interested to see, if manipulating MMPs will decrease age-related inflammation. Knocking out MMP-9, as well as MMP-12, proved to have an effect on microglia proliferation and activation. Microglia density (fig. 3. 2A, G) was significantly lower in MMP-9 KO mice (mean = 83.28, Dunn's post hoc $p = 0.0461$) and

MMP-12 KO (mean = 71.37, Dunn's post hoc $p = 0.0011$) comparing to wildtype control mice (mean = 109.3), when male and female mice are analyzed together. The same

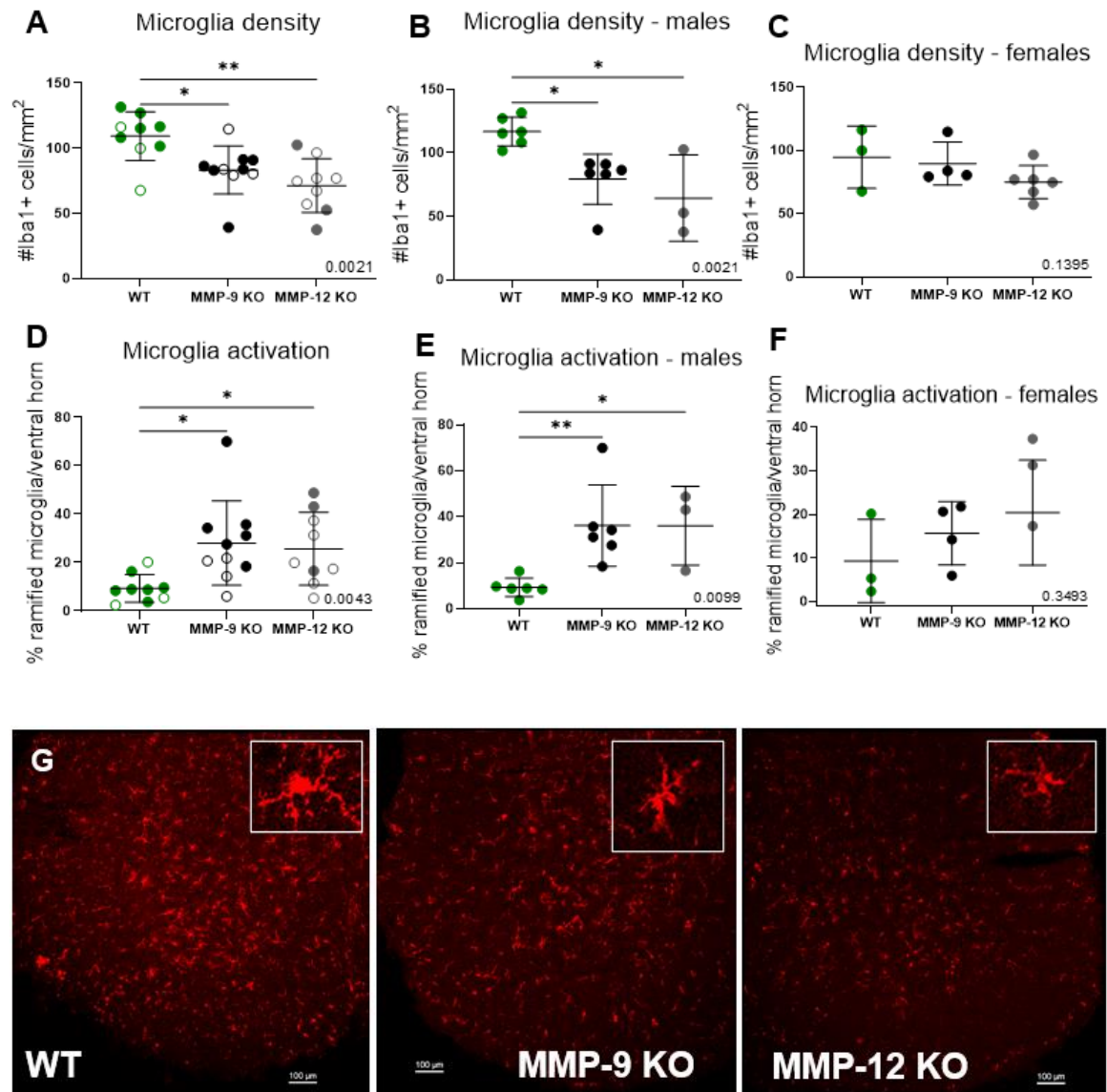


Figure 3.2 MMP-9 and MMP-12 KO decrease microglia activation. **A** – the number of microglial cells is lower in both knockouts comparing to wildtype control mice (Kruskal-Wallis $p = 0.0021$; MMP-9 KO: $n = 10$, mean $\pm$ SD = 83.28 ± 18.48 , Dunn's post hoc $p = 0.0461$; MMP-12 KO: $n = 9$, mean $\pm$ SD = 71.37 ± 20.57 , Dunn's post hoc $p = 0.0011$; WT: $n = 9$, mean $\pm$ SD = 109.3 ± 18.78 ; open circles represent females); **B** – both MMP-9 and MMP-12 KO male mice exhibit lower microglia density (Kruskal-Wallis $p = 0.0021$; MMP-9 KO: $n = 6$, mean $\pm$ SD = 79.13 ± 19.79 , Dunn's post hoc $p = 0.0134$; MMP-12 KO: $n = 3$, mean $\pm$ SD = 64.23 ± 33.99 , Dunn's post hoc $p = 0.0307$) comparing to wildtype males ($n = 6$, mean $\pm$ SD = 116.7 ± 11.20); **C** – female MMP-9 and MMP-12 KO mice do not show a decrease in microglia density (Kruskal-Wallis $p = 0.1395$; MMP-9

KO: $n = 4$, mean $\pm$ SD = 89.51 ± 16.91 , Dunn's post hoc $p > 0.9999$; MMP-12 KO: $n = 6$, mean $\pm$ SD = 74.94 ± 13.00 , Dunn's post hoc $p = 0.2302$) comparing to wildtype controls ($n = 3$, mean $\pm$ SD = 94.62 ± 24.70); **D** – MMP-9 and MMP-12 KO mice exhibit decrease microglia activation, as they have more non-activated, ramified microglia (Welch's ANOVA $p = 0.0043$; MMP-9 KO: $n = 10$, mean $\pm$ SD = 28.01 ± 17.41 , Dunnett's post hoc $p = 0.0165$; MMP-12 KO: $n = 9$, mean $\pm$ SD = 25.69 ± 15.04 , Dunnett's post hoc $p = 0.0240$; WT: $n = 9$, mean $\pm$ SD = 9.358 ± 5.740 ; open circles represent females); **E** – both MMP-9 and MMP-12 KO male mice exhibit lower microglia activation (ordinary one-way ANOVA $p = 0.0099$; MMP-9 KO: $n = 6$, mean $\pm$ SD = 36.21 ± 17.67 , Dunnett's post hoc $p = 0.0099$; MMP-12 KO: $n = 3$, mean $\pm$ SD = 36.15 ± 17.21 , Dunnett's post hoc $p = 0.0313$) comparing to wildtype males ($n = 6$, mean $\pm$ SD = 9.357 ± 4.041); **F** – female MMP-9 and MMP-12 KO mice do not show a decrease in microglia activation (ordinary one-way ANOVA $p = 0.3493$; MMP-9 KO: $n = 4$, mean $\pm$ SD = 15.72 ± 7.27 , Dunnett's post hoc $p = 0.6285$; MMP-12 KO: $n = 6$, mean $\pm$ SD = 20.47 ± 12.05 , Dunnett's post hoc $p = 0.2537$) comparing to wildtype controls ($n = 3$, mean $\pm$ SD = 9.361 ± 9.538); **G** – a representative image of a ventral horn in the lumbar spinal cord of an old WT, MMP-9 KO, and MMP-12 KO mouse, stained for Iba1. Insets show morphology of a single cell.

trend is true for male mice (MMP-9 KO: 79.13, Dunn's post hoc $p = 0.0134$; MMP-12 KO: 64.23, Dunn's post hoc $p = 0.0307$, WT males: 116.7; fig. 3.2 B). However, the difference was not seen in females (MMP-9 KO: 89.51, Dunn's post hoc $p > 0.9999$; MMP-12 KO: 74.94, Dunn's post hoc $p = 0.2302$, WT females: 94.62; fig. 3.2 C). Similarly, microglia activation was significantly decreased in both sexes combined and male, but not female mice, when analyzed separately. MMP-9 and MMP-12 KO mice have more non-activated, ramified microglia (MMP-9 KO: 28.01, Dunnett's post hoc $p = 0.0165$; MMP-12 KO: 25.69, Dunnett's post hoc $p = 0.0240$; WT: 9.358; fig. 3.2 D), and this trend is even stronger in males alone (MMP-9 KO: 36.21, Dunnett's post hoc $p = 0.0099$; MMP-12 KO: 36.15, Dunnett's post hoc $p = 0.0313$; WT males: 9.357; fig. 3.2 E). Female MMP-9 and MMP-12 KO mice exhibited a trend towards a decrease in microglia activation (an increase in the number of non-activated microglia), however it did not reach statistical significance (MMP-9 KO: 15.72, Dunnett's post hoc $p = 0.6285$; MMP-12 KO: 20.47, Dunnett's post hoc $p = 0.2537$; WT females: 9.361, fig. 3.2 F). There is overall a negative correlation between the α -MN number and microglia density (Pearson $r = -0.32$), slightly stronger in male ($r = -0.37$) than female mice ($r = -0.26$; Appendix B

Supplemental Figure B.1 A). Interestingly, the direction of the correlation is reversed in MM-12 KO mice ($r = 0.29$), while in MMP-9 KO and WT mice the correlation is still

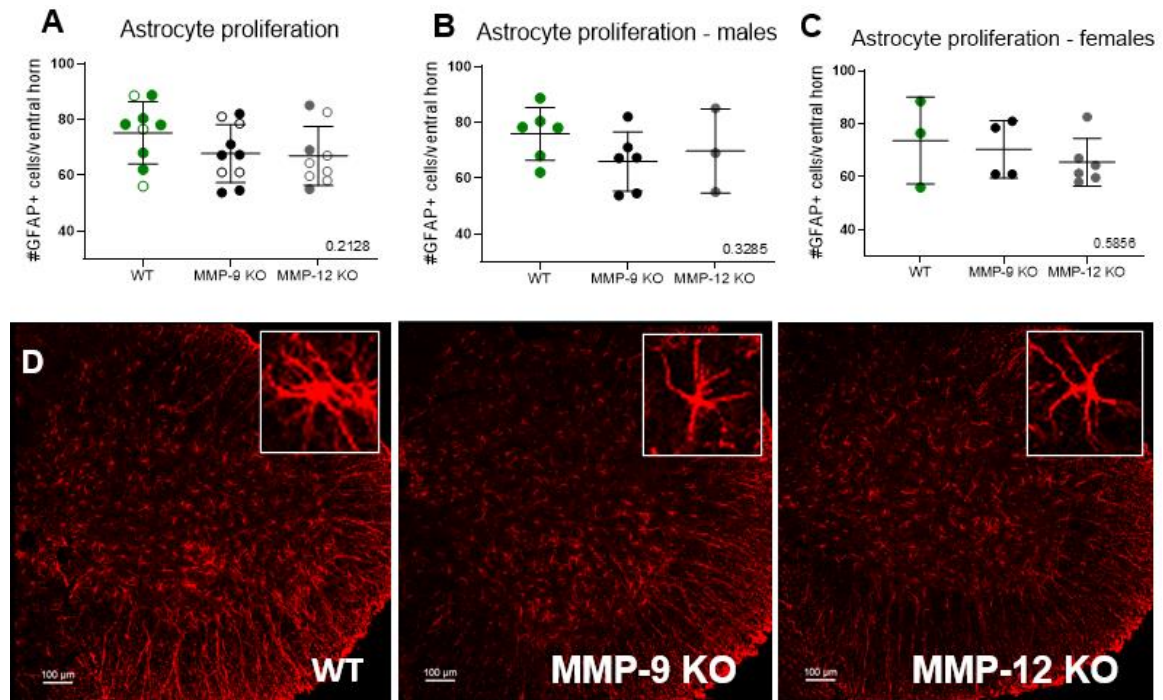


Figure 3.3 Knocking out MMP-9 and MMP-12 does not affect astrocyte proliferation. **A** – the number of GFAP-positive cells is not significantly different at 25 months of age (ordinary one-way ANOVA $p = 0.2128$; MMP-9 KO: $n = 10$, mean $\pm$ SD = 67.73 ± 10.37 , Dunnett's post hoc $p = 0.2428$; MMP-12 KO: $n = 9$, mean $\pm$ SD = 66.88 ± 10.54 , Dunnett's post hoc $p = 0.1953$; WT: $n = 9$, mean $\pm$ SD = 75.14 ± 11.17 ; open circles represent females); **B** – knocking out MMP-9 or MMP-12 does not have effect on astrocyte activation in male mice (ordinary one-way ANOVA $p = 0.3285$; MMP-9 KO: $n = 6$, mean $\pm$ SD = 65.96 ± 10.64 , Dunnett's post hoc $p = 0.2516$; MMP-12 KO: $n = 3$, mean $\pm$ SD = 69.67 ± 15.01 , Dunnett's post hoc $p = 0.6623$) comparing to wildtype controls ($n = 6$, mean $\pm$ SD = 75.88 ± 9.468); **C** – knocking out MMP-9 or MMP-12 does not have effect on astrocyte activation in female mice (ordinary one-way ANOVA $p = 0.5649$; MMP-9 KO: $n = 4$, mean $\pm$ SD = 70.38 ± 10.87 , Dunnett's post hoc $p = 0.8940$; MMP-12 KO: $n = 6$, mean $\pm$ SD = 65.49 ± 8.985 , Dunnett's post hoc $p = 0.4980$) comparing to wildtype controls ($n = 3$, mean $\pm$ SD = 73.67 ± 9.488); **D** – a representative image of a ventral horn of lumbar spinal cord of an old WT, MMP-9 KO, and MMP-12 KO mouse, immunostained for GFAP; the insets show a morphology of a single astrocyte.

negative ($r = -0.27$ and $r = -0.082$, respectively). As expected, the overall correlation between the α -MN number and the percentage of non-activated microglia (as a measure of microglia activation) is positive ($r = 0.4609$; males: $r = 0.4880$, females: $r = 0.63559$),

however in MMP-12 KO mice, the correlation approaches zero ($r = 0.06$; MMP-9 KO: $r = 0.49$, females: $r = 0.16$; Appendix B Supplemental figure. B.1 B). Both knock-outs though did not exhibit any changes in astrocyte activation in neither male, females, nor when combined (fig. 3.3). On average, MMP-9 KO mice had 67.73 GFAP-positive cells per ventral horn, while MMP-12 KO had 66.88, and WT 75.14.

Knocking down MMP-12, but not MMP-9, decreases BSCB permeability

BSCB permeability increases with age (Piekarz et al., 2020), which contributes to the age-related α -MN loss. We previously showed that an “anti-aging” treatment with OKN-007 decreases BSCB permeability in lumbar spinal cord of old WT mice, which was reflected in higher α -MN counts in treated animals (ref. OKN aging paper), therefore we wanted to see, if manipulating MMPs can affect BSCB integrity. Interestingly, BSCB permeability to HRP test revealed that there is a significant decrease in the number of HRP extravasation regions in the MMP-12 KO mice (on average 0.4722) comparing to WT controls (2.350, ordinary one-way ANOVA $p = 0.0114$, Dunnett’s post hoc $p = 0.0062$), but not in MMP-9 KO mice (1.442, Dunnett’s post hoc $p = 0.2227$. fig. 3.4 A and B). Consistently with the results of BSCB permeability assay, western blotting revealed that claudin-5 is more abundant in MMP-12 KO mice (on average 1.458), comparing to WT controls (0.8975, ordinary one-way ANOVA $p = 0.0587$, Dunnett’s post hoc $p = 0.0404$), which indicates that there is a positive effect on tight junctions. MMP-9 KO mice, however, do not show statistically significant increase in claudin-5 levels (1.073, Dunnett’s post hoc $p = 0.1895$; fig. 3.4 C and D).

MMP-12 KO affects gastrocnemius muscle mass in aging females, but not males, without affecting remaining hindlimb muscles

Sarcopenia is a common component of aging (Waltz et al., 2018) and the loss of innervation is one of the reasons behind it (Muller et al, 2007). Since we observed that manipulating MMPs influenced α -MN survival in aging, we wanted to see, if decreased

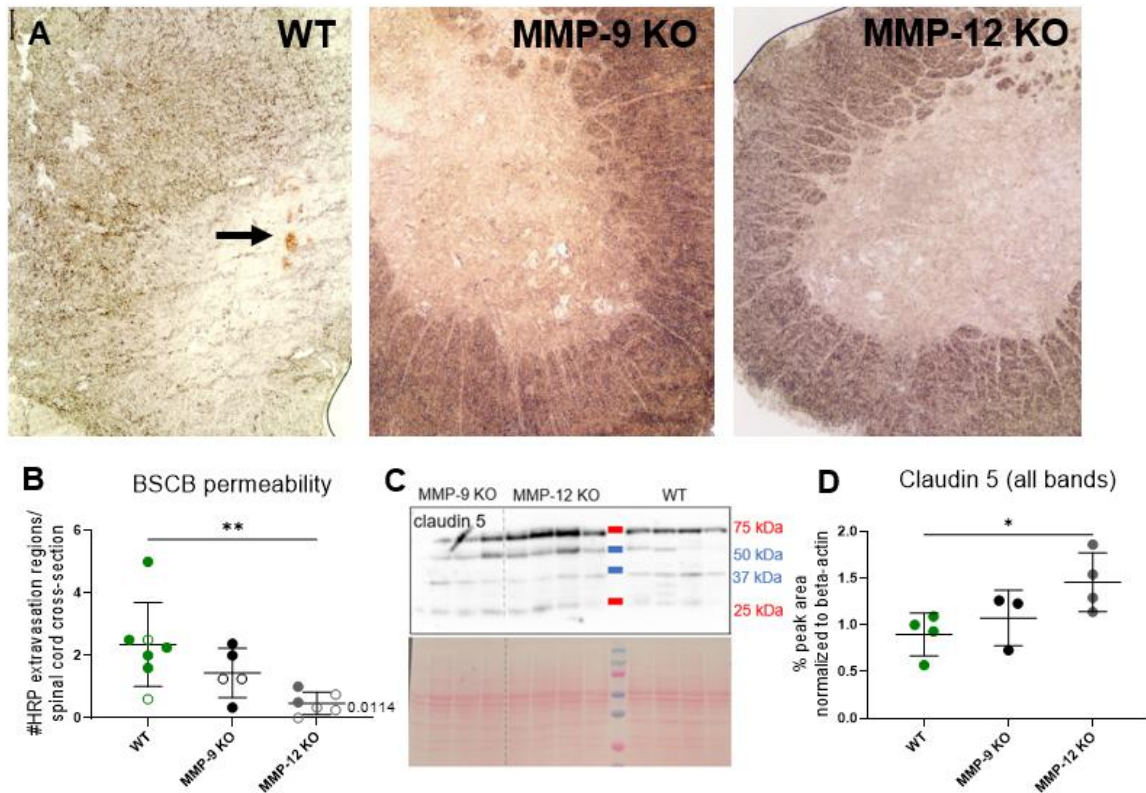


Figure 3.4 Knocking down MMP-12, but not MMP-9, decreases BSCB permeability. **A** – a representative image of a ventral horn in lumbar spinal cord of old WT, MMP-9 KO, and MMP-12 KO mice after injecting HRP; **B** – there is a significant decrease in the number of HRP extravasation regions in the MMP-12 KO mice ($n = 6$, mean $\pm$ SD = 0.4722 ± 0.36) comparing to WT controls ($n = 7$, mean $\pm$ SD = 2.350 ± 1.343 , ordinary one-way ANOVA $p = 0.0114$, Dunnett's post hoc $p = 0.0062$), but not in MMP-9 KO mice ($n = 5$, mean $\pm$ SD = 1.442 ± 0.7882 ; Dunnett's post hoc $p = 0.2227$); **C** – western blot probed for claudin-5 (top panel) and corresponding Ponceau stain (bottom); **D** – quantification of all claudin-5 bands; claudin-5 is more abundant in MMP-12 KO mice ($n = 4$, mean $\pm$ SD = 1.458 ± 0.315), comparing to WT controls ($n = 4$, mean $\pm$ SD = 0.8975 ± 0.2279 , ordinary one-way ANOVA $p = 0.0587$, Dunnett's post hoc $p = 0.0404$; MMP-9 KO: $n = 3$, mean $\pm$ SD = 1.073 ± 0.2977 , $p = 0.1895$).

α -MN loss translates into maintaining muscle mass with age. Only MMP-12 KO female mice showed significant difference in gastrocnemius muscle weight, in both absolute

weight (ordinary one-way ANOVA $p = 0.0315$; Dunnett's post hoc $p = 0.0191$) and after normalizing to body weight (ordinary one-way ANOVA $p = 0.0359$; Dunnett's post hoc $p = 0.0250$; fig. 3.5 A and B). Other hindlimb muscles, such as soleus, quadriceps, TA, and EDL, remained unaffected, so did body weight, except for MMP-12 KO males

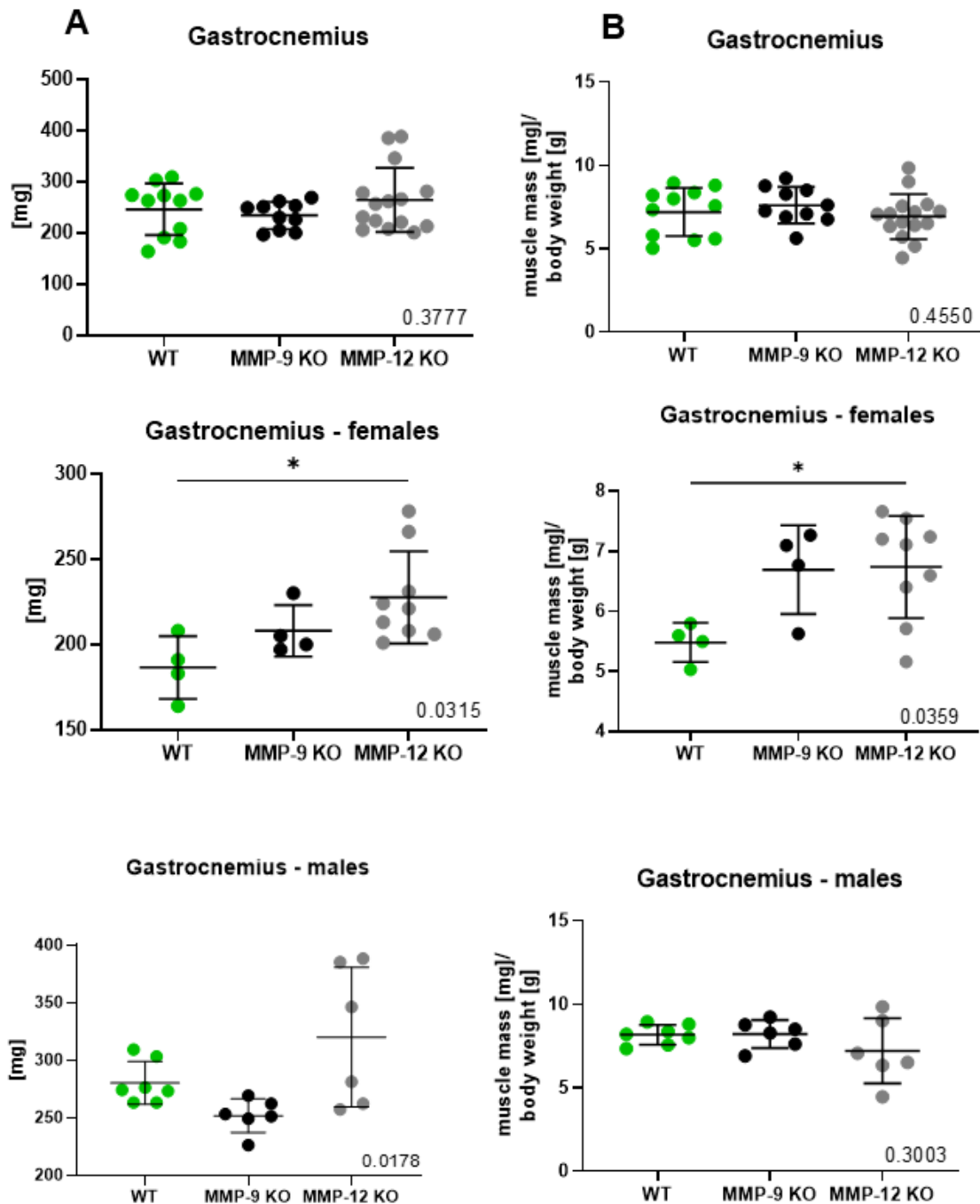


Figure 3.5 MMP-12 KO affects gastrocnemius muscle mass in aging females, but not males. **A** – absolute muscle weight in mg show that knocking out MMP-12 but not MMP-9 increases gastrocnemius muscle mass in females, but not in males. Both males and females: Kruskal-Wallis $p = 0.3777$; Dunn's post hoc WT vs MMP-9 KO $p = 0.9599$, WT vs MMP-12 KO $p > 0.9999$; females: ordinary one-way ANOVA $p = 0.0315$; Dunnett's post hoc WT vs MMP-9 KO $p = 0.3349$, WT vs MMP-12 KO $p = 0.0191$; males: ordinary one-way ANOVA $p = 0.0178$; Dunnett's post hoc WT vs MMP-9 KO $p = 0.3052$, WT vs MMP-12 KO $p = 0.1235$; **B** – after normalizing to body weight, MMP-12 KO female mice still exhibit higher gastrocnemius muscle mass comparing to female WT mice. Both males and females: ordinary one-way ANOVA $p = 0.4550$; Dunn's post hoc WT vs MMP-9 KO $p = 0.6944$, WT vs MMP-12 KO $p = 0.8235$; females: ordinary one-way ANOVA $p = 0.0359$; Dunnett's post hoc WT vs MMP-9 KO $p = 0.0649$, WT vs MMP-12 KO $p = 0.0250$; males: ordinary one-way ANOVA $p = 0.3003$; Dunnett's post hoc WT vs MMP-9 KO $p = 0.9971$, WT vs MMP-12 KO $p = 0.3041$.

(Kruskal-Wallis $p < 0.0001$, Dunn's post hoc $p = 0.0486$; Appendix B Supplemental Figure B.2).

Discussion

In our study we see that knocking out MMP-12 and MMP-9 has a positive effect on α -MN survival in aging, when data from male and female mice are analyzed together. When the sexes are analyzed separately though, only MMP-9 KO males showed higher α -MN number comparing to age and sex-matched wildtype controls. MMPs have been shown before to influence motor neuron health, as *dMMP1* expression increases with age in *Drosophila* motor neurons and was linked to motor function impairment (Azpurua et al., 2018). The study shows that reactivation of MMPs in aging has deleterious effects, even though MMPs are important for proper development, and later in life play crucial role in wound healing. Some reports mention sex-specific differences in MMPs expression, but not in context of aging. For instance, men have higher levels of plasma MMP-3 upon infection (Collazos et al., 2015), as well as higher levels of MMP-8 in tuberculosis (Sathyamoorthy et al., 2015). However, aortas of female rats have lower levels of MMP-9, which corresponds also to decreased inflammation in terms of macrophage infiltration, and better vascular outcomes in abdominal aortic aneurysm (Ailawadi et al., 2004; see

Boese et al, 2017 for review of sex differences in vascular pathology). There are also sex differences in MMP-2 expression in smooth muscle progenitor cells (Li et al., 2017)

The immunostaining for Iba1 revealed that both MMP-9 KO and MMP-12 KO mice had lower inflammation levels, in terms of decreased microglia density and activation (males and females combined), however after looking at males and females separately, we saw sex-specific response, as only MMP-9 and MMP-12 KO males showed lower microglia density and activation. None of the groups though exhibited any differences in astrocyte proliferation comparing to wildtype controls. Interestingly, knocking out MMP-12 affected the relationship between α -MN number and microglia activation. Our results show that there is a general negative correlation between α -MN number and microglia density, however MMP-12 KO mice exhibit a reverse trend, and in this case the correlation becomes positive. Also, we noticed a positive correlation between α -MN number and the percentage of non-activated, ramified microglia, which is not present in MMP-12 KO mice. Those changes would suggest that in absence of MMP-12 microglia activation is less harmful to α -MNs. Additionally, as also decreased microglia density and activation in male MMP-12 KO mice suggests, MMP-12 seems to be important in microglia activation. Hansmann et al. (2019) reached a similar conclusion in their study of Theiler's murine encephalomyelitis in MMP-12 KO mice (sex not specified).

We also see an improvement in BSCB integrity in MMP-12 KO mice, as shown in BSCB permeability to HRP assay. BSCB allows for maintaining the healthy environment for α -MNs, as toxic metabolites present in the blood, cannot infiltrate the parenchyma. Literature reports that blood-brain barrier (BBB) permeability increases with age, leads to hyperactivation of transforming growth factor β pathway, and contribute to neuronal dysfunction (Senatorov Jr. et al., 2019). TGF- β signaling in turn was shown to influence

MMP expression, especially MMP-12 (Towner et al., 2019). MMPs were also linked to BBB disruption in focal ischemia in rat (Yang et al., 2007; Yang and Rosenberg, 2011). An increase abundance of claudin-5 in MMP-12 KO mice that we observe in this study is another indication that increased levels of MMP-12 in aging have a deleterious effect on BSCB integrity. Claudin-5 downregulation was previously linked with MMP-induced decrease in BBB integrity (Yang et al., 2007; Yang and Rosenberg, 2011). It is expressed by endothelial cells and it's necessary for the maintenance of tight junctions. Decreased abundance of claudin-5 was linked to neuroinflammation and neurodegenerative diseases (Greene et al., 2019). Claudin-5 is commonly shown to produce a single band on western blots, however the multiple bands that we observe in claudin-5 western blot could be oligomers or proteins with different post-translational modifications, as well as different isoforms, as for example McCaffrey et al. (2007) report multiple claudin-5 isoforms that span molecular weights of 32-155 kDa.

MMP levels were shown to be affected by resistance training in aging muscle in rats (especially MMP-2, de Sousa Neto et al., 2018), as well in a pilot study on humans (MMP-2, MMP-9, MMP-15, Wessner et al., 2018), suggesting that MMPs play an important role in muscle remodeling in exercise and aging. Our study shows that knocking out MMP-12 has a protective effect on gastrocnemius muscle in female, but not male mice, in terms of decreasing age-related loss of muscle mass. Also, old MMP-12 KO males had higher body weight than age-matched male wildtype mice. Since knocking out MMP-9 has a beneficial effect on α -MNs and inflammation in aging spinal cord in males, while knocking out MMP-12 seems to affect more muscle mass in females, it would be interesting to see in a future study, whether MMP-9 and MMP-12 double KO could have a synergistic effect on both neuronal and muscle health in aging in both sexes. Moreover, a future lifespan study would demonstrate, if manipulating

MMPs has a beneficial effect on lifespan in mice. Furthermore, sacrificing animals later than at 25 months of age could reveal more MMP-mediated changes in muscle mass.

In conclusion, our study shows that manipulating MMPs has a sex-dependent protective effect on aging spinal cord, since α -MNs are more protected in males (MMP-9 KO), which also have decreased inflammation, in terms of decreased microglia density and activation (both MMP-9 KO and MMP-12 KO), as well as higher body weight (MMP-12 KO). However, only females show increase in hindlimb muscle mass (gastrocnemius in MMP-12 KO) at the time of the sacrifice.

Materials and Methods

1. Animals

All experiments were approved by Oklahoma Medical Research Foundation (OMRF) Institutional Animal Care and Use Committee. MMP-9 (stock No: 007084) and MMP-12 KO (stock No: 004855) mice were purchased from the Jackson Laboratory and bred in OMRF's animal facility, in specific pathogen free environment, with standard 12 h light/dark cycle. Mice had unrestricted access to chow and water. All mice were aged in our colony and sacrificed when they reached 25 months of age. The knock-outs were verified by genotyping.

2. Immunofluorescence

Spinal cord immunostaining was performed as previously described (Piekarz et al., 2020) with the use of the following antibodies were used: anti-NeuN – Cell Signaling #D3S3I, rabbit mAB, 1:500; anti-Iba1 – Wako #011-27991, goat, 1:250; anti-GFAP – Abcam, ab7260, rabbit, 1:1000; Cy3-conjugated AffiniPure goat anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #111-165-003, 1:200; Alexa Fluor 488- conjugated AffiniPure donkey anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #711-545-152,

1:200; CyTM3 AffiniPure donkey anti-goat IgG (H+L) – Jackson ImmunoResearch, #705-165-147, 1:200. The images were taken on the Nikon C2 confocal microscope. Alpha-motor neurons were counted as described previously (Piekarz et al., 2020; total MNs counted – 1,664). Microglia proliferation was measured by counting all Iba1-positive cells in both gray and white matter of the ventral horn (10,280 cells in total) and normalized to the area. Microglia activation was assessed by counting the number of non-activated, ramified microglia and expressed as the percentage of all Iba1-positive cells per ventral horn. Astrocyte activation was measured by counting all GFAP-positive protoplasmic astrocytes in gray matter per ventral horn (7,191 cells in total). The images were analyzed blindly.

3. Immunoblotting

Immunoblotting was performed as previously described (Piekarz et al., 2020). The following antibodies were used: anti-claudin-5 (A-12) mouse monoclonal antibody, Santa Cruz sc-374221, 1:100; anti- β -actin, rabbit, CST #4967, 1:1000; goat anti-mouse IgG-HRP, Santa Cruz, sc-2005, 1:5,000; mouse anti-rabbit IgG-HRP, Santa Cruz, sc-2357, 1:5,000. The blots were analyzed using ImageJ software by measuring peak area and normalizing to β -actin.

4. Blood-spinal cord barrier permeability assay

BSCB permeability assay to HRP consisted in injecting HRP (type II, #P8250, Sigma, St. Louis, MO, 75 mg/kg in 0.9% saline) intraperitoneally into anaesthetized mice as previously described (Piekarz et al., 2020). HRP extravasation regions per spinal cord hemi-section were then counted using a brightfield microscope.

5. Statistical analysis

Statistical analyses were performed using Graphpad Prism 9. After testing for normality, either ordinary two-way ANOVA (with Welch's correction when necessary) with Dunnett's post hoc multiple comparisons test or Kruskal-Wallis with Dunn's post hoc test was used. The post hoc multiple comparisons were performed pairwise relative to the WT control.

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CHAPTER IV

OKN-007 DECREASES AGE-RELATED ALPHA-MOTOR NEURON LOSS AND IMPROVES MUSCLE FUNCTION IN WILDTYPE MICE

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CONTRIBUTIONS BY DOCTOR OF PHILOSOPHY CANDIDATE:

KMP created the experimental design together with HVR; KMP performed the experiments, analyzed the data and created the figures 4.1, 4.2 A-D, 4.3 A-C, 4.4, 4.5, Supplemental Figure C.1; plotted the results in figures 4.2 E, 4.6; created Table 4.1, Supplemental Table C.1; and drafted the manuscript.

Abstract

Aging is accompanied by neuronal loss and neuroinflammation, as well as by a gradual decline in muscle mass and function. In the present study we treated wildtype male mice with OKN-007, a compound with anti-oxidative, neuroprotective and anti-inflammatory properties, starting at 16 months of age until the sacrifice at 25 months of age. The treated mice exhibited a decrease in age-related α -motor neuron (α -MN) loss (on average 15.45 α -MNs per ventral horn comparing to 11.44 in untreated controls), as well as lower abundance of cleaved caspase-3, suggesting a decrease in apoptosis. The treatment also decreased neuromuscular junction (NMJ) fragmentation and down-regulated markers of denervation, as well as significantly improved muscle contractile force, without affecting hindlimb muscle mass. However, the treated mice did not exhibit the age-related lean mass loss. OKN-007 decreased microglia proliferation and

activation, but did not reduce astrocyte proliferation in the lumbar spinal cord of old mice. The treatment improved as well blood-spinal cord barrier (BSCB) integrity, which was shown to be affected in aging. Moreover, RNA-seq analysis revealed that the transcriptome of treated mice is altered and the changes involve the transcripts linked to pathways and diseases such as fibrotic pathway, organismal death, cell death, movement disorders, and quantity of neurons. Our findings suggest that OKN-007, due to its neuroprotective and anti-inflammatory effect on the aging spinal cord, could be a potential therapeutic that can improve α -MN health and muscle function in aging. Further studies are needed to pinpoint the precise mechanism of action of OKN-007 in aging.

Significance Statement

Age-related neuronal loss and decline in muscle mass and function are common issues that contribute to frailty and motor deficits in elderly population. In the present study we show that OKN-007 decreases age-related α -motor neuron loss, improves muscle function, as well as decreases inflammation and blood-spinal cord barrier permeability. Furthermore, the compound has a high translational potential due to the fact that it can be administered in drinking water and the lack of harmful side effects. Currently OKN-007 is already in multiple clinical trials as a treatment for glioblastoma, and has been administered to humans for several years without adverse effects.

Introduction

Aging is associated with neuronal loss and decline of muscle mass and function, also known as sarcopenia (Waltz et al., 2018). It leads to a decrease in activity, loss of independence and lower quality of life, which have significant implications for society and healthcare costs (Beaudart et al., 2014; Steffl et al., 2017). Our laboratory has demonstrated that loss of innervation is a critical factor in the initiation of sarcopenia (Muller et al, 2007) and in the previous experiments we observed a 41% decrease in α -

motor neuron number in 26-27 months old wildtype C57Bl/6 mice comparing to 3-6 months old control group (Piekarz et al., 2020).

OKN-007 (Oklahoma nitrone 007) is a nitrone-based compound widely investigated glioblastoma research (Floyd et al., 2013; Towner et al., 2019) for which is currently in multiple clinical trials (clinicaltrials.gov: NCT03649464, NCT03587038, NCT01672463), as it decreases decrease tumor volume and increases patients' survival (de Sousa et al., 2015a; Towner et al., 2019). OKN-007 has anti-oxidative (Floyd et al., 2011; de Souza et al., 2015b; Altinoz and Elmaci, 2018), as well as anti-inflammatory properties (Fan et al., 2010) and was shown to be neuroprotective in traumatic brain injury (Clausen et al., 2008). OKN-007 parent compound, PBN, normalized neuronal structural changes in APP/PS1 mice (Garcia-Alloza et al., 2010), and in a previous study we observed a beneficial effect of OKN-007 on disease progression and neuronal survival in G93A ALS mouse model (ref. ALS paper), which shows that OKN-007 could be a potential therapeutic for neurodegeneration. It was suggested that OKN-007 can have a beneficial effect in aging as well (Floyd et al., 2013), due to the fact that aging has been connected to neuroinflammation and increased oxidative stress, however to our knowledge, the effect of OKN-007 in aging hasn't been investigated yet and our study is the first to show the neuroprotective effect of treatment with OKN-007 in aging.

In the present study, we aimed to determine, if the treatment with OKN-007 can be beneficial in the spinal cord and muscle aging. Because OKN-007 was shown to have neuroprotective, anti-oxidative and anti-inflammatory properties, we hypothesized that it can decrease age-related α -MN loss and therefore improve muscle health. We treated WT male mice with OKN-007 in drinking water (150 mg/kg/day) starting at 16 months of age, which is before we observe a significant loss of α -MNs. The treatment lasted 9 months, until the sacrifice when the animals reached 25 months of age. We then

assessed the number of α -MNs, muscle mass and function (contractile force), the levels of inflammation and apoptosis, as well as NMJ structure, BSCB permeability and changes in transcriptome. Here we report that the treatment with OKN-007 decreased age-related α -MN loss and NMJ fragmentation, as well as improved muscle function and age-related loss of lean mass, without affecting hindlimb muscle mass at the time of the sacrifice. Treated old mice exhibited lower microglia proliferation and activation, but not astrocyte proliferation, and had lower BSCB permeability. Given our results, OKN-007 is a promising compound for preventing age-related α -MN loss and improving muscle function.

Results

OKN-007 decreases age-related α -motor neuron loss and apoptosis at 25 months of age

In a previous study we reported an age-related α -MN loss and identified increased apoptosis as one of the reasons behind it (Piekarz et al., 2020). Here we observed a significantly higher number of α -MNs at 25 months of age in OKN-007-treated mice comparing to untreated controls (on average 15.45 and 11.44 MNs/ventral horn, respectively; $p = 0.0028$; fig. 4.1 A and B). Since the treatment affected neuronal survival, we were interested to see, if caspase-3 activation is decreased in the treated group. Indeed, the abundance of cleaved caspase-3 is lower in OKN-007 treated group ($p = 0.0322$), suggesting a decrease in apoptosis (fig. 4.1 C and D).

OKN-007 decreases NMJ fragmentation and down-regulates denervation markers

Since the treatment with OKN-007 increases MN survival, we were wondering, if it contributes also to improved NMJ health. Neuromuscular junction staining revealed that, similarly to OKN-007 effect on NMJ in ALS (Chapter V), the treatment decreased NMJ fragmentation, without affecting NMJ area (fig.4.2 A, C-D). On average, 8.36% of NMJs

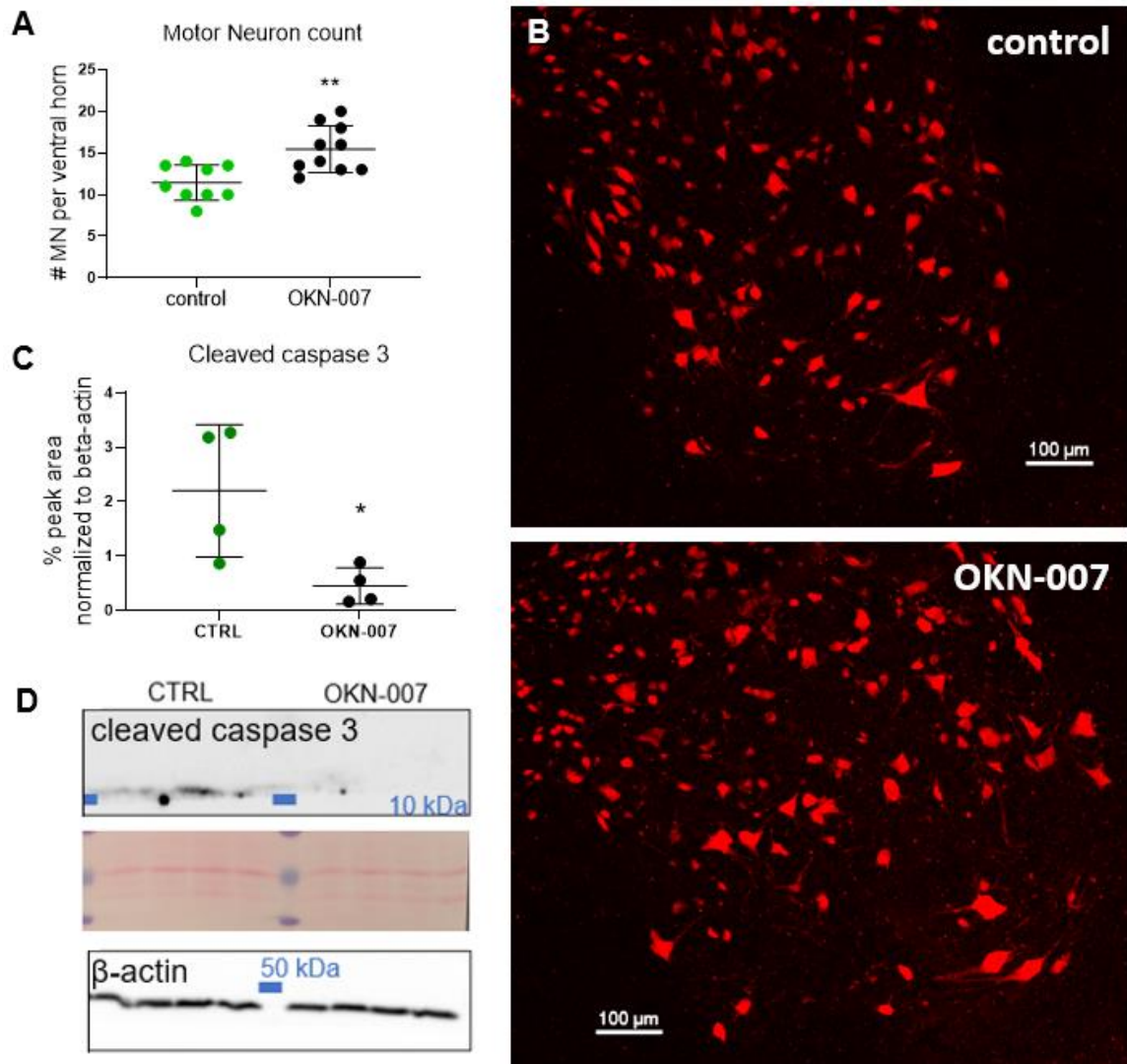


Figure 4.1 OKN-007 decreases age-related α -motor neuron loss. **A** – α -MN counts; at 25 months of age the number of motor neurons is higher in treated mice ($n = 10$, mean $\pm$ SD = 15.45 ± 2.79) comparing to untreated controls ($n = 9$, mean $\pm$ SD = 11.44 ± 2.11 ; unpaired two-tailed t-test $p = 0.0028$); **B** – a representative image of a lumbar spinal cord ventral horn of an old untreated WT mouse (control) and old OKN-007-treated WT mouse (OKN-007); **C** – quantification of western blot probed for cleaved caspase-3. There is a significant decrease in cleaved caspase-3 abundance in old treated WT mice ($n = 4$, mean $\pm$ SD = 0.45 ± 0.335) comparing to old controls ($n = 4$, mean $\pm$ SD = 2.189 ± 1.214 ; unpaired two-tailed t-test $p = 0.0322$); **D** – western blot membrane showing cleaved caspase-3 band (top panel), Ponceau stain (middle panel), and beta-actin band (bottom panel).

are fragmented in treated mice, while in untreated controls the percentage of fragmented NMJs reaches 21.33%. There is a trend towards an increase in the number of fully innervated NMJs in treated old mice comparing to control (48.34% and 32.05%, respectively), however due to a high variance, often noted in old mice, it does not reach statistical significance (fig.4.2 B). Despite the absence of clear structural effect on NMJ denervation, the treatment significantly down-regulates the transcripts of 4 out of 5 denervation markers (GADD45 α , AChR subunit α , RunX1, and MuSK, but not SLN; fig. 4.2 E).

The treatment prevents age-related decrease in lean mass, as well as improves muscle function, without affecting hindlimb muscle mass

Because the treatment improves neuronal health and NMJ fragmentation and function, we hypothesized that the neuroprotective effect will be reflected in an improved muscle health. The treatment does not affect body weight at the time of sacrifice (fig. 4.3 A), but QMR revealed that there is a significant increase in lean mass at this time point (Šídák's post hoc test $p = 0.0056$; fig. 4.3 B). Fat mass is not affected by the treatment, only by time (time effect $p = 0.0018$; fig. 4.3 C). However, there is no difference in dissected hindlimb muscle mass (Appendix C Supplemental Figure C.1) between treated and untreated groups. Interestingly, there is a significant improvement in muscle function assessed by measuring muscle contraction force resulting from the stimulation of increasing frequency (1-300Hz; fig. 4.3 D). For both middle-aged WT mice (16 mo) and old untreated mice, as well as old OKN-007-treated mice and untreated mice, the absolute force starts to diverge at 120Hz (Tukey's post hoc middle-aged vs old control $p = 0.0043$, old control vs old treated $p = 0.0492$; 200Hz: middle-aged vs old control $p = 0.0009$, old control vs old treated $p = 0.0303$; 250Hz: middle-aged vs old control $p =$

0.0002, old control vs old treated $p = 0.0348$; 300Hz: middle-aged vs old control $p = 0.0002$, old control vs old treated $p = 0.0136$).

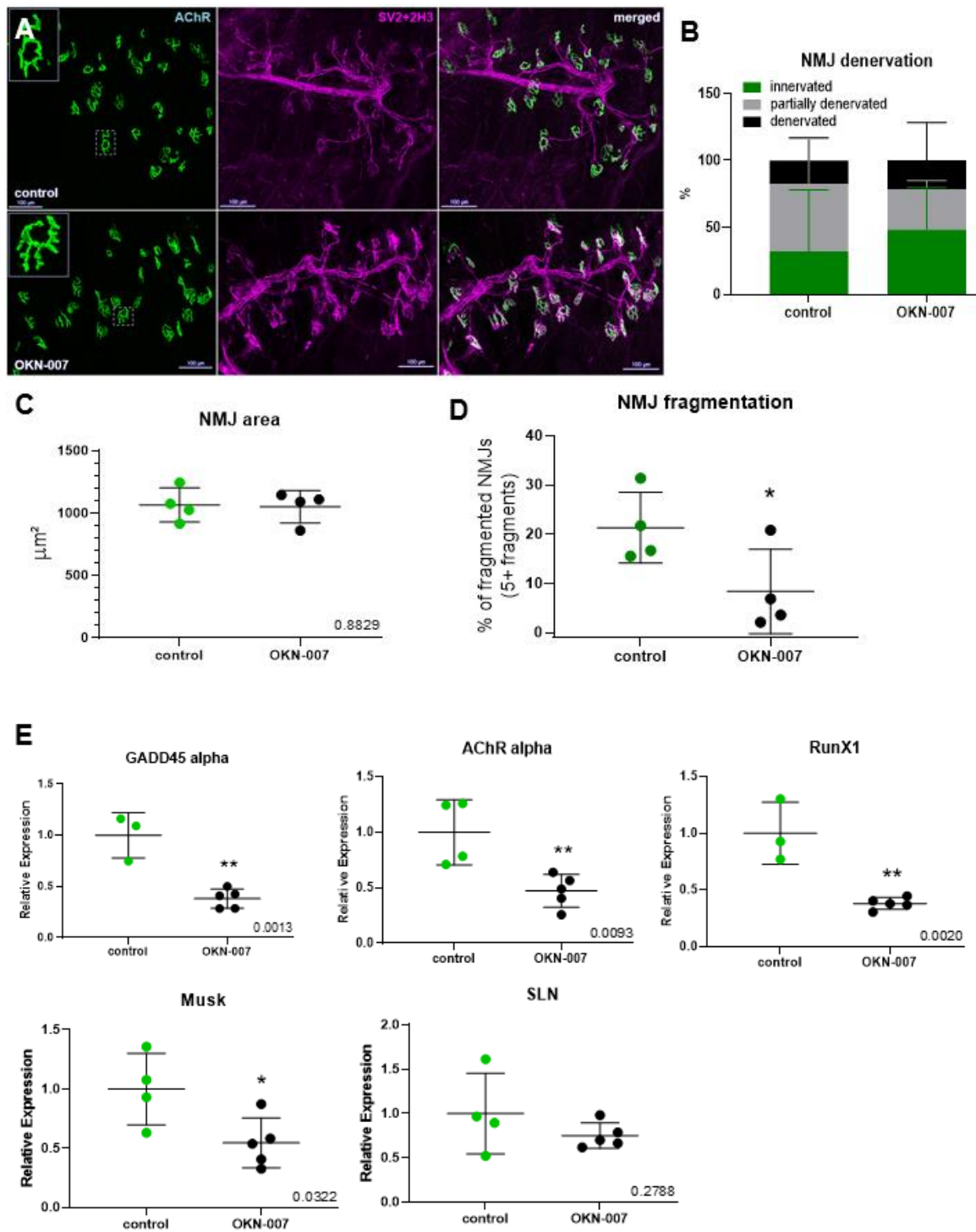


Figure 4.2 OKN-007 improves NMJ structure and down-regulates denervation markers. **A** – a representative image of NMJ staining for OKN-007 treated and untreated old mice, immunostained for neurofilaments and synaptic vesicles (SV2 and 2H3; in magenta), and for acetylcholine receptors (bungarotoxin conjugated to a fluorophore (in green); the inset shows the morphology of a single AChR; **B** – OKN-007 treatment does not change NMJ denervation defined as the level of overlap between presynaptic and postsynaptic sides; **C** – NMJ (AChR) area does not differ in both treated and untreated WT mice at 25 months of age (mean $\pm$ SD = 1051 ± 129.5 and 1066 ± 137.6 , respectively); **D** – the treatment decreases NMJ (AChR) fragmentation in old mice ($n = 4$, mean $\pm$ SD = 8.362 ± 8.5), comparing to the untreated group ($n = 4$, mean $\pm$ SD = 21.33 ± 7.2 ; unpaired one-tailed t-test $p = 0.0297$); **E** – the transcript levels of most denervation markers are lower in OKN-007-treated group compared to the control (GADD45 α : treated – $n = 5$, mean $\pm$ SD = 0.38 ± 0.09 , untreated – $n = 3$, mean $\pm$ SD = 1 ± 0.22 ; unpaired two-tailed t-test $p = 0.0013$; AChR α : treated – $n = 5$, mean $\pm$ SD = 0.47 ± 0.15 , untreated – $n = 4$, mean $\pm$ SD = 1 ± 0.29 ; unpaired two-tailed t-test $p = 0.0093$; RunX1: treated – $n = 5$, mean $\pm$ SD = 0.38 ± 0.05 , untreated – $n = 3$, mean $\pm$ SD = 1 ± 0.27 ; unpaired two-tailed t-test $p = 0.0020$; MuSK: treated – $n = 5$, mean $\pm$ SD = 0.55 ± 0.21 , untreated – $n = 4$, mean $\pm$ SD = 1 ± 0.30 ; unpaired two-tailed t-test $p = 0.0322$; SLN: treated – $n = 5$, mean $\pm$ SD = 0.75 ± 0.14 , untreated – $n = 4$, mean $\pm$ SD = 1 ± 0.45 ; unpaired two-tailed t-test $p = 0.28$).

If it comes to the specific force, middle-aged mice and old controls start to diverge at 80Hz ($p = 0.0193$), while old controls and treated old mice begin to diverge at 120Hz ($p = 0.0109$, middle-aged vs old control $p = 0.0002$; 200Hz: middle-aged vs old control $p = 0.0003$, old control vs old treated $p = 0.0398$; 250Hz: middle-aged vs old control $p = 0.0001$, old control vs old treated $p = 0.0456$; 300Hz: middle-aged vs old control $p < 0.0001$, old control vs old treated $p = 0.0235$).

OKN-007 decreases microglia proliferation and activation, without affecting astrocyte proliferation

Since inflammation often accompanies the process of aging and we previously reported increased microglia and astrocyte activation in the lumbar spinal cord of old WT mice (Piekarz et al., 2020), OKN-007, due to its anti-inflammatory properties, seemed a reasonable candidate to decrease inflammation in aging spinal cord. Immunostaining revealed a decrease in both microglia proliferation ($p = 0.0216$) and activation ($p = 0.0091$; fig. 4.4 A-C) in the spinal cord of treated old WT mice. However, as published

before (Chapter V), the treatment had no effect on astrocyte proliferation ($p = 0.2803$; fig. 4.4 D and E).

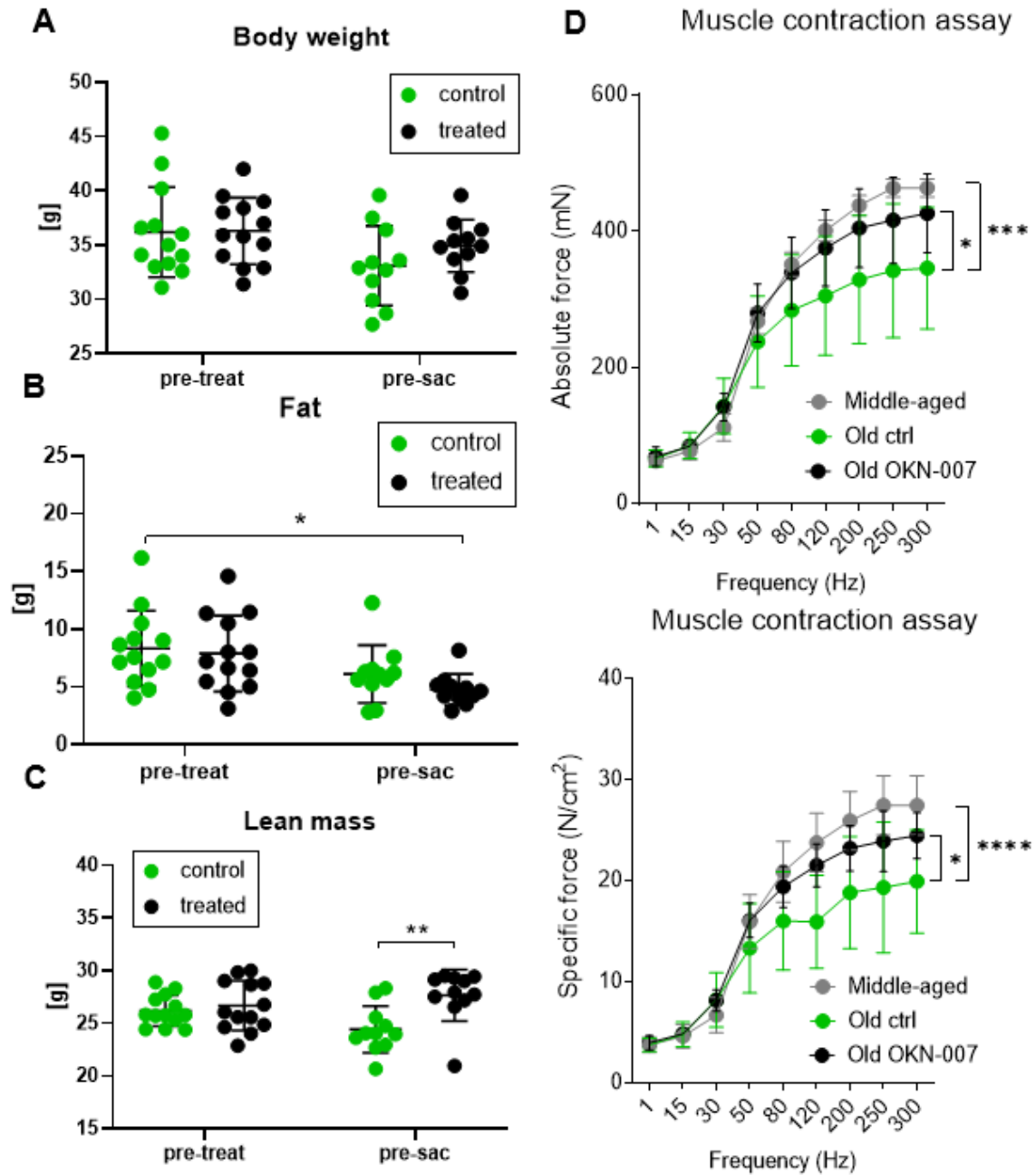
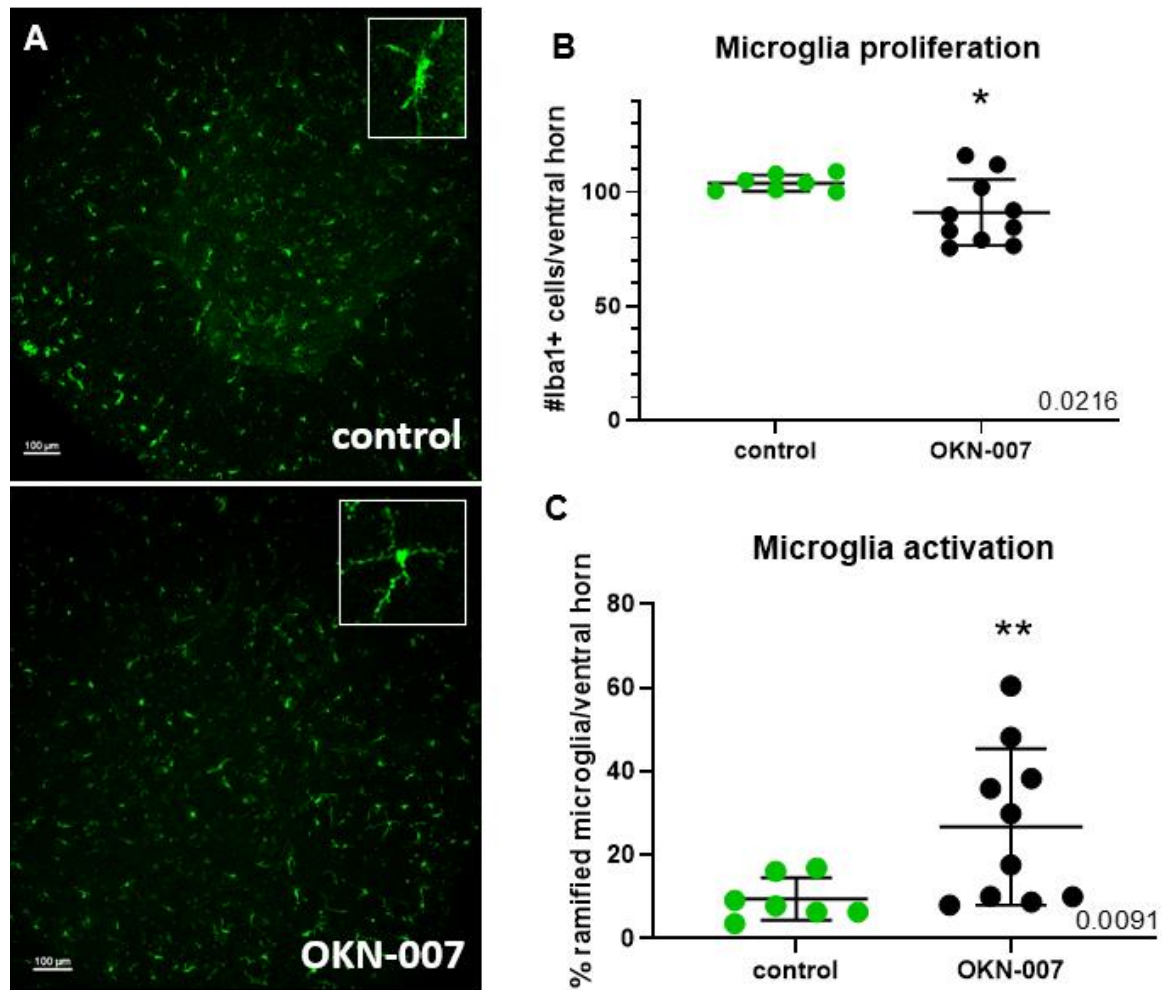


Figure 4.3 OKN-007 prevents age-related decrease in lean mass and improves muscle function. **A** – the treatment does not affect body weight which is influenced only by time (two-way ANOVA, time effect $p = 0.0292$, treatment effect $p = 0.3349$, interaction $p = 0.3870$); **B** – there is no difference in fat mass (established by QMR) between treated and untreated old WT mice at each timepoint (two-way ANOVA, time effect $p = 0.0018$, treatment effect $p = 0.2675$, interaction $p = 0.5567$); **C** – there is a significance

difference in lean mass pre-sacrifice (25 mo) between treated and untreated WT mice (Šídák's post hoc test for control vs treated $p = 0.0056$, two-way ANOVA, time effect $p = 0.5295$, treatment effect $p = 0.0045$, interaction $p = 0.0310$); D – muscle contraction assay revealed differences in both absolute (top; two-way ANOVA, frequency effect $p < 0.0001$, treatment effect $p < 0.0001$, interaction $p = 0.0093$) and specific force (bottom; two-way ANOVA frequency effect $p < 0.0001$, treatment effect $p < 0.0001$, interaction $p = 0.0036$) between middle-aged WT mice (16 mo) and old untreated mice, as well as between old OKN-007-treated mice and untreated ones. The stars on the graph indicate significance of the difference at the highest frequency of stimulation (300Hz; Tukey's multiple comparisons test).



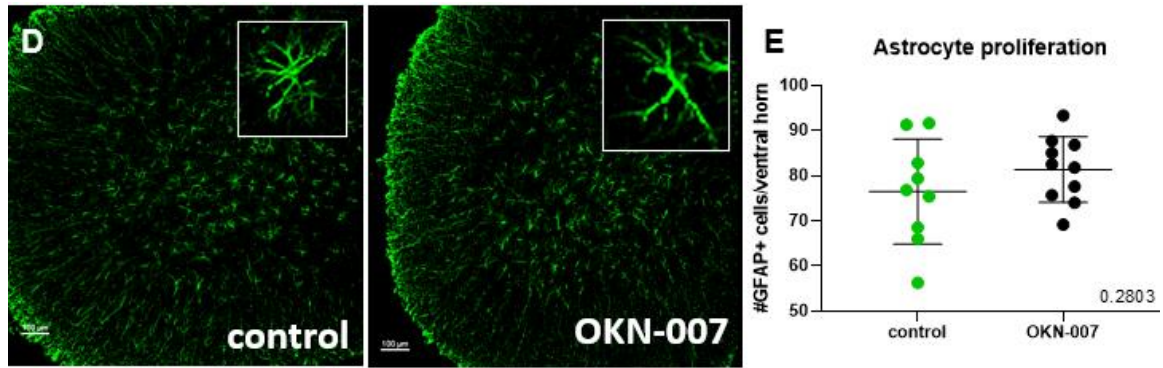


Figure 4.4 OKN-007 decreases microglia proliferation and activation, without affecting astrocyte proliferation. **A** – a representative image of immunostaining for Iba1 in lumbar spinal cord ventral horn of treated and untreated old wildtype mice; inset shows the morphology of a representative microglial cell; **B** – quantification of microglia proliferation in terms of the number of Iba1-positive cells per ventral horn; the treated group has less microglia (treated: $n = 10$, mean $\pm$ SD = 91.05 ± 14.49 ; untreated: $n = 7$, mean $\pm$ SD = 103.9 ± 3.63 unpaired t-test with Welch's correction $p = 0.0216$); **C** – quantification of microglia activation – the number of non-activated microglia (expressed as a percentage of all Iba1-positive cells per ventral horn) is higher in treated group ($n = 10$, mean $\pm$ SD = 26.72 ± 18.67) comparing to the control ($n = 7$, mean $\pm$ SD = 9.488 ± 5.08 ; unpaired t-test with Welch's correction $p = 0.0091$); **D** – a representative image of anti-GFAP immunostaining to visualize astrocytes in ventral horn of treated and untreated wildtype mice; the inset shows a close-up of a representative astrocyte; **E** – quantification of GFAP proliferation. The treatment does not affect the number of GFAP-positive cells (treated: $n = 10$, mean $\pm$ SD = 81.36 ± 7.27 ; untreated: $n = 9$, mean $\pm$ SD = 76.46 ± 11.59 ; two-tailed unpaired t-test, $p = 0.2803$).

OKN-007 decreases blood-spinal cord barrier (BSCB) permeability

As we previously reported, the BSCB permeability increases with age (Piekarczyk et al., 2020). Inflammation and increased blood-brain barrier permeability go hand in hand (Kim et al., 2012), so we hypothesized that since OKN-007 has an anti-inflammatory effect, it could improve BSCB integrity. Moreover, OKN-007 was shown to down-regulate MMP expression (Towner et al., 2019), and MMPs were connected to a decrease in barrier integrity (Yang et al., 2007; Yang and Rosenberg, 2011). The BSCB permeability to HRP test revealed that the treatment indeed decreases BSCB permeability, as there is a decrease in the number of HRP extravasation regions in the treated group in ($p = 0.0491$; fig. 4.5).

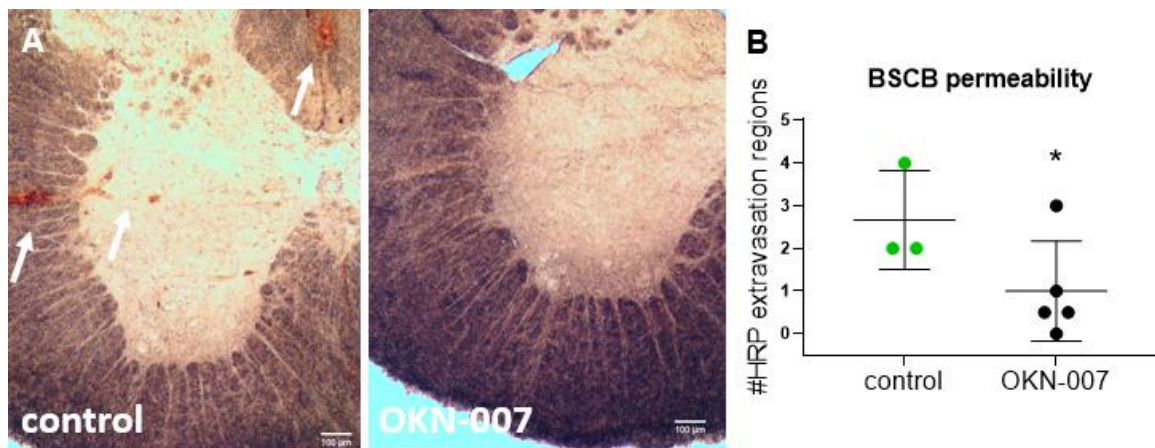


Figure 4.5 OKN-007 decreases BSCB permeability. **A** – a representative image of a lumbar spinal cord ventral horn for treated and untreated old WT mice after HRP injection and visualization with DAB; **B** – there is a decrease in the number of HRP extravasation regions in the treated group ($n = 5$, mean $\pm$ SD = 1 ± 1.17) comparing to untreated controls ($n = 3$, mean $\pm$ SD = 2.667 ± 1.155 , one-tailed unpaired t-test, $p = 0.0491$).

The treatment with OKN-007 changes the spinal cord transcriptome

RNA-seq analysis (fig. 4.6) revealed that there are 2257 differentially expressed genes (DEGs) in young vs old dataset, 2670 DEGs in young vs old OKN-007-treated, 119 in old OKN-007-treated vs old untreated and 133 genes that differ in young vs untreated old and that shift back towards the “youthful” state when OKN-007 is given. The most down-regulated transcript in treated vs untreated old mice is solute carrier family 22 organic cation transporter member 1 (Slc22a1), while the most downregulated one is immunoglobulin heavy constant gamma 2B (Ighg2b; Table 4.1). The list of all differentially expressed genes in treated vs untreated old mice can be found in Appendix C Supplemental Table C.1. IPA analysis shows that the top diseases affected by the treatment with OKN-007 influence, among other, organismal death, concentration of triacylglycerol, glucose tolerance, cell death, movement disorders, quantity of neurons and quantity of cells. Among the most enriched canonical pathways affected by the treatment, there are fibrotic pathway, GPCR-mediated nutrient sensing, GP6 signaling pathway, endocannabinoid neuronal synapse pathway, corticotropin releasing hormone

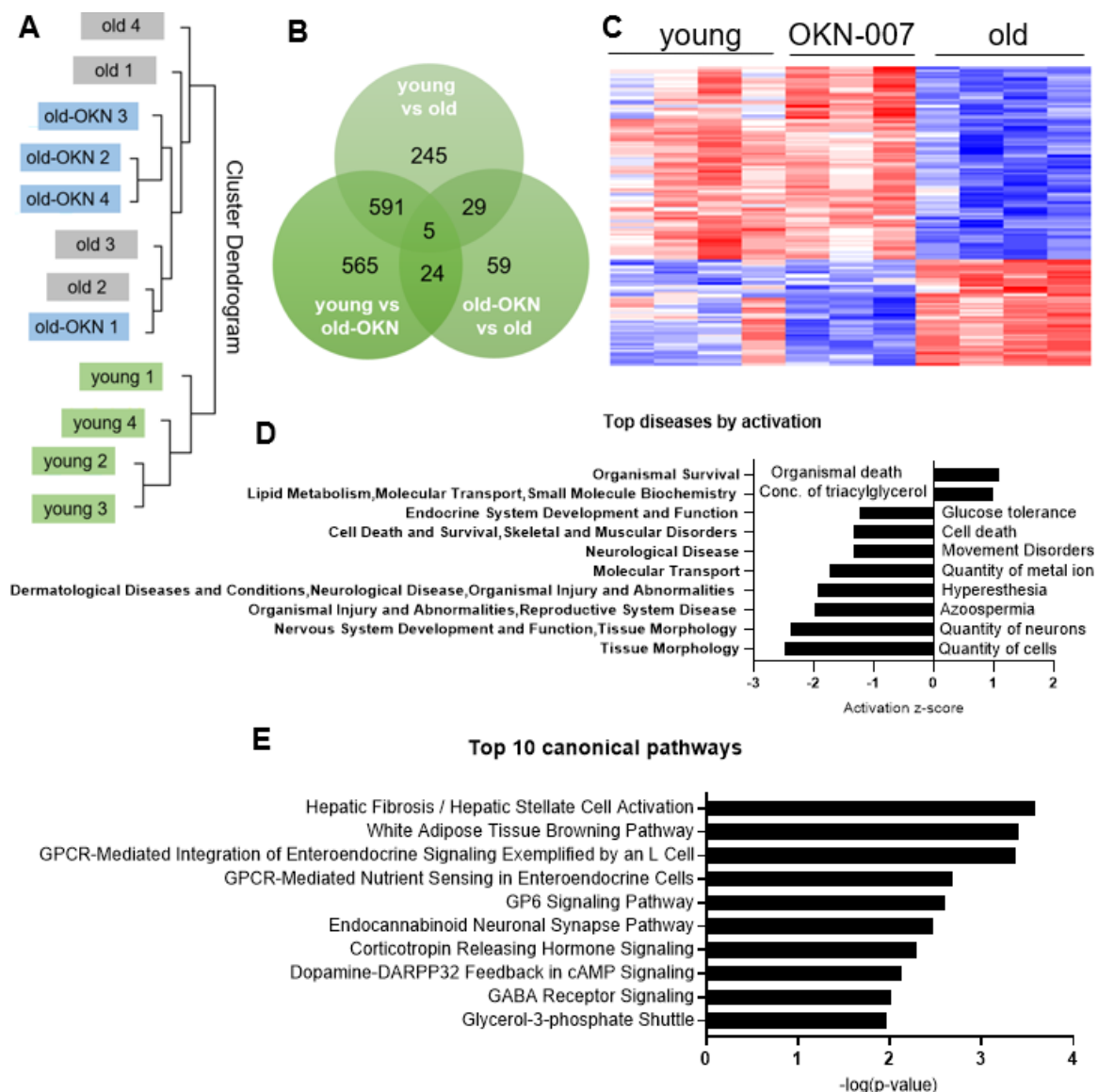


Figure 4.6 The treatment with OKN-007 affects the spinal cord transcriptome. **A** – dendrogram showing clustering of samples after RNA-seq analysis (young – 3-4 mo; old untreated – 25 mo, old OKN-007-treated – 25 mo); sample old-OKN 1 clustered with old untreated ones and was dropped from further analysis as a “non-responder” **B** – Venn diagrams showing overlap in gene expression revealed by RNA-seq analysis; **C** – the heatmap showing 133 genes that are differentially expressed in young vs old mice and which OKN-007 brings back to the “youthful” state; **D** – top ten diseases affected by OKN-007 in aging (old untreated vs old OKN-007-treated) sorted by the activation score, identified by IPA analysis; **E** – top ten canonical pathways affected by OKN-007 (IPA analysis; old untreated vs old OKN-007-treated), ordered by p-value (with B-H correction).

signaling, dopamine-DARPP32 feedback in cAMP signaling, GABA receptor signaling, and glycerol-3-phosphate shuttle. Interestingly, 22 out of 119 DEGs in treated vs untreated old groups are poorly described predicted genes, which makes the results less informative for the purpose of elucidating the mechanism behind the neuroprotective effect of OKN-007 in spinal cord aging.

Discussion

Our study shows that OKN-007 has a neuroprotective and anti-apoptotic effect on aging spinal cord, which is in agreement with the previous reports regarding the neuroprotective properties of OKN-007 (Clausen et al., 2008; Ewert et al., 2012; Du et al., 2016). We saw a decrease in age-related α -MN loss and decreased caspase-3 activation in old WT mice treated for 9 months, starting at 16 months of age. As discussed previously (ref. ALS paper), increasing OKN-007 dosage to for example 300 mg/kg could potentially bring further benefits. Also, it remains to be established, if starting the treatment at an earlier timepoint could increase the beneficial effect on the aging spinal cord. The only NMJ structural improvement that we observed was decreased AChR fragmentation, however OKN-007-treated old mice showed functional improvement both at the level of neuron (down-regulation of denervation markers GADD45 α , AChR subunit α , RunX1, and MuSK, but not sarcolipin) and muscle (increased contraction force), in the absence of effect on hindlimb muscle mass or on the degree of physical overlap between pre- and post-synaptic sides at the time of sacrifice. Therefore, OKN-007 may confer functional benefits that can compensate for detrimental age-related structural changes. We did not see either any effect on NMJ area that was shown to increase with age (Piekarz et al., 2020). However, comparing to our previous study (Piekarz et al., 2020) NMJ area did not change with age. Here, the NMJ area in old untreated group is comparable to the young one from our previous

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
-0.47	-4.71	-4.24	0.000078	Ighg2b	immunoglobulin heavy constant gamma 2B
-5.89	-3	2.88	0.000036	Slc22a1	solute carrier family 22 (organic cation transporter), member 1
-5.38	-2.52	2.86	0.00014	Gm48478	predicted gene, 48478
-1.18	1.66	2.83	8.8E-06	Gm38372	predicted gene, 38372
-5.89	-3.21	2.68	0.00011	Gm10581	predicted gene 10581
-5.53	-2.88	2.65	0.000038	Gm16202	predicted gene 16202
-5.89	-3.26	2.62	4.3E-06	Gm3448	predicted gene 3448
-2.12	-4.63	-2.51	0.00012	Slc5a12	solute carrier family 5 (sodium/glucose cotransporter), member 12
-3.38	-5.89	-2.5	0.000069	Dcpp3	demilune cell and parotid protein 3
-5.89	-3.45	2.44	0.00014	Gm34176	predicted gene, 34176
-3.5	-5.89	-2.38	0.000057	Gm48773	predicted gene, 48773
-3.53	-5.89	-2.36	5.5E-06	Wnt3a	wingless-type MMTV integration site family, member 3A
-3.73	-5.89	-2.16	0.00011	Gm9451	predicted gene 9451
-3.96	-5.89	-1.93	0.000043	4930583P06Rik	RIKEN cDNA 4930583P06 gene
1.27	3.14	1.87	0.000019	Gucy1a2	guanylate cyclase 1, soluble, alpha 2
-0.08	-1.93	-1.85	0.00013	Cxcl13	chemokine (C-X-C motif) ligand 13
-1.5	0.32	1.82	0.000092	Gm28322	predicted gene 28322
-4.09	-5.89	-1.79	0.00013	AI646519	expressed sequence AI646519
-4.09	-5.89	-1.79	0.00013	Gm48407	predicted gene, 48407
0.84	2.5	1.66	0.00008	Gm26945	predicted gene, 26945

Table 4.1 Top 20 differentially expressed molecules in old treated vs old control

study, so there was no pathology for OKN-007 to reverse in this case. It might be because our second cohort (from NIA) did not show much of sarcopenic phenotype at the time of sacrifice, and a future lifespan study could clarify the issue. However, the

functional improvement shows that OKN-007 can also support proper neurotransmission at the time of the sacrifice. Similarly to what we observed regarding NMJ, we saw an improvement in muscle function, without changes in hindlimb muscle mass, however the treatment prevented the loss of the lean mass with age, as revealed by the QMR. There is a significant improvement in muscle contractile force, both absolute and specific after stimulation of 120Hz and above.

Consistently with the previous reports (e.g., Floyd et al., 2000; Fan et al., 2010) and our previous study on effects of OKN-007 in ALS progression in G93A mouse model (ref. ALS paper), our results reveal an OKN-007 mediated anti-inflammatory effect, as we saw a decrease in microglia activation, as well as proliferation, without changes in astrocyte activation. The decreased inflammation has a beneficial effect on MN survival. Another factor that contributes to the maintenance of healthy MN environment is BSCB integrity. The literature reports that the treatment with OKN-007 normalized increased blood-brain barrier (BBB) permeability in a BBB in vitro model (Culot et al., 2009), so we were curious, if the treatment can also normalize the increased blood-spinal cord barrier (BSCB) permeability that we observe in the spinal cord from old mice (Piekarz et al., 2020). Indeed, the treatment with OKN-007 improves the integrity of the barrier, as there are less HRP extravasation regions in BSCB permeability to HRP assay. It creates a healthier MN environment that is more conducive to MN survival, as the toxic metabolites present in the blood cannot penetrate parenchyma as easily as when the barrier is more permeable. It could also decrease the influx of immune system cell from the bloodstream. Senatorov Jr. et al (2019) linked age-related increase in BBB permeability to hyperactivation of transforming growth factor β (TGF- β) pathway in astrocytes and neuronal dysfunction. Interestingly, TGF- β signaling was proposed to be behind the OKN-007 beneficial effect in glioblastoma (Towner et al., 2019), as the

treatment inhibited TGF- β 1, which down-regulated matrix metalloproteinases (MMPs), especially MMP-12. Also, TGF- β is linked to microglia maintenance (Butovsky et al, 2014), and to decreased microglia activation and proliferation (Suzuruma et al., 1993). However, OKN-007 mechanism of action in aging may differ from the one in tumors. Further studies are needed to elucidate the mechanism behind OKN-007 neuroprotective effect in aging, however, as discussed before (ref. ALS paper), the fact that OKN-007 affects microglia but not astrocytes, may be a hint pointing towards the mechanism of action.

RNA-seq analysis revealed changes in transcriptome in the spinal cord of treated mice in pathways related to neurons (such as quantity of neurons, dopamine-DARPP32 feedback in cAMP signaling, GABA receptor signaling), as well as in transcripts connected to cell death, movement disorders, quantity of cells, fibrotic pathway, GPCR-mediated nutrient sensing, etc. Interestingly some of the transcripts affected are connected to microglial activation, such as Cxcl13 (Esen et al., 2014) and blood-brain barrier such as SLC22a1 (aka OCT1), which is expressed in rat brain microvessels (Lin et al., 2010) and it's important for transport across the barrier. The authors report that silencing SLC22a1 has a protective effect in a model of Parkinson's disease, as it caused a decreased MPTP uptake, and we saw that the SLC22a1 was the most down-regulated transcript in the OKN-007-treated group. SLC22a1 was also shown to be produced by microglia (Fan et al., 2018). On the other hand, the most upregulated transcript, Ighg2b, was shown to be present in remyelinated lesions in progressive multiple sclerosis (Elkjaer et al, 2019), suggesting that perhaps OKN-007 could alleviate demyelination that can accompany aging (Piekarz et al., 2020). We must point out though, that one of the treated mice clustered with untreated old mice (old-OKN 1) and was subsequently dropped from the further analyses as a non-responder, as it could not

provide information about the effect of OKN-007 on the transcriptome then. It suggests that some individuals may not respond to the treatment. However, aging is not a homogenous process, the population of aged mice is more heterogeneous than for example a tumor mouse model, so certain variance in response to OKN could be expected. Surprisingly high number of genes differentially expressed in treated old mice comparing to untreated old controls (22 out of 119; 13 out of top 20) remains unidentified at this point, as they constitute predicted genes that are poorly described.

In conclusion, our study shows that treatment with OKN-007 decreases age-related α -MN loss, affects inflammation in terms of decreasing microglia activation and proliferation, as well as decreases NMJ fragmentation and improves muscle function.

Materials and Methods

1. Animals

The experimental protocols were approved by the Oklahoma Medical Research Foundation (OMRF) Institutional Animal Care and Use Committee. Male C57BL/6J from OMRF colony and NIA were treated with OKN-007 in drinking water (150 mg/kg/day) from 16 months of age until sacrifice at 25 months of age. The mice were maintained on a 12 h light/dark cycle, and were fed ad libitum normal chow diet. Their water consumption and body weight were monitored weekly to allow for dose adjustments, if necessary, and to determine how much of the drug was actually consumed (on average 150.58 mg/kg/day).

2. Immunofluorescence

Spinal cord immunofluorescent staining was done as previously described (Piekarz et al., 2020). The following antibodies were used: anti-NeuN – Cell Signaling #D3S3I, rabbit mAB, 1:500; anti-Iba1 – Wako #011-27991, goat, 1:250; anti-GFAP – Abcam,

ab7260, rabbit, 1:1000; Cy3-conjugated AffiniPure goat anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #111-165-003, 1:200; Alexa Fluor 488- conjugated AffiniPure donkey anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #711-545-152, 1:200; CyTM3 AffiniPure donkey anti-goat IgG (H+L) – Jackson ImmunoResearch, #705-165-147, 1:200. The slides were imaged using Nikon C2 confocal microscope. Alpha-motor neuron were counted as described previously (Piekarz et al., 2020; total number of cells counted – 3,293). Microglia proliferation was assessed by counting all Iba1-positive cells per ventral horn (both gray and white matter; 6,825 cells in total) and microglia activation was determined by counting the number of non-activated, ramified microglia and expressed as the percentage of all Iba1-positive cells per ventral horn. Astrocyte proliferation was measured by counting all GFAP-positive cells per ventral horn (protoplasmic astrocytes in gray matter; 7,346 cells in total). Neuromuscular junction (NMJ) staining was performed as previously described (Piekarz et al., 2020), with a modification consisting in using frozen gastrocnemius muscle, instead of fresh one, as described in Chapter V. Antibodies used: anti-SV2, DSHB, #2315387, mouse monoclonal, 1:50; anti-2H3, DSHB, #2314897, mouse monoclonal, 1:50; α -BTX-Alexa 488, Invitrogen, #B13422, 1:1000; goat anti-mouse Cy3, Jackson ImmunoResearch, #115-165-146, 1:250. The scorer was blinded to the identity of the samples.

3. Immunoblotting

Western blotting was done as previously described (Piekarz et al., 2020) with the use of the following antibodies: anti-cleaved caspase-3, (Asp175, 5A1E) rabbit mAb CST #9664, 1:1000; goat anti-rabbit IgG-HRP, Santa Cruz, sc-2004, 1:10,000.

4. Muscle contraction assay

Contractile properties of extensor digitorum longus (EDL) muscle were measured in vitro as previously described (Ahn et al., 2019). Mice were sacrificed using gaseous carbon dioxide, and EDL muscle was immediately excised and prepared for functional assays in a bicarbonate-buffered solution (in mmol/l: 137 NaCl, 5 KCl, 1 MgSO₄, 1 NaH₂PO₄, 24 NaHCO₃, and CaCl₂) gassed with a mixture of 95% O₂ and 5% CO₂ at room temperature. After preparation, we placed the EDL muscle in an organ bath containing bicarbonate-buffered solution at room temperature and determined the length that induces maximal twitch force, i.e. optimal length (L_O). To find L_O, we preloaded the muscle with 25-30 mN and stimulated at 1Hz (600 mA current, 0.25ms pulse). The stimulations were repeated in 0.2- to 0.3 mm shortening steps until reaching maximal twitch force. We then increased the temperature to 32°C for 20 min of thermos-equilibration followed by the force-frequency protocol, consisting of stimulus frequencies of 1-300 Hz interspersed by 1 min intervals. Isometric force data were normalized by estimated muscle cross-sectional area (CSA; N/cm²) for calculation of specific force. To estimate the muscle CSA, we divided the muscle weight (g) by L_O and multiplied to muscle specific density (1.056 g/cm³; Close 1972). All data were recorded and analyzed using commercial software (DMC and DMA, Aurora Scientific).

5. qPCR

RNA was isolated from 20 mg of previously frozen spinal cord using standard TRIzol protocol and converted to cDNA using Superscript III 1st Strand Synthesis Kit (Invitrogen, Cat #18080-051), according to manufacturer's instructions. The results were normalized to the expression of 18s gene. The following primers were used: AChR- α : forward ACCTGGACCTATGACGGCTCT, reverse AGTTACTCAGGTCGGGCTGGT; GADD45- α : forward AGACCGAAAGGATGGACACG, reverse GTACACGCCGACCGTAATG; MuSK: forward TCCTGCGTGCTCCTGAATC, reverse

TGCAGCGTAGGGTTACAAAGG; Runx1: forward GATGGCACTCTGGTCACCG,
reverse GCCGCTCGGAAAAGGACAA, SLN: forward
GAGGTGGAGAGACTGAGGTCCTTGG, reverse GAAGCTCGGGGCACACAGCAG;
18s: forward GTGGAGCGATTTGTCTGGTT, reverse
GACGGTGATGGTCAGAGTGAAGC.

6. Blood-spinal cord barrier permeability assay

HRP (type II, #P8250, Sigma, St. Louis, MO, 75 mg/kg in 0.9% saline) was injected IP into anaesthetized mice as previously described (Piekarz et al., 2020). HRP extravasation regions per spinal cord hemi-section were then counted.

7. RNA-seq

Total RNA was isolated from 30 mg of the spinal cord of young, old control mice, and old OKN-007-treated mice using TRIzol reagent (Invitrogen, CA, USA) according to the manufacturer's protocol. The samples were prepared then and processed by the Clinical Genomics Center at OMRF (omrf.org/research-faculty/core-facilities/next-generation-sequencing/). TruSeq Stranded mRNA Library Kit (Illumina) was used for library preparation and the samples were sequenced on Illumina NextSeq 500. The results were then processed and analyzed by OMRF Genomics and Data Science group. One OKN-007-treated mouse clustered with untreated controls and was dropped from further analysis as a non-responder, since it would not provide information on OKN-007-induced transcriptional changes.

8. Statistical analysis

Statistical analyses were done in Graphpad Prism 9. To compare two groups, after confirming normality, unpaired two-tailed t-test was used, while to assess the difference

between treatments and timepoints, ordinary two-way ANOVA with post hoc multiple comparisons test was used.

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CHAPTER V

OKN-007 TREATMENT SLOWS DOWN DISEASE PROGRESSION IN SOD1 G93A ALS MOUSE MODEL BY DECREASING ALPHA-MOTOR NEURON LOSS AND MICROGLIA ACTIVATION, WITHOUT AFFECTING MUSCLE MASS AT P145

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CONTRIBUTIONS BY DOCTOR OF PHILOSOPHY CANDIDATE:

KMP created the experimental design together with HVR; KMP performed the experiments, analyzed the data and created the figures 5.1-5.5, 5.6 D-G, Supplemental Figure D.1; plotted the results in figure 5.6 A-C; and drafted the manuscript.

Abstract

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that leads to death of motor neurons in the brain and the spinal cord. In our study, we investigated whether treatment with OKN-007, a compound with antioxidant and anti-inflammatory properties, can modulate ALS disease onset and progression. We treated female and male SOD1 G93A mutant mice and wildtype mice with OKN-007 in drinking water starting at P60, before the mice become symptomatic, until sacrifice around day P145, just prior to end-stage. Disease progression was monitored and scored on a scale of 0-5 based on hindlimb tremor and progressing hindlimb dysfunction leading to paralysis. The treatment did not delay the disease onset, however, it significantly slowed down the disease progression. At sacrifice, the treated group did not progress beyond the score of 3, and majority of the mice reached only the score of 2, whereas the highest score in the untreated group was 4 and the most frequent score was the score of 3. The

treatment prevented the alpha-motor neuron loss without affecting body weight or the hindlimb muscle mass. The treated G93A mice had also healthier neuromuscular junctions, as the treatment decreased the fragmentation of acetylcholine receptors. OKN-007 did not affect astrocyte proliferation, while it had a significant effect on microglia activation. Moreover, perfusion MRI revealed that OKN-007 normalized the blood flow in G93A mice. Our results suggest that, due to its neuroprotective and anti-inflammatory effect, OKN-007 may be a potential treatment aiming at slowing down ALS progression.

Significance Statement

ALS is a progressive neurodegenerative disease that leads to death of motor neurons, paralysis and death. No cure exists and currently there are only few options, if it comes to disease modifying treatments. Our study shows that OKN-007 could be a potential therapeutic for ALS, as it slows down the disease progression in G93A ALS mouse model and prevents alpha-motor neuron loss, as well as decreases inflammation. Further studies are needed to show, whether the treatment can also increase lifespan.

Introduction

Amyotrophic lateral sclerosis (ALS), known as Lou Gehrig's disease, is a progressive neurodegenerative disease that leads to death of motor neurons in the brain and the spinal cord, which causes paralysis and eventually death due to the respiratory failure. There are currently four FDA-approved treatments for ALS: Riluzole (Rilutek) has an antiglutamatergic effect and increases patients' survival (by ~90 days); Edaravone (Radicava) is a free radical scavenger which slows down decline of motor function (a 33% decrease after 6 months of treatment; Jaiswal 2019); Nuedexta aims at treating pseudobulbar affect that accompanies the disease (Smith et al., 2017), and Tiglutik is a

liquid formulation of Riluzole for patients with difficulties swallowing, and there is still no cure for the disease (Dyer and Smith, 2016).

OKN-007 (OKlahoma Nitrone 007; disodium 4-(tert-butyl-imino) methyl benzene-1,3-disulfonate N-oxide, also known as NXY-059 and HPN-07) is a nitrone-based molecule that is being investigated as a treatment for glioblastoma (GBM; Towner et al., 2019; Floyd et al., 2013). It is a disulfonyl-phenyl derivative of PBN (N-tert-Butyl- α -phenylnitrone; Altinoz and Elmaci, 2018), which was first investigated in 1985-1988 in brain ischemia/reperfusion injury in gerbils, however the nitrones were used in free radical-related research since the 1960s (Floyd et al., 2013). OKN-007 was reported to increase survival and decrease tumor volume in GBM xenografts (de Sousa et al., 2015a; Towner et al., 2019), as well as in liver cancer (Yoon et al., 2018). It is currently in multiple clinical trials for the treatment of glioblastoma (e.g., NCT03649464, NCT03587038, NCT01672463 - clinicaltrials.gov). OKN-007 suppresses free radical production in model of cerebral ischemia (Altinoz and Elmaci, 2018), and in F98 rat glioma model (de Souza et al., 2015b) through spin trapping (Floyd et al., 2011), Nitrone compounds, besides scavenging free radicals, have been shown to have an anti-inflammatory effect, as the OKN-007 parent compound, PBN, prevented microglia activation and loss of NMDA receptors in kainic acid-induced rat epilepsy model (Floyd et al., 2000) and in IL-1 β -induced acute brain injury (Fan et al., 2010). PBN was reported to alleviate neuritic abnormalities in APP/PS1 mice, suggesting that nitrones can influence neuronal plasticity, however without affecting amyloid-beta plaques (Garcia-Alloza et al., 2010). Because oxidative stress and neuroinflammation were shown to contribute to cognitive decline in neurodegenerative diseases and in aging, and since nitrones have an anti-inflammatory effect and have free radical scavenging ability, it was proposed that PBN-nitrones can have a therapeutic effect in those ailments (Floyd et al.,

2013), however the potential therapeutic effect of OKN-007 in neurodegenerative diseases and aging hasn't been fully investigated. So far, OKN-007 (denominated in the study as HPN-07) in combination with N-acetylcysteine (NAC) was shown to decrease tau immunoreactivity in rat model of blast-induced traumatic brain injury (Du et al., 2016), and had a protective effect on the inner ear and cochlear nucleus in rats exposed to noise (Lu et al., 2014) and in the rat blast injury model (Ewert et al., 2012). Also, monotherapy with OKN-007 (denominated in the study NXY-059) was shown to be neuroprotective in rat model of fluid percussion TBI (Clausen et al., 2008).

In our study, we investigated whether treatment with OKN-007 can modulate ALS disease onset and progression. Since OKN-007 is a compound with antioxidant, anti-inflammatory and neuroprotective properties, we hypothesized that it will preserve α -MN number and muscle mass, as well as decrease neuroinflammation, which would influence the disease progression. We treated SOD1 G93A mutant mice and wildtype (WT) mice with OKN-007 in drinking water before they became symptomatic (starting at P60), tracked the disease progression, and after sacrificing the animals around day P145, assessed alpha-motor neuron number, inflammation and muscle mass. Here we report that the treatment with OKN-007 slowed down the disease progression, however without delaying the disease onset. The treated G93A mice had higher α -MN count, as well as exhibited lower microglia activation, however the treatment did not affect muscle mass or astrocyte proliferation.

Results

OKN-007 slows down disease progression without affecting disease onset

Disease progression was monitored and scored on a scale of 0-5 based on hindlimb tremor and progressing hindlimb dysfunction leading to paralysis. The treatment did not

delay the disease onset, as the treated G93A mice became symptomatic (reached the score of 1) on day 94 (median), while the untreated G93A mice reached that point on day 90 (Gehan-Breslow-Wilcoxon test $p = 0.4311$, fig. 5.1 A), however, in time both groups started to diverge as the disease progressed. The treatment significantly slowed down the disease progression, as the treated G93A mice reached a score of 2 on day 121, while untreated G93A mice progressed to stage 2 on day 114 ($p = 0.0347$, fig. 5.1 B). Not enough treated mice reached the score of 3 to calculate the median, however the control mice kept progressing and the median for reaching the score of 3 in this group was 141 days ($p = 0.0079$, fig. 5.1 C). One untreated mouse reached the score of 4, while none of the treated mice progressed to that stage by the time of sacrifice (fig. 5.1 D).

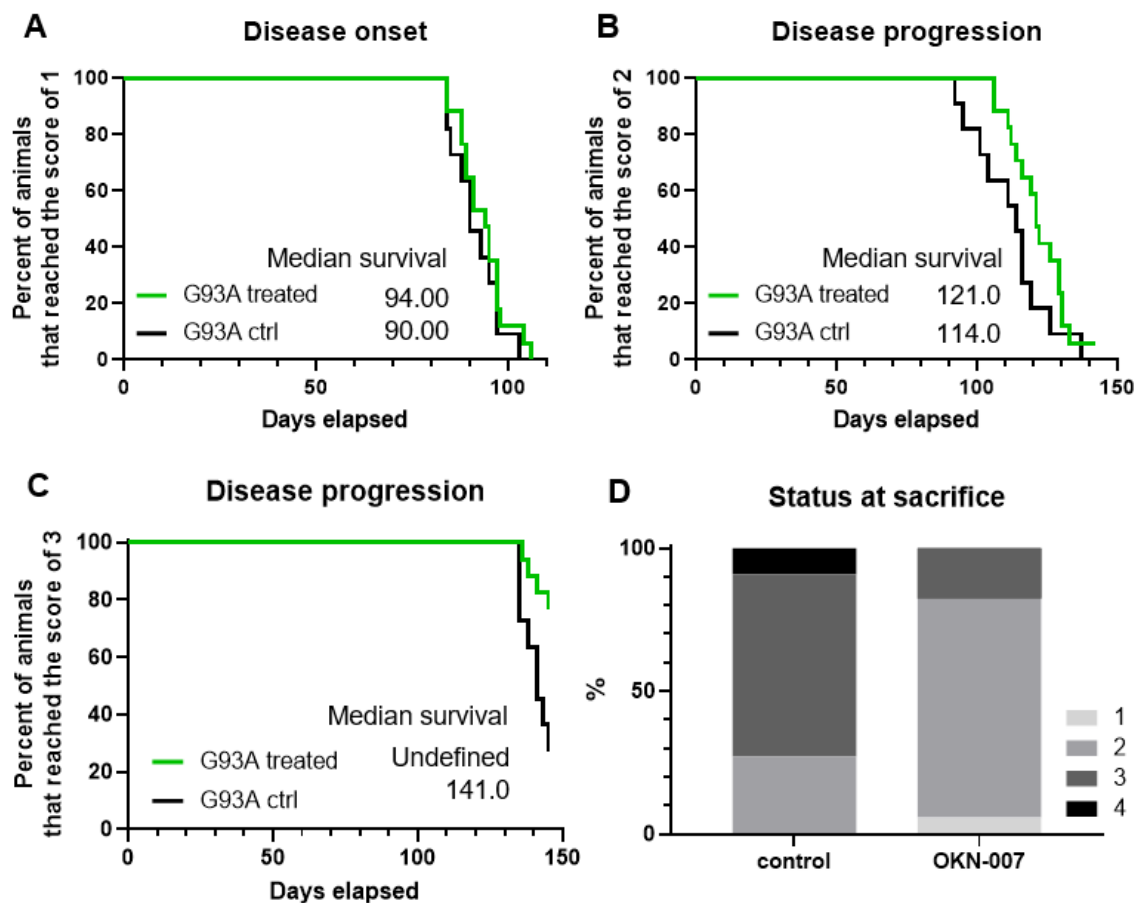


Figure 5.1 OKN-007 slows down disease progression without affecting disease onset. **A** – OKN-007 does not affect the disease onset (Gehan-Breslow-Wilcoxon test $p = 0.4311$); median day when untreated G93A mice become symptomatic (reach the score of 1, the scoring system is described in Methods section) is 90 days ($n = 11$), while for the treated ones it's 94 days ($n = 17$); **B** – treated mice progress to the score of 2 on average 7 days later than controls (G93A treated – 121 days, untreated – 114; $p = 0.0347$); **C** – the untreated mice reach the score of 3 at day 141, however the median is not reached by the treated group ($p = 0.0079$); **D** – mice status at the time of sacrifice (around day 145).

OKN-007 inhibits α -motor neuron loss

Since we saw that the treatment slows down the disease progression, we were wondering, if it's due to increased alpha-motor neuron (MN) survival and/or decreased muscle degeneration. Motor neuron counts revealed higher MN number in treated G93A mice comparing to untreated controls (fig. 5.2). Control G93A mice had 12.80 ± 3.37 alpha-MNs per ventral horn, while the treated G93A mice had 18.54 ± 3.88 alpha-MNs (Šídák's post hoc test $p = 0.0017$), which was comparable to the counts in WT mice (untreated: 20.98 ± 5.02 , treated: 20.98 ± 5.02 ; two-way ANOVA, genotype effect $p < 0.0001$, treatment effect $p = 0.0002$).

The treatment doesn't affect muscle mass at the time of sacrifice

Surprisingly, there was no difference between treated and untreated mice in muscle mass in none of the dissected hindlimb muscles at the time of sacrifice, there was only the genotype effect (Appendix D Supplemental Figure D.1). However, there is a trend towards body weight and lean mass preservation in treated G93A mice, while fat mass is affected only by time ($p < 0.0001$; fig. 5.3). The omnibus test shows a treatment effect on body weight ($p = 0.0225$), as well as time x treatment x genotype interaction ($p = 0.0121$), however there is no difference in post hoc test between treated and untreated G93A mice at the last timepoint. The lean mass follows a similar trend, where untreated

G93A mice start declining at the last timepoint, while the treated mice lean mass does not. Here as well

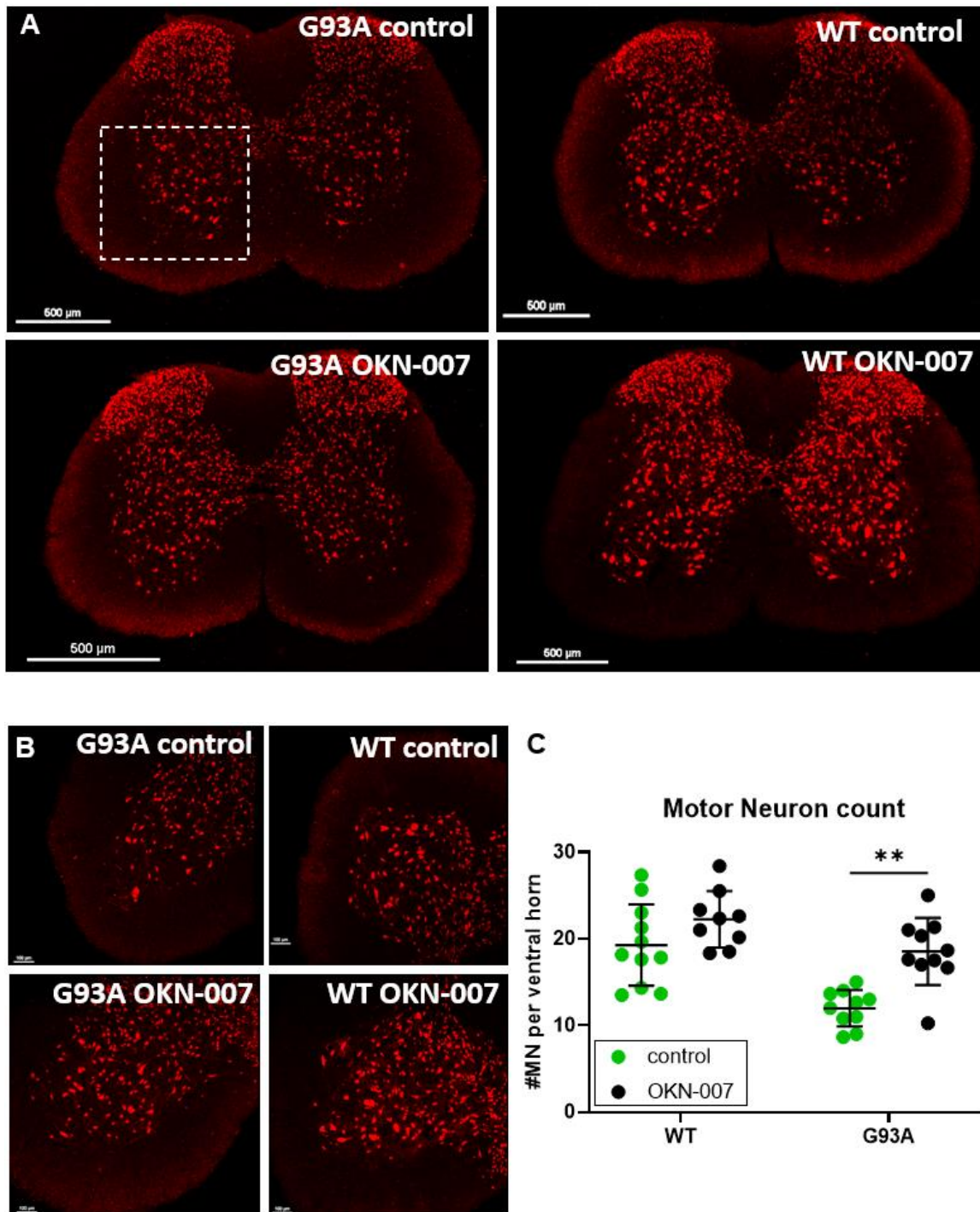


Figure 5.2 OKN-007 prevents α -motor neuron loss. **A** – a representative image of anti-NeuN immunostaining of a lumbar spinal cord cross-section (4x) with ventral horn; **B**

– a representative image of a ventral horn (10x) of treated and untreated wildtype and G93A mice (all groups 5 mo), used for α -MN counts; **C** – quantification of neuronal staining. There is a significant α -MN loss (two-way ANOVA, genotype effect $p < 0.0001$, treatment effect $p = 0.0002$) in untreated G93A mice ($n = 10$, mean $\pm$ SD = 12.80 ± 3.37) but not in the OKN-007 treated G93A group ($n = 10$, mean $\pm$ SD = 18.54 ± 3.88 ; Šídák's multiple comparisons post hoc test: G93A treated vs G93A control $p = 0.0017$). Both treated wt ($n = 9$, mean $\pm$ SD = 20.98 ± 5.02) and wt controls ($n = 11$, mean $\pm$ SD = 19.27 ± 4.68) significantly differ from untreated G93A mice (Šídák's post hoc test: wt control vs G93A control $p = 0.0003$, wt treated vs G93A control $p < 0.0001$); Prism identified two outliers (Grubbs, $\alpha = 0.1$), which were removed from the analysis; scorer was blinded to the group identity.

the omnibus test reports time $\times$ treatment $\times$ genotype interaction ($p = 0.0187$), but the post hoc test does not show any difference between treated and untreated G93A mice at the last timepoint, where masses start to diverge between those groups.

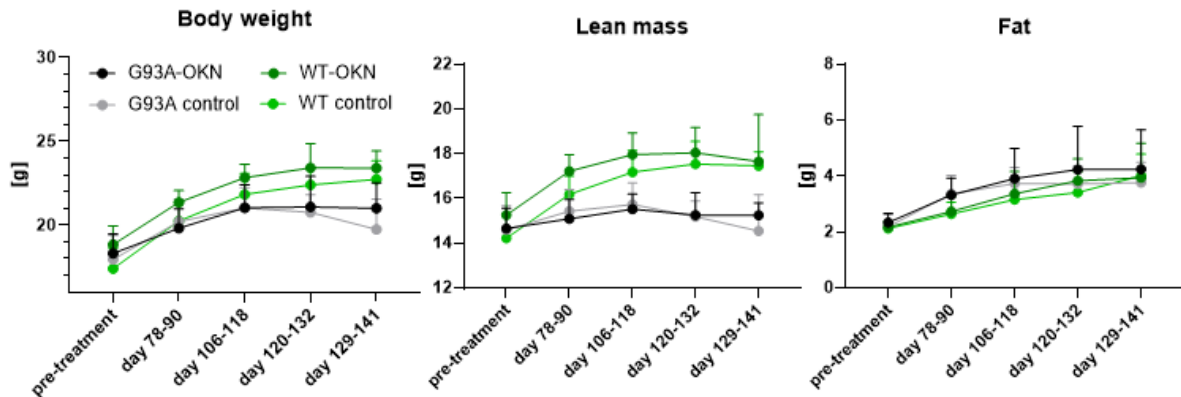


Figure 5.3 OKN-007 affects body weight and lean mass, without having effect on fat mass. **A** – there is a strong time ($p < 0.0001$) and genotype effect ($p = 0.0006$) on body weight, as well as time $\times$ genotype interaction ($p < 0.0001$). The treatment effect is weaker ($p = 0.0225$), there is also time $\times$ treatment $\times$ genotype effect ($p = 0.0121$; mixed-effects model, matching factor: time); **B** – there is a time $\times$ treatment $\times$ genotype interaction ($p = 0.0187$) in lean mass, as well as a time ($p < 0.0001$) and genotype effect ($p < 0.0001$), and time $\times$ genotype interaction ($p < 0.0001$); **C** – the fat mass is only influenced by time ($p < 0.0001$).

OKN-007 decreases neuromuscular junction (NMJ) fragmentation

Since we observed an increased MN survival in treated G93A mice, we asked the question whether the higher MN counts are reflected in preservation of muscle innervation, since MN loss and NMJ disruption are common in ALS mouse models (e.g., in *tdp-43* mutant and *zfp106* KO mice – Orr et al., 2020). OKN-007 treatment does not change NMJ (AChR) area (two-way ANOVA, treatment effect $p = 0.68$), however there is also no genotype effect ($p = 0.8167$), so there was no deficit there for OKN-007 to improve (fig. 5.4 A and B). There is a significant difference though in NMJ (AChR) fragmentation. On average 35% of NMJs in untreated G93A mice are fragmented, while the treatment with OKN-007 decreases NMJ fragmentation to 5% (Tukey's post hoc test $p = 0.0004$; fig. 5.4 C, left panel). There is also a clear difference in the distribution of the number of NMJ fragments between untreated G93A mice and the remaining groups (fig. 5.4 C, right panel). Unfortunately, due to an unusually high background in NMJ staining in untreated G93A mice (fig. 5.4 B, lower left panel), it was impossible to assess denervation, as the nerve terminals were indistinguishable from the background in that group.

OKN-007 decreases microglia activation, without affecting astrocyte proliferation

Because ALS has a strong inflammatory component that affects MN survival, we wanted to establish, if treatment with OKN-007 would affect inflammation in the spinal cord. OKN-007 treatment did not affect astrocyte proliferation (two-way ANOVA, genotype effect $p < 0.0001$), as shown by anti-GFAP immunostaining (fig. 5.5 D and E; Tukey's post hoc test for treated vs untreated G93A mice $p = 0.35$), nor the microglia proliferation measured as the number of Iba1-positive cells (two-way ANOVA, genotype effect $p < 0.0001$, treatment effect $p = 0.0547$ Tukey's post hoc between treated and untreated G93A mice $p = 0.18$, fig. 5.5 A and B). However, the treatment strongly decreased

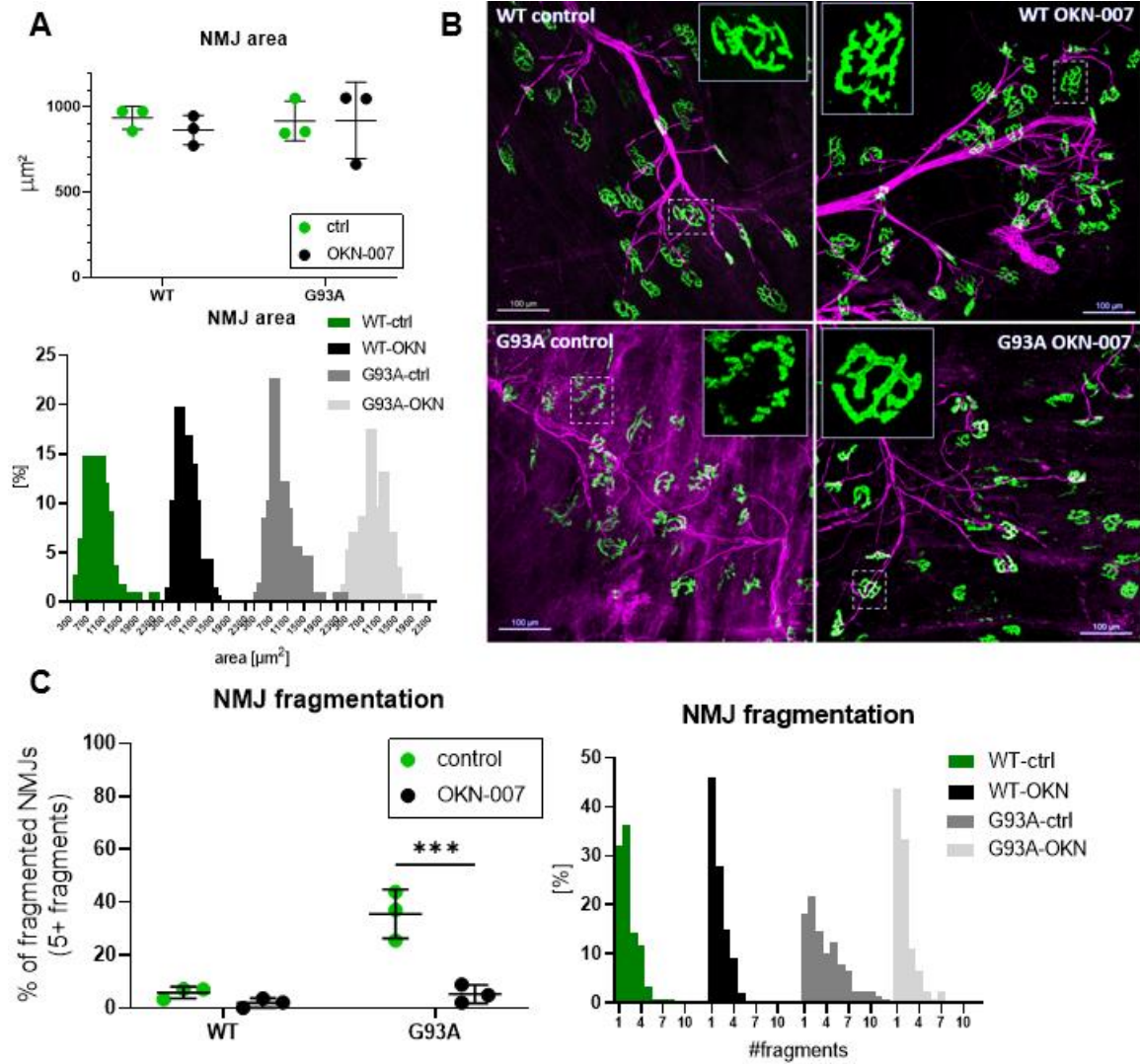


Figure 5.4 OKN-007 decreases NMJ fragmentation. **A** – top panel: OKN-007 treatment does not change NMJ (AChR) area ($n = 3$ per group, two-way ANOVA, treatment effect $p = 0.68$), there is also no genotype effect ($p = 0.8167$); bottom panel: a comparison of NMJ area distribution between both genotypes and treatments; **B** – a representative image (20x) of NMJ of treated and untreated wt and G93A mice, immunostained for SV2 and 2H3 (pre-synaptic part; in magenta), and with bungarotoxin conjugated to a fluorophore (post-synaptic part – AChR; in green), as well as a close-up of a single AChR (inset); **C** – There is a significant decrease in NMJ fragmentation in G93A treated group compared to untreated controls (Tukey's post hoc test $p = 0.0004$; two-way ANOVA, genotype effect $p = 0.0005$, treatment effect $p = 0.0004$, interaction $p = 0.0022$), there is also a significant difference between wt control and G93A control (Tukey's post hoc test $p = 0.0005$), as well as between treated wt and untreated G93A mice (Tukey's post hoc test $p = 0.0002$); the right panel shows the distribution of NMJ fragment number, with a clear right-shift in G93A control mice.

microglia activation in G93A mice, expressed both as the number of non-activated, ramified microglia per ventral horn (two-way ANOVA, genotype effect $p < 0.0001$, interaction $p = 0.0008$; Tukey's post hoc test for treated G93A vs untreated $p = 0.0212$) and as the percentage of all Iba1-positive cells per ventral horn (two-way ANOVA, genotype effect $p < 0.0001$, treatment effect $p < 0.0001$, interaction $p = 0.0012$; Tukey's post hoc for treated G93A vs untreated $p < 0.0001$; fig. 5.5 C).

The treatment with OKN-007 normalizes perfusion in the lumbar spinal cord of G93A mice, without changing blood vessel density, and affects myo-inositol levels

ALS affects spinal cord tissue structure, as the apparent diffusion coefficient (ADC) measured using diffusion-weighted MRI (DWI) was higher in the lumbar spinal cord in control G93A mice (two-way ANOVA, genotype effect $p = 0.0026$, fig. 5.6 A). The treatment decreases ADC in G93A mice, but does not reach the statistical significance (treatment effect $p = 0.0815$; Tukey's post hoc for G93A treated vs untreated $p = 0.1261$). Perfusion MRI revealed that the relative tissue blood flow (rTBF, fig. 5.6 B) in the spinal cord is significantly decreased in control G93A mice comparing to WT (two-way ANOVA, genotype effect $p < 0.0001$, Tukey's post hoc test $p < 0.0001$), suggesting a decreased vascularization or functional deficit, and the OKN-007 treatment significantly increased the blood flow in G93A mice (interaction effect $p = 0.0004$, Tukey's post hoc test G93A treated vs untreated $p = 0.0057$). Despite striking difference in perfusion rate in treated and control G93A mice, there is no significant difference in blood vessel density though, as revealed by anti-CD31 immunostaining (fig. 5.6 D-G). Moreover, the MR spectroscopy data (fig. 5.6 C) indicates that the level of myo-inositol decreases in control G93A mice, which is indicative of neuroinflammation, and the treatment increases the level of this metabolite (two-way ANOVA, treatment effect $p =$

0.0351, interaction $p = 0.0040$, Tukey's post hoc test G93A treated vs untreated $p = 0.0049$).

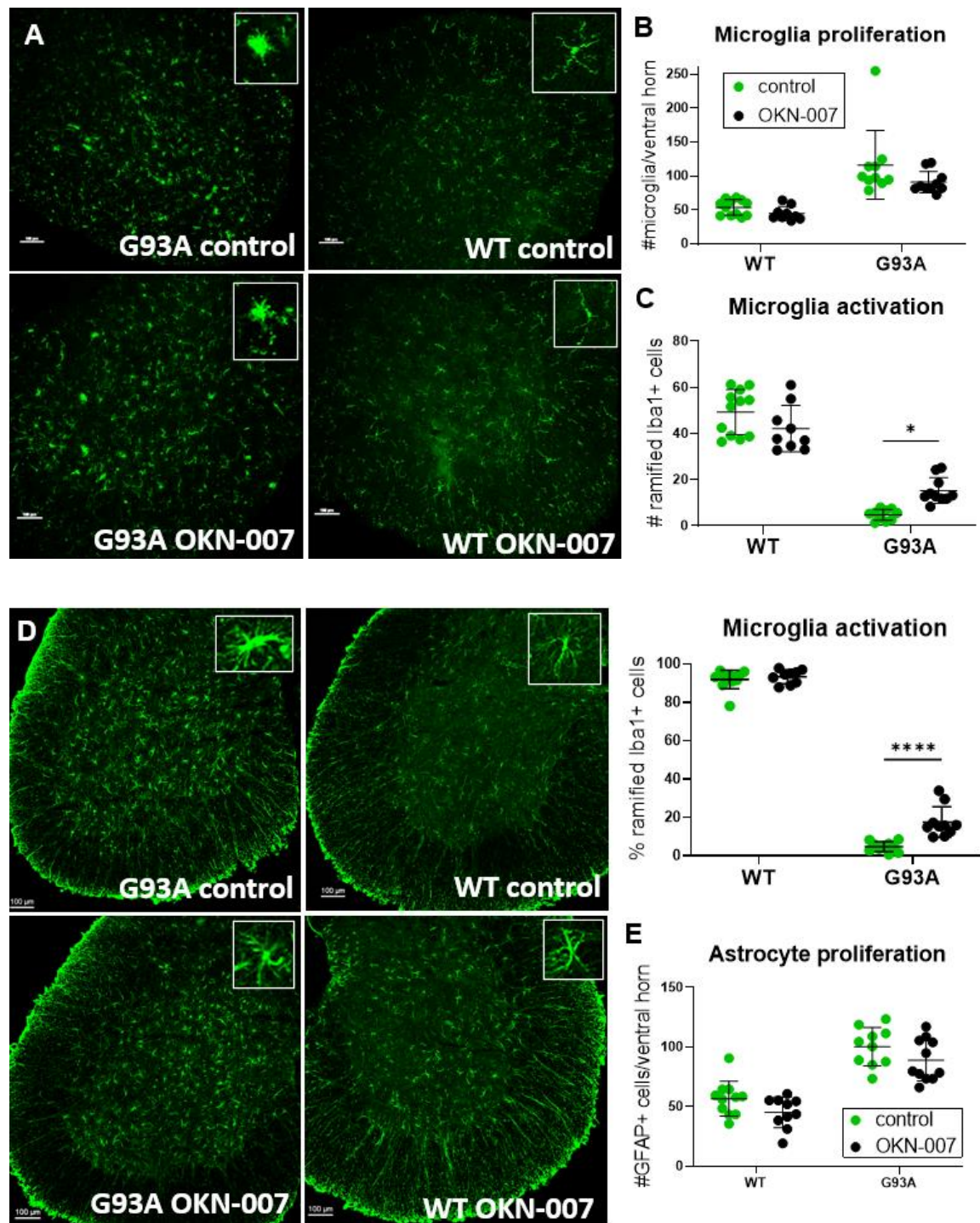


Figure 5.5 OKN-007 decreases microglia activation, but does not affect astrocyte proliferation. A – a representative image of a lumbar spinal cord ventral horn stained

for Iba1 in treated and untreated wildtype and G93A mice; inset shows a single cell morphology; **B** – quantification of Iba1-positive microglial cells per ventral horn; there is a strong genotype effect (two-way ANOVA, $p < 0.0001$) on microglia proliferation (treatment effect $p = 0.0547$, no significant difference in Tukey's post hoc between treated and untreated G93A mice); **C** – quantification of microglia activation; top panel: the number of non-activated, ramified microglia per ventral horn. There is a strong genotype effect (two-way ANOVA, $p < 0.0001$) and interaction ($p = 0.0008$). OKN-007 decreases microglia activation in G93A mice (Tukey's post hoc test for treated G93A vs untreated $p = 0.0212$; untreated wt: $n = 12$, mean $\pm$ SD = 49.27 ± 9.7 ; treated wt: $n = 9$, mean $\pm$ SD = 42.10 ± 10.05 ; untreated G93A: $n = 10$, mean $\pm$ SD = 4.811 ± 2.362 ; treated G93A: $n = 10$, mean $\pm$ SD = 15.28 ± 5.604); bottom panel: non-activated, ramified microglia count expressed as a percentage of all Iba1-positive cells per ventral horn; there is a strong genotype effect (two-way ANOVA, $p < 0.0001$), treatment effect ($p < 0.0001$), as well as interaction ($p = 0.0012$). OKN-007 decreases microglia activation in G93A mice (Tukey's post hoc test for treated G93A vs untreated $p < 0.0001$; untreated wt: $n = 12$, mean $\pm$ SD = 91.93 ± 4.8 ; treated wt: $n = 9$, mean $\pm$ SD = 93.39 ± 3.6 ; untreated G93A: $n = 10$, mean $\pm$ SD = 4.746 ± 2.6 ; treated G93A: $n = 10$, mean $\pm$ SD = 17.55 ± 7.9); **D** – a representative image of GFAP immunofluorescence in ventral horn of treated and untreated wildtype and G93A mice with an inset showing a close-up of a single cell; **E** – quantification of GFAP proliferation. There is a strong genotype effect (two-way ANOVA, genotype effect $p < 0.0001$) on GFAP-positive cell number. Although the omnibus test indicates the presence of treatment effect ($p = 0.0221$), Tukey's post hoc test shows no significant difference between treated and untreated G93A mice ($p = 0.35$).

Discussion

The properties of OKN-007 make it an ideal candidate for a disease modifying therapeutic for ALS. Similarly to Radicava, it is a free radical scavenger, it also has an anti-inflammatory effect, which is important, since ALS has a strong inflammatory component. Furthermore, neuronal loss is another feature of ALS (Ragagnin et al., 2019), and OKN-007 was shown to be neuroprotective (Clausen et al., 2008; Ewert et al., 2012; Lu et al., 2014; Du et al., 2016).

Our study shows that treatment with OKN-007 modifies the disease progression. It did not delay the disease onset, however the disease progressed more slowly in treated mice. At this point it is unknown, whether the treatment can increase lifespan as well. To answer this question, we plan to conduct in the future a lifespan study that would also show, if the treated and untreated groups diverge even more, if it comes to the disease progression at late end-stage of the disease which we didn't reach in our

current study. It also remains to be established, if the treatment could affect the disease onset as well, if it was started at an earlier timepoint (before P60 that was the starting

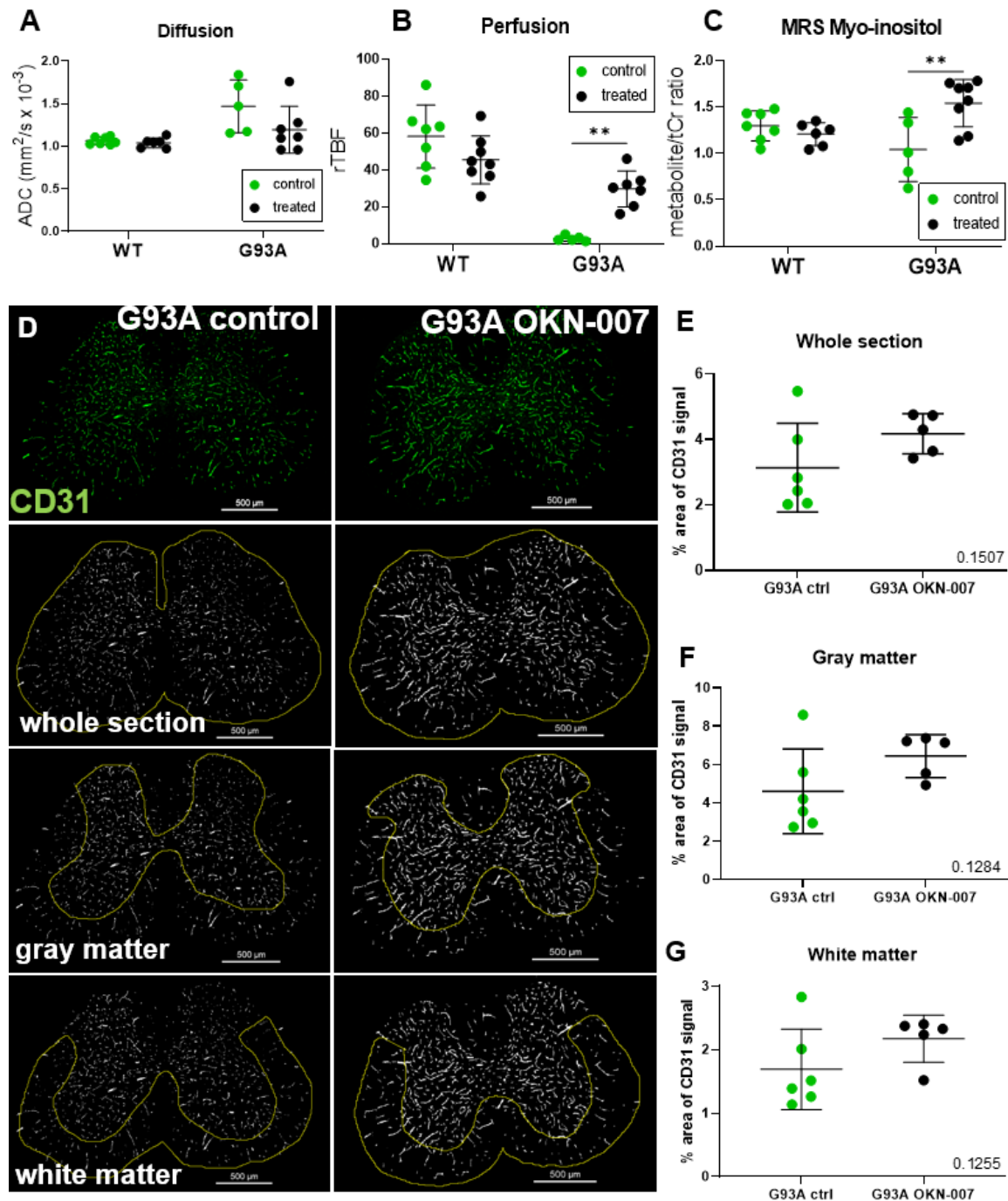


Figure 5.6 The treatment normalizes perfusion in the lumbar spinal cord, affects the levels of myo-inositol, but does not affect vasculature density. A – Diffusion (measured by MRI) is increased in G93A mice (two-way ANOVA, genotype effect $p = 0.0026$). There is no difference in diffusion after treatment with OKN-007 treatment effect $p = 0.0815$, interaction $p = 0.14$); **B –** Perfusion is significantly decreased in untreated

G93A mice ($n = 5$, mean $\pm$ SD = 2.794 ± 1.431) comparing to wt controls ($n = 7$, mean $\pm$ SD = 58.09 ± 17.09 ; two-way ANOVA, genotype effect $p < 0.0001$, Tukey's post hoc test $p < 0.0001$) and the treatment with OKN-007 increases significantly the relative tissue blood flow ($n = 7$, mean $\pm$ SD = 29.74 ± 9.716 ; interaction effect $p = 0.0004$, Tukey's post hoc test G93A treated vs untreated $p = 0.0057$). OKN-007 has no effect on wt mice perfusion ($n = 8$, mean $\pm$ SD = 45.42 ± 12.99 , Tukey's post hoc test wt treated vs untreated $p = 0.2239$); **C** – OKN-007 increases the levels of myo-inositol in G93A mice ($n = 8$, mean $\pm$ SD = 1.542 ± 0.2540 ; two-way ANOVA, treatment effect $p = 0.0351$, interaction $p = 0.0040$; Tukey's post hoc test G93A treated vs untreated $p = 0.0049$) comparing to untreated G93A mice ($n = 5$, mean $\pm$ SD = 1.043 ± 0.3451); **D** – a representative image of a cross section of lumbar spinal cord of treated and untreated G93A mice stained for CD31 to visualize blood vessels (z-stack of 11, maximum intensity projection – first row) and binary mask created by thresholding showing blood vessel density with selected region of interest: whole spinal cord section (second row), gray matter (third row), and white matter (bottom row). Gray and white matter boundary was established by overlaying the CD31 image with fluoromyelin staining (not shown); **E-G** – quantification of blood vessel density as a percentage of total area of the region of interest (whole section, gray matter and white matter, respectively). There is a trend of increase in treated G93A mice, however it doesn't reach statistical significance (two-tailed unpaired t-test for whole section and gray matter, Mann-Whitney for white matter).

point in this study). Another consideration is the dose. Our dose was selected based on the previously published studies in mice (150 mg/kg/day; e.g., Towner et al., 2019), but much higher dose of 300mg/kg was also used in published animal studies (Du et al., 2016; Ewert et al., 2017). Possibly, using a higher dose could potentially slow the disease progression further and influence also the disease onset time.

Since we saw that the treatment slows down the disease progression, we were wondering, if it's due to an increased MN survival that could contribute also to maintaining muscle mass in treated G93A mice. Indeed, OKN-007-treated G93A mice had higher MN counts at the time of sacrifice, which is consistent with previous reports of OKN-007-mediated neuroprotective effect. OKN-007 can also support proper neurotransmission, since our NMJ staining demonstrates that the treatment significantly decreases NMJ fragmentation. High background in muscles from untreated G93A mice that made assessing denervation impossible could be due to the increased muscle stress and degeneration in that model (damaged muscle fibers cause more

autofluorescent background in immunostaining). What is interesting, we did not see such a high background in OKN-007-treated G93A mice, suggesting that the compound can possibly improve muscle health. Surprisingly though, we did not see any difference in hindlimb muscle mass in treated vs untreated G93A mice. QMR revealed a trend towards body weight and lean mass preservation in treated G93A mice, however the post hoc test did not show statistical significance. Perhaps it's because we sacrificed animals right at the beginning of end-stage, at which point the difference may not be as pronounced. The possible difference in muscle mass could be more visible at the later timepoint, so a lifespan study would be more conclusive in this matter. Also, a higher dose and starting the treatment earlier could influence muscle mass more.

Consistently with the previous literature, our results demonstrate that OKN-007 affects microglia but not astrocytes, which may offer a clue as to mechanism of action. Decreased microglia activation creates less inflammatory environment, which is more conducive to MN survival. MRI revealed also a decrease in perfusion in untreated G93A mice, and the treatment significantly increased the perfusion rate in G93A mice, which can also contribute to increased MN survival, by providing more oxygen and nutrients. However, we did not see any difference in blood vessel density in anti-CD31 immunostaining, which may suggest that the improvement in perfusion rate in the treated G93A mice can be functional, not structural, at least at the time of sacrifice.

Several mechanisms of action for OKN-007 were proposed in the literature. OKN-007 is believed to work by inhibiting TGF- β 1, which also effects extracellular matrix (ECM) and downregulates matrix metalloproteinases (MMPs), especially MMP-12, as indicated by an RNA microarray analysis (Towner et al., 2019). OKN-007 is also SULF2 inhibitor, which is a therapeutic target in liver cancer treatment (Yoon et al., 2018; Lanzi et al., 2017; Floyd et al., 2011). Hedgehog/GLI1 signaling has been reported to play role as well in OKN-007 therapeutic effect in hepatocellular carcinoma (Zheng et al., 2013).

Another reported neuroprotective mechanism of action was inhibiting the release of cytochrome C from mitochondria by maintaining Akt phosphorylation in rat cerebral ischemia model (NXY-059; Yoshimoto et al., 2002). Also, OKN-007 suppresses the expression of VEGFR-2 (de Souza et al., 2015c) and basic fibroblast growth factor (bFGF; Floyd et al., 2013). However, all those reports focus mostly on OKN-007 effects in cancer, while the mechanism of action in neurodegeneration may differ. For example, in tumors OKN-007 has a pro-apoptotic effect and contributes to tumor cell death, while in neurodegeneration it is neuroprotective and decreases cell death. In the light of findings by Orr et al. (2020) that the presynaptic homeostatic plasticity (PHP) compensates for synaptic loss and allows to maintain neurotransmission despite NMJ degeneration, it's interesting to note that the level of NMJ disruption (fragmentation) that we observe in control G93A mice is close to the threshold described by the authors, at which the PHP cannot compensate for synaptic loss anymore. Since we see an effect of OKN-007 on NMJs, perhaps the treatment possibly contributes to the protective phenomenon of PHP and protects synapses (NMJs), which delays the disease progression. Further studies are needed to pinpoint precisely the mechanism behind OKN-007 neuroprotective effect in ALS mouse model, to which our study offers preliminary clues. All in all, our study shows that OKN-007 is a promising candidate for ALS treatment, as it slows down the disease progression and increases MN survival.

Materials and Methods

1. Animals

All experiments were approved by the Oklahoma Medical Research Foundation (OMRF) Institutional Animal Care and Use Committee. Female and male SOD1 G93A and wildtype littermate control mice of C57BL/6J background were used for the experiments and treated with OKN-007 in drinking water (150 mg/kg/day) from 60 days of age till the

sacrifice around day 145. The mice were housed in specific pathogen-free barrier environment facility at OMRF, maintained on a 12 h light/dark cycle, and were fed ad libitum normal chow diet. Water consumption and body weight were monitored weekly to adjust the dose as needed, and to establish how much of the drug was consumed (on average 147.6 mg/kg/day). All data shown is from both male and female mice, except for MRI, muscle weights and QMR data, which is from females.

2. Disease progression scoring

Mice were weighed weekly and the disease progression was assessed twice a week using a previously described disease progression scoring system (Evans et al., 2015), with modifications. The scale of 0-5 was used to describe the disease progression as follows: 0 – nonsymptomatic; 1 – hind limb tremor when mouse suspended from tail; 2 – hind limb tremor and difficulties separating hind limbs when suspended from tail; 3 – difficulties walking; 4 – mouse unable to walk on all four legs or dragging hind limbs; 5 – mouse unable to right itself within 30s, euthanasia required (end-stage).

3. Immunofluorescence

The lumbar part of the spinal cord was processed as previously described (Piekarz et al., 2020). Shortly, the tissue was fixed in 4% PFA overnight, washed in PBS, and incubated overnight in 30% sucrose in PBS. The tissue was incubated in OCT (Sakura Finetek, Torrance, CA) for 1h, then embedded in a mixture of OCT and TFM (EMS, Hatfield, PA) 1:1 and frozen on dry ice. The following day, the blocks were sectioned on cryotome (Leica; 18 µm thick) and mounted on Superfrost Plus Slides (VWR, Radnor, PA). For immunostaining, the frozen sections were rehydrated, washed three times for 5 min in 0.3% Triton X-100 in PBS then blocked for 2h at RT in blocking buffer (5% normal goat or donkey serum, 1% BSA and 0.3% Triton X-100 in PBS), then incubated

overnight at 4°C with primary antibodies in the blocking solution (anti-NeuN – Cell Signaling #D3S3I, rabbit mAB, 1:500; anti-Iba1 – Wako #011-27991, goat, 1:250; anti-GFAP – Abcam, ab7260, rabbit, 1:1000; anti-CD31 antibody was developed by Xia lab at OMRF, Armenian hamster, 1:100). The next day the slides were washed 3x5 min in 0.3% Triton X-100 in PBS and incubated with secondary antibodies in blocking buffer for 2h at RT (Cy3-conjugated AffiniPure goat anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #111-165-003, 1:200; Alexa Fluor 488- conjugated AffiniPure donkey anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #711-545-152, 1:200; CyTM3 AffiniPure donkey anti-goat IgG (H+L) – Jackson ImmunoResearch, #705-165-147, 1:200; CyTM3 AffiniPure goat anti-armenian hamster IgG (H+L) – Jackson ImmunoResearch, #127-165-160, 1:200). The slides were then washed again in 0.3% Triton X-100 in PBS 3x 5 min, and mounted using VectaShield mounting medium (Vestor Laboratories, Inc., Burlingame, CA). The cells were then visualized using Nikon C2 confocal microscope. Alpha-motor neuron counts were performed as described previously (Piekarz et al., 2020; total number of cells counted – 3,293). Microglia proliferation was assessed by counting all Iba1-positive cells per ventral horn (12,463 cells in total). Also, microglia activation was measured by counting the number of non-activated, ramified microglia and expressed both in absolute number and as the percentage of all Iba1-positive cells per ventral horn. Astrocytes proliferation was established by counting all GFAP-positive cells per ventral horn (14,068 cells in total). CD31 staining was analyzed with ImageJ software. Binary mask of blood vessels was created after thresholding and the area covered by the blood vessels was measured and expressed as a percentage of the total area of the region of interest (either whole spinal cord cross-section, just gray matter or just white matter). Neuromuscular junction (NMJ) staining was performed as previously described (Piekarz et al., 2020), but instead of using fresh muscle, the staining was done on frozen tissue. Mouse gastrocnemius

muscle was dissected, cleaned from connective tissue and cut in small flat pieces at the time of the sacrifice. Muscle pieces were then fixed (1h RT in 10% StüMOL in ddH₂O), washed twice in PBS (5 min each time), cryoprotected by incubating overnight in 30% sucrose in PBS, frozen on dry ice and stored at -80°C. On the day of staining, tissues were thawed, and staining was continued as previously described, starting from the permeabilization step. The person that analyzed all IF images was blinded to the identity of the samples.

4. MRI

The MRI measurements were taken as described previously (Piekarz et al., 2020) on female mice using a 7 Tesla 30 cm-bore Bruker Biospec MRI system (Bruker Biospin Corporation, Billerica, MA). The following scans were done: Diffusion-weighted imaging (DWI), ¹H Magnetic Resonance Spectroscopy (¹H-MRS), pre- and post-contrast, and enhanced T1-weighted rapid acquisition with relaxation enhancement imaging (CE T1 RARE). A sagittal multislice DWI sequence covering the thoracic and lumbar region of spinal cord was performed using echo planar imaging (EPI) with the following parameters: repetition time/echo time (TR/TE) = 3000/30.50 ms, diffusion gradient duration = 4.5 ms, diffusion gradient separation = 10.60 ms, 8 averages, 10-25 slices with thickness of 0.8-1.0 mm, matrix size 96 × 96, and a FOV of 4.5 × 4.5 cm².

5. Statistical analysis

The results were analyzed using Graphpad Prism software (versions 8 and 9). To assess the difference between genotypes and treatments, ordinary two-way ANOVA with post hoc multiple comparisons test was used. For comparisons between two groups, after confirming normality, unpaired two-tailed t-test was used.

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CHAPTER VI:

TROPONINS ARE EXPRESSED IN ALPHA-MOTOR NEURONS OF WILDTYPE MOUSE AND CAN TRANSLOCATE TO THE NUCLEUS TO POTENTIALLY ACT AS TRANSCRIPTION FACTORS

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CONTRIBUTIONS BY DOCTOR OF PHILOSOPHY CANDIDATE:

KMP created the experimental design together with HVR; KMP performed all the experiments, analyzed the data, created the figures and tables, and drafted the manuscript.

Abstract

Troponins are commonly associated with cardiac and skeletal muscle tissue, where they form a complex that plays a critical role in muscle contraction. Although there is evidence for expression of troponins in non-skeletal muscle cells, including cancer cell lines (troponin T, C and I) and chicken embryonic dorsal root ganglion (troponin C), troponins' function in this non-canonical context still remains to be understood. Here we report for the first time the presence of troponins I, T and C in α -motor neurons (α MNs) of wildtype C57Bl/6J mouse spinal cord. RNA-seq analysis revealed a 40-fold increase in troponin I mRNA levels in spinal cord from old mice compared to the young mice. The abundance of troponin I and C proteins is also elevated with age, 3.77- and 5.16-fold respectively, as measured by western blot in spinal cord lysates. Importantly, nuclear isolation and immunoblotting indicate that troponin I and T translocates to the nucleus in aged but not young α -motor neurons, suggesting that it may act as a transcription factor

that potentially initiate gene expression changes associated with spinal cord aging. In support of this idea, bioinformatic analysis described in the literature predicts that troponin I and T, but not C, can bind DNA, and additional reports confirm the ability of troponins to associate with DNA and to regulate it. Thus, we propose that troponins I and T are a promising candidate for a transcription factor that may regulate age related changes in spinal cord. To confirm the presence of troponins in the nucleus and to determine their binding sites in the genome, we performed CUT&RUN-seq. Elucidating the role troponins plays in α -motor neuron aging could allow us to identify the mechanisms that drive age-related α -motor neuron loss, and to develop a strategy aiming at preventing neuronal loss in aging and potentially ameliorate age-related loss of muscle mass and function that results from the loss of innervation.

Introduction

Our previous RNA-seq analysis showed an increase in troponin I in the spinal cord of old wildtype mice (4th most altered transcript, Supplementary Data Table 1, sheet “Molecules”, Piekarz et al., 2020). We were surprised to see a troponin transcript among the results, because traditionally troponins are not associated with the nervous system. Classically, troponins are found in skeletal and cardiac muscles where they form a complex necessary for muscle contraction (Takeda et al., 2003; see Gomes et al., 2002 for review). Troponin I is an inhibitory subunit, troponin T binds tropomyosin, while troponin C has calcium binding capability. Together they regulate muscle physiology and a disruption of the complex leads to the development of various myopathies (e.g., Dai et al., 2020; see Chang et al., 2008 for review). However, there have been reports on the non-canonical localization of troponins, for instance in tumors (troponin I; Casas-Tintó et al., 2016), as well as in healthy tissue, such as liver (Grewal and Bag, 1996; Berezowsky and Bag, 1992). Non-canonical activity of troponins is often linked to an increased cell

death or involves interactions with DNA (see Johnston et al., 2017 for review). For example, the fragments of troponin T that are not associated with myofilaments were shown to be cytotoxic induce apoptosis in C₂C₁₂ myoblasts and HEK cells (Jeong et al., 2009). Troponins have been shown to translocate to the nucleus, as Sahota et al. (2009) demonstrated the presence of troponin I in nuclei of cultured S2 cells and that troponin I SUMOylation is necessary for the translocation, while nuclear localization of troponin T was linked to apoptosis in aging skeletal muscle (Zhang et al., 2013a and b). Moreover, troponins were reported to interact with DNA, as ChIP-seq experiment by Nunez Lopez et al. (2018) revealed that troponin T binds to DNA within the p53 binding motif. Bioinformatic analysis confirms that troponins I and T contain DNA-binding domains (Johnston et al., 2017). Troponins can also regulate chromosomal stability, as troponin I and tropomyosin complex mutant flies exhibited chromosomal loss and abnormal cell division, as well as disruption of cell polarity during development in *Drosophila* (Sahota et al. 2009). Troponin T was also shown to act as a transcription factor that binds *Cacna1* promoter region in skeletal muscle, and therefore to regulate expression of voltage sensing calcium channel subunit $\alpha 1$ (Cav1.1; Zhang et al., 2016).

The presence of troponins was also reported in the central nervous system. In 1975 Fine et al. showed that troponin C is present in the brain. Another report came out in 1983 (Roisen et al.), demonstrating the presence of troponin C in cultured dorsal root ganglion neurons of chicken and rat. Also Berezowsky and Bag (1992) reported the presence of slow troponin C mRNA in the brain, while Lowe et al. (2007) described ketamine-dependent expression of troponin T in hippocampus, amygdala, hypothalamus, cortex and cerebellum of mouse brain. Furthermore, Lyckman et al. (2008) demonstrated that cardiac troponin C is present in layer 4 pyramidal neurons of visual cortex.

Here we followed up on our surprising discovery that troponin I transcript is expressed in the spinal cord of wildtype mouse and that it is upregulated with age, and we report the presence of all three troponins in the spinal cord. The troponins localize to alpha-motor neurons, and the abundance of troponin I and C increased with age. We also demonstrate the troponins I and T ability to translocate to the nucleus and we perform CUT&RUN-seq analysis to determine their binding sites. We propose that troponins can act as transcription factors relevant to the α -MN aging.

Results

Troponins are expressed in the spinal cord and the abundance of troponin I and C, but not T, changes with age

In our previously published RNA-seq data (Piekarz et al., 2020, Supplementary Data Table 1, sheet: Molecules) we saw that troponin I fast skeletal type is expressed in the spinal cord of wildtype mice and that it is upregulated with age (26-27 mo vs 9 mo, fig. 6.1 A). To confirm the surprising presence in the spinal cord of a protein that is classically associated with muscle, we did immunoblotting, using whole spinal cord lysates, which confirmed that all three troponins were present in the spinal cord (fig. 6.1 B-E). Moreover, the abundance of troponin I and C increases with age in the spinal cord (troponin I: two-tailed unpaired t-test $p = 0.0035$, 3.77-fold increase; troponin C: $p = 0.0019$, 5.16-fold increase), while the abundance of troponin T does not change with age in the spinal cord (t-test $p = 0.3933$).

Troponins colocalize to alpha-motor neurons

To further confirm the presence of the troponins in the spinal cord and to verify their localization, we performed immunostaining on the lumbar spinal cord frozen sections from young and old mice, which revealed that troponins colocalized to the α MNs (fig 6.2). Similarly to what we saw in immunoblotting, troponins C and I were more abundant

in αMNs of the old mice, while the troponin T signal intensity was similar in both young and old mice.

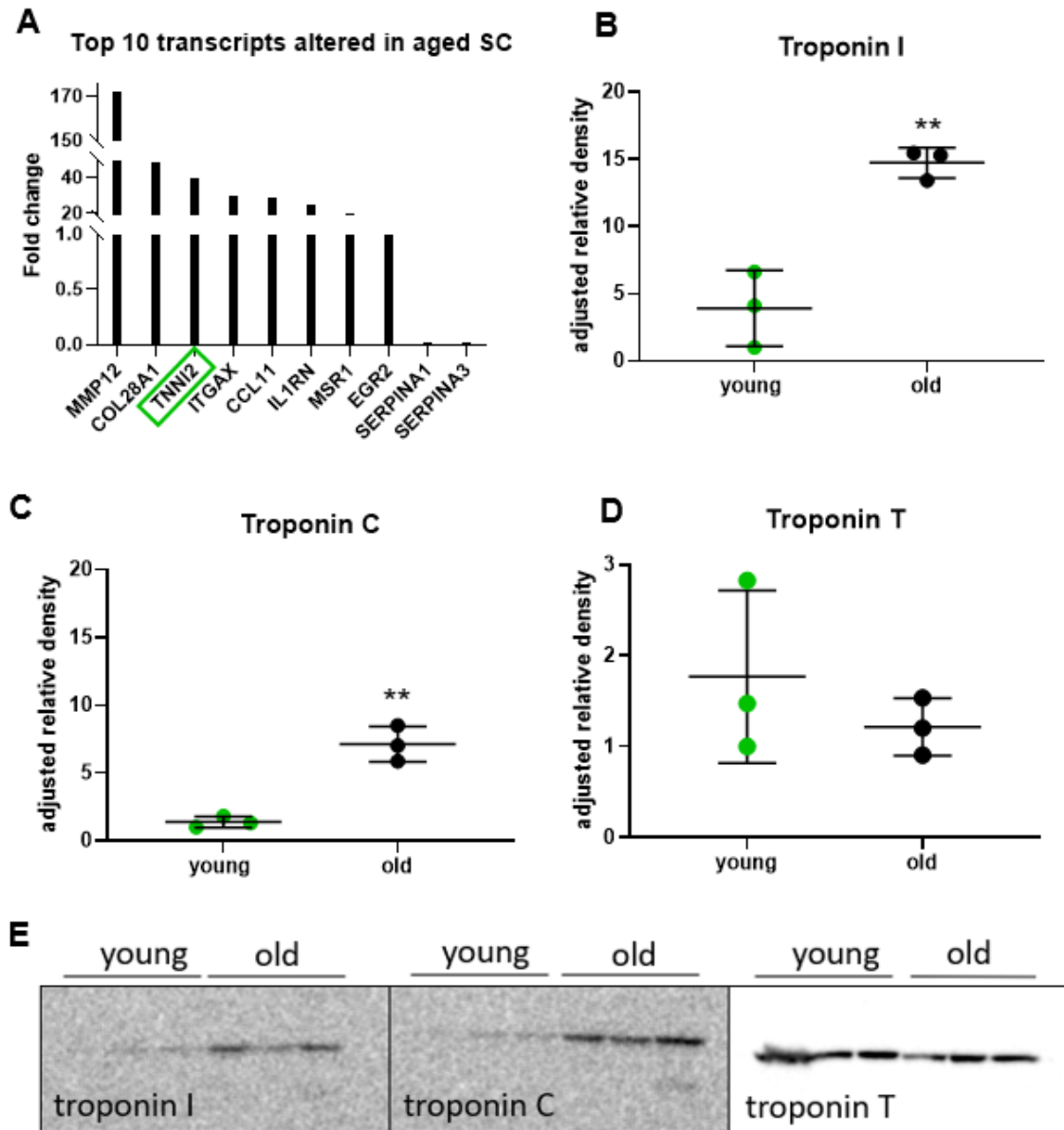


Figure 6.1 Troponins are expressed in the spinal cord. **A** – our previously published RNA-seq data (Piekarz et al., 2020, Supplementary Data Table 1, sheet: Molecules) reveals that troponin I fast skeletal type is expressed in spinal cord of wildtype mice and it's upregulated with age (26-27 mo vs 9 mo); **B** – the abundance of troponin I increases with age in the spinal cord (two-tailed unpaired t-test $p = 0.0035$; young: 6-7 mo, mean $\pm$ SD = 3.91 ± 2.815 , old: 27-28 mo, mean $\pm$ SD = 14.74 ± 1.134); **C** – the abundance of troponin C is higher in old mice spinal cord (two-tailed unpaired t-test $p = 0.0019$; old: 27-28 mo, mean $\pm$ SD = 7.133 ± 1.314) comparing to the young mice (young: 6-7 mo, mean $\pm$ SD = 1.383 ± 0.41); **D** – the abundance of troponin T does not change with age

in the spinal cord (two-tailed unpaired t-test $p = 0.3933$; young: 3 mo, mean $\pm$ SD = 1.768 ± 0.949 , old: 25 mo, mean $\pm$ SD = 1.216 ± 0.316); **E** – western blot probed for troponins I, C and T for young and old wildtype mice (whole spinal cord lysate).

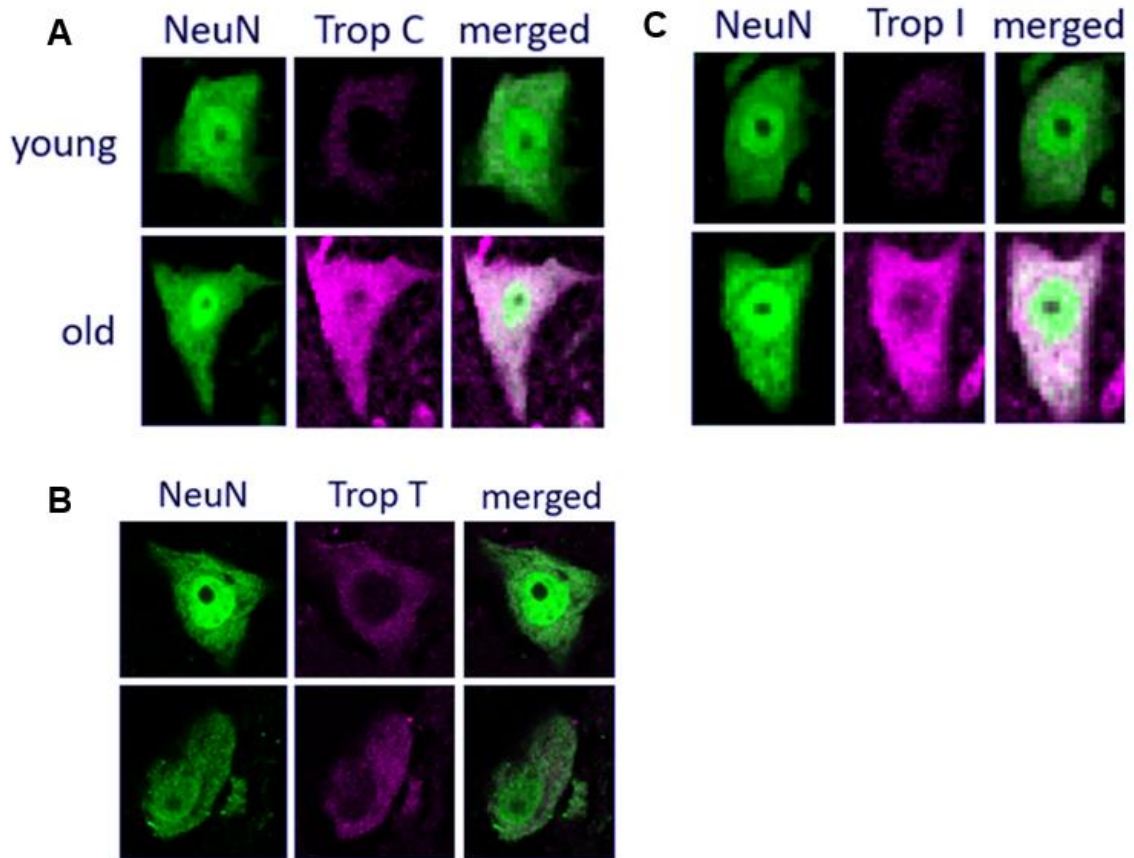


Figure 6.2 Troponins colocalize to alpha-motor neurons. **A** – the immunofluorescence reveals that troponin C (magenta) is expressed in αMNs (green) in spinal cord of young (3-7 mo) and old (29 mo) wildtype mice; **B** – troponin T (magenta) is expressed in αMNs (green) in spinal cord of young (3-7 mo, top panel) and old (29 mo, bottom panel) wildtype mice; **C** – troponin I (magenta) is also expressed in αMNs (green) in spinal cord of young (3-7 mo, top) and old (29 mo, bottom) wildtype mice.

Troponin T and I translocate to the nucleus and are present in motor neurons in models of accelerated MN loss

Since there were reports of non-canonical presence of troponins (Fine et al., 1975, Roisen et al., 1983) and their translocation to the nucleus with age, which was linked to skeletal muscle aging (Zhang et al., 2013), we were interested, if also in αMNs troponins

can translocate to the nucleus with age, where they can potentially act as transcription factors that might drive α MN aging. To answer that question, we isolated α MNs from the spinal cord of young (6mo) and old (26 mo) wildtype mice and subjected them to nuclear isolation and immunoblotting. The results indicate that troponins I and T, but not C, are present in the nucleus of old α MNs (fig 6.3 A). If nuclear translocation of troponins in α MNs favors neuronal death, we hypothesized that the troponins will also be detectable in α MN nuclei of SOD1 G93A (ALS mouse model; Gurney et al., 1994) and SOD1 KO mice (considered a model of accelerated aging; Muller et al., 2006), which both exhibit accelerated α MN loss. Indeed, we saw that troponin T, and to a lesser degree, troponin I are present in the nuclei of α MNs in both G93A (5 mo) and SOD1 KO (4.7 mo) mice. The presence of troponin T in those models was further confirmed by immunostaining (fig. 6.3 B).

Troponins I and T are present in the nucleus and bind DNA in both young and old wildtype mice

The nuclear translocation of troponin T and I in α MNs suggests that those proteins could perhaps act as transcription factors related to α MN aging. To ascertain that, we isolated α MNs from spinal cords of young (4-8 mo) and old (25-29 mo) wildtype mice and performed CUT&RUN-seq (fig. 6.4 A). CUT&RUN-seq is a relatively new technique, similar to ChIP-seq, however it is optimized for low input (Skene and Henikoff, 2017; Meers et al., 2019), which makes it a preferable approach in our case. Validation of a successful α MN isolation was done by immunoblotting, which showed enrichment of ChAT and depletion of GFAP in the motor neuron fraction (MN) comparing to the whole spinal cord lysate (SC) control, as well as by culturing α MNs up to 5 DIV and staining the cultures for ChAT (fig. 6.4 B). Surprisingly, CUT&RUN-seq revealed that besides being present in the nucleus of old mice, both troponins I and T are also present in the nucleus

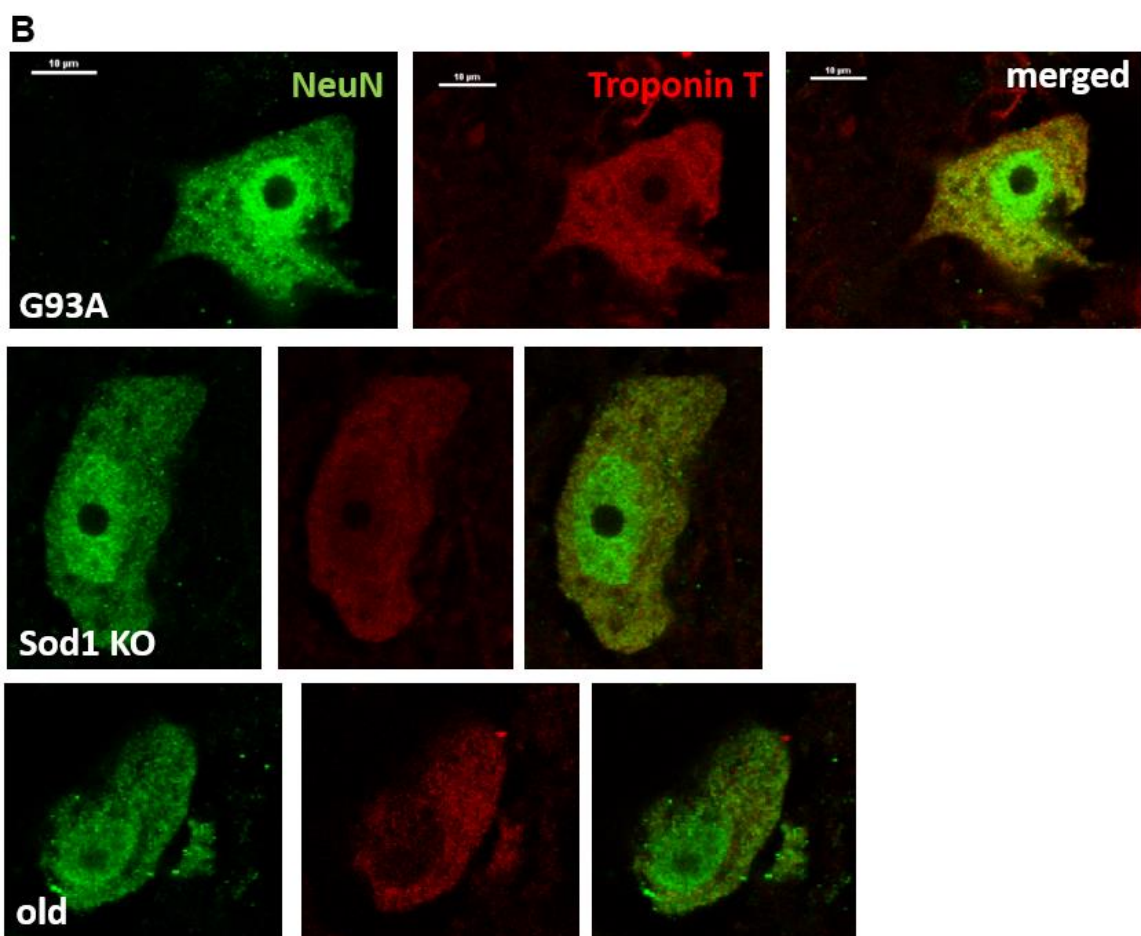
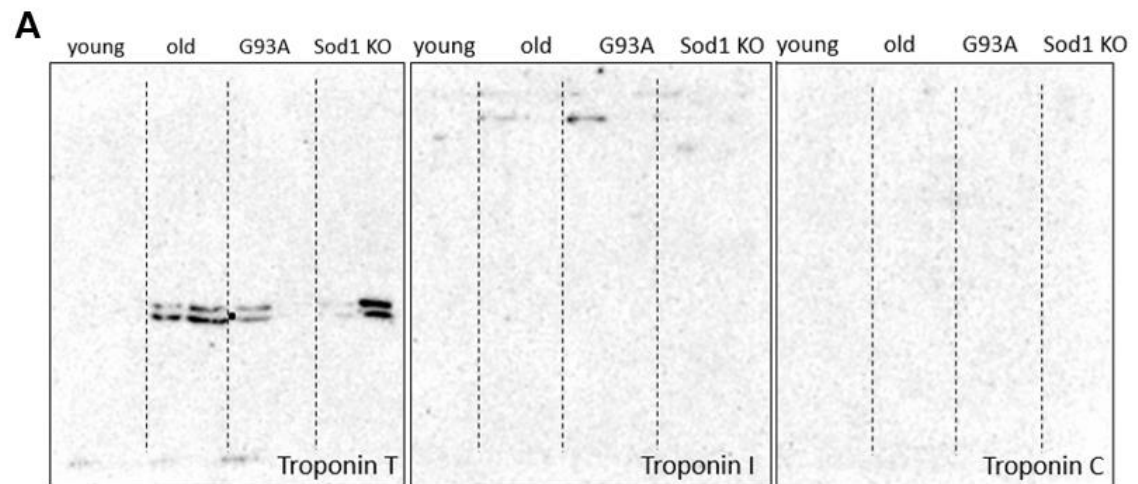


Figure 6.3 Troponin T and I translocate to the nucleus and are present in motor neurons in models of accelerated MN loss. A – the immunoblotting following the nuclear isolation from the isolated α MNs of young (6 mo) and old (26 mo) wildtype mice, as well as from SOD1 G93A (5mo) and SOD1 KO (4.5 mo) mice, probed for all three troponins; **B** – immunofluorescent staining for troponin T (red) and co-staining with NeuN

(green) of SOD1 G93A (top), SOD1 KO (middle), and old wildtype mouse for reference (29 mo, bottom panel).

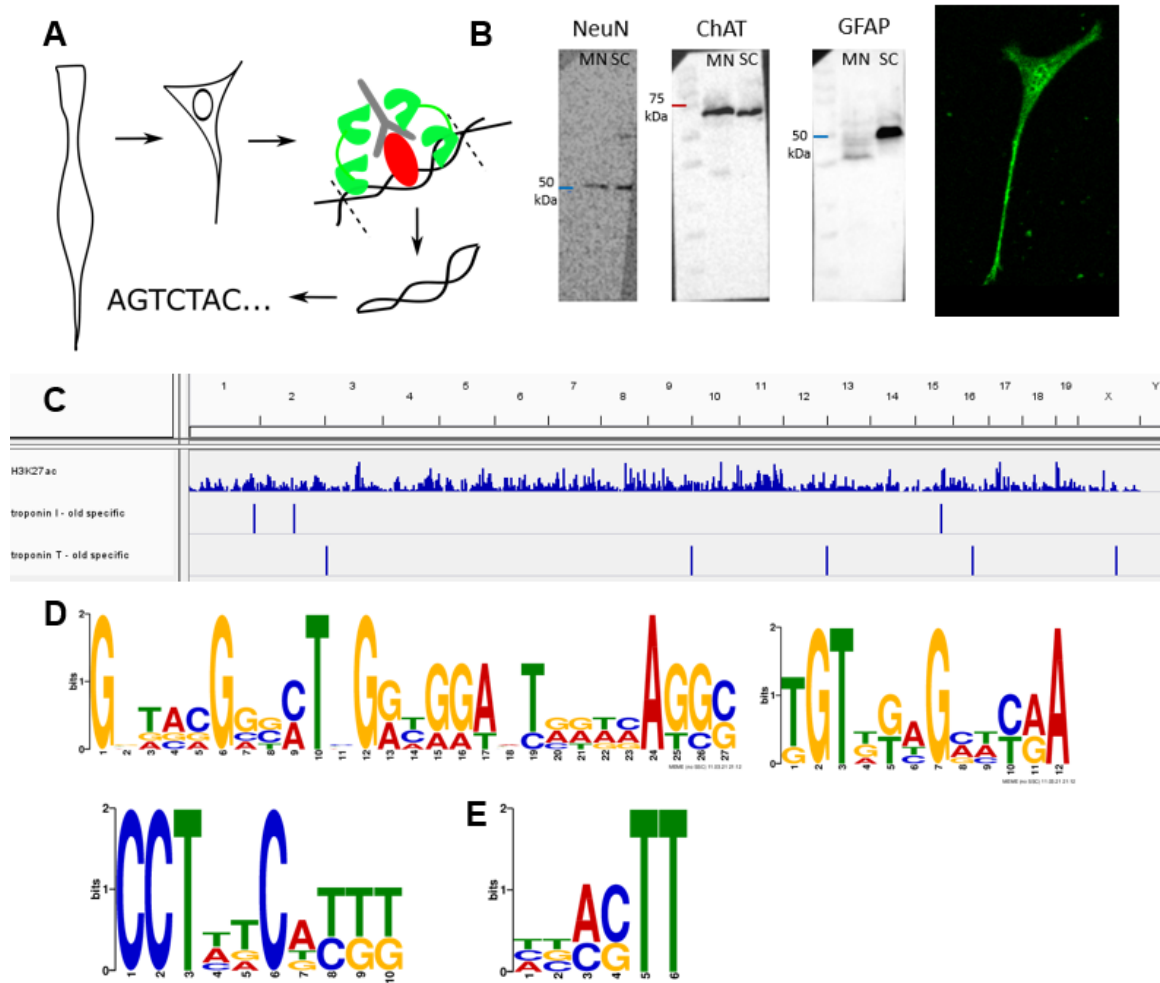


Figure 6.4 Troponin T and I binding sites in old wildtype mice. **A** – a schematic of the CUT&RUN-seq experiment. Shortly, the α MNs were isolated from the spinal cords of young and old wildtype mice and served as an input for CUT&RUN-seq. During CUT&RUN procedure, the cells were bound to concavilin-A-coated magnetic beads, permeabilized with digitonin and incubated with anti-troponin I or T antibodies (in the schematics the red oval represents troponins, anti-troponin antibody is shown in grey). Next, protein A/G fused to MNase (green) was added and the DNA cleavage was induced by adding Ca^{2+} , then the protein A/G-antibody-cleaved DNA complex is released to the supernatant, from which the DNA is isolated and then sequenced; **B** – validation of α MN isolation from the spinal cord of wildtype mice by immunoblotting (3 mo; MN – isolated motor neuron fraction; SC – whole spinal cord lysate control; probed for NeuN, ChAT and GFAP), and by culturing the α MNs and staining them for ChAT; **C** – the genomic location of peaks specific for old mice visualized using IGV genome browser (top panel – H3K27ac control, middle – old-specific peaks for troponin T, bottom – old-specific peaks for troponin I; **D** – DNA motifs shared by the sequences to which troponin T binds (old-specific peaks); **E** – shared DNA motif for troponin I binding sites (old-specific peaks).

and bind DNA in young mice (full list of peaks can be found in Appendix E Supplemental Tables 1 and 2). There is a partial overlap between binding sites for troponin I and troponin T, probably due to their structural similarity. In case of both troponins, there is also a partial overlap between binding sites in young and old mice, however there are several peaks that are only present in old mice (three peaks for troponin I and five peaks for troponin T; fig. 6.4 C and Table 6.1), which shows that troponins' binding to DNA changes with age. Five out of eight peaks in total are located in intergenic regions, one in gene exon and two in an intron (Table 6.1). Three common motifs were identified that are shared by all old mice-specific troponin T binding sites (fig. 6.4 D), from which KGTDKHHG and CCTHDC exhibit the highest conservation. Comparing those motifs to the binding sequences of transcription factors in HOCOMOCO database shows that the first motif shows resemblance to forkhead box (FOX) family of transcription factors binding site, especially foxd3 binding sequence TGTTTn (Kulakovskiy et al., 2018), while CCT motif appears often in the binding sequences of zinc finger proteins, such as Zfx, Zic2, Zic3, Zkscan1 etc. In case of troponin I, one motif was identified (fig. 6.4 E), which shows high conservation of TT nucleotides in position 5 and 6.

Discussion

To our surprise, our previous RNA-seq analysis revealed the presence of troponin I fast skeletal type in the spinal cord and an increase in its abundance with age (~40-fold increase; Piekarz et al., 2020). To follow up, we confirmed the presence of all three troponins in the spinal cord of wildtype mice. Furthermore, our immunoblots demonstrated that the abundance of both troponin I and C, but not T, increases with age. Intrigued by the discovery, we further confirmed the presence of troponins in the spinal cord by performing immunostaining. Immunofluorescence showed that the troponins colocalized to α MNs. The so-called non-canonical localization of troponins was

Peaks specific for old (old-young)					
chromosome	start	stop	gene	upstream of	downstream of location
Troponin T					
chr3	5860425	5860754		1700008P02Rik	Pex2 intergenic
chr9	123371012	123371252	Lars2		exon
chr13	3372266	3372484		2810429I04Rik	Speer6-ps1 intergenic
chr16	57391513	57391781	Cmss1, Filip1l		intron
chrX	109484401	109484742	Apool	Satl1, Zfp711	intergenic
Troponin I					
chr1	183299200	183299209	Aida		intron
chr2	98666409	98666417		Gm33780	Lrrc4c intergenic
chr15	75086410	75086419		Ly6c2	Ly6c1 intergenic

Table 6.1. Old mice-specific CUT&RUN peaks for troponins T and I.

suggested before in the literature, as for example Fine et al. (1975) demonstrated the presence of troponin C in the brain, while Roisen et al. (1983) reported the presence of troponin C in cultured neurons of chicken's and rat's dorsal root ganglion, however, to our best knowledge, we report here for the first time the presence of all three troponins in α MNs of wildtype mouse. Roisen et al. (1983) hypothesized that calcium sensing troponin C might play role in the intra-axonal movement, but the role of troponins in neuronal cells remains mostly unknown.

Since troponin T was shown to translocate to the nucleus and play role in skeletal muscle aging (Zhang et al., 2013a), and troponin T fragments can induce apoptosis (Jeong et al., 2009, Zhang et al., 2013b), we wanted to see, if troponins can translocate to the nucleus in α MNs. Our nuclear isolation and immunoblotting shows that troponin I and T, but not C, translocate to the nucleus in α MNs of old wildtype mice, as well as in the mouse models that exhibit accelerated α MNs loss, such as SOD1 G93A and SOD1 KO mouse models. Those results are in agreement with previously published bioinformatic analysis that predicts that both troponins I and T, but not C, can bind DNA

(Johnston et al., 2017; Table 2). Even though the overall troponin T abundance in the whole spinal cord lysate does not change with age, it is the troponin that seems to translocate the most to the nucleus of α MNs in aging and disease. We further confirmed the presence of troponin T in α MNs in SOD1 G93A and SOD1 KO mouse models by immunofluorescent staining. As shown by Jeong et al. (2009), in physiological conditions, fragments of troponin T that are not associated with myofilaments are efficiently degraded, however when troponin T was overexpressed, it exceeded the degradation capacity and became cytotoxic. Perhaps a similar process takes place in the α MNs in aging, and in this way the age-related increase in troponin C and I abundance leads to cytotoxicity. Alternatively, the degradation process may become less efficient with age, causing troponin T to become more toxic, even though there is no age-related increase in troponin T levels in the spinal cord.

We then performed CUT&RUN-seq to establish where in the genome troponins bind. Despite the fact that nuclear isolation and western blotting revealed the presence of troponins in the nuclei of old, but not young wildtype mice, CUT&RUN-seq experiment showed that both troponins I and T bind to DNA in young mice as well. The reason behind this discrepancy may be the difference in sensitivity of both methods. The nuclear abundance of troponins as transcription factors is very low and might be then undetectable by western blotting, which is not the case when performing CUT&RUN-seq. There is age-dependent difference in troponin binding and the binding sites are mostly found in the intergenic regions. Among the genes located in the vicinity of the binding sites (here defined as the closest genes upstream and downstream of the binding site), some of them seem to be of a special interest, given their links to neurons or aging process. For instance, peroxisomal biogenesis factor 2 (Pex2) gene was linked to a decrease in lifespan in yeast (GenAge database; Tacutu et al., 2018) and to Zellweger syndrome that affects neuronal migration (Faust et al., 2001) and myelination

(Faust and Hatten, 1997), while a mutation in *Lars2* was associated with increased lifespan in *C.elegans* (GenAge database; Tacutu et al., 2018). Filip1l overexpression was linked to an increased apoptosis (Kwon et al., 2008) and Zfp711 is a transcriptional regulator necessary for proper brain development (GeneCards – the human gene database, Stelzer et al., 2016). Another gene that is relevant the neuronal function is *Lrrc4c*, which encodes a protein important for maintaining proper synaptic structure (Choi et al., 2019). Lastly, both *Ly6c1* and *Ly6c1* have acetylcholine receptor inhibitor activity (BioGPS database, Wu et al., 2009). The common motif analysis revealed that the old mice-specific troponin T binding sites are similar to the binding site of another transcription factor, *foxd3*. *Foxd3* itself was shown to have an age-related differential binding, which influences Wnt pathway in hindlimb skeletal muscle of wildtype mouse (Ryu et al., 2011). Perhaps, the reason behind the age-dependent differential binding is the competition for binding sites with another transcription factor that starts binding additional sites with age, as for example troponin T does. We hypothesize then that besides potentially influencing the expression of the genes located in the vicinity of the binding sites, or the genes located elsewhere in the genome but that are in the physical contact with the binding sites due to the three-dimensional chromatin structure, the age-dependent differential binding of troponins can also influence expression of other genes by competing for the binding sites with other transcription factors that bind sites of a similar sequence, and therefore inhibit their activity.

In conclusion, we demonstrate that troponins I, T and C are expressed in α MNs and that the abundance of troponin I and C increases with age. Both troponin I and T can translocate to the nucleus of α MNs in wildtype mice, as well as in the mouse models that exhibit accelerated α MNs loss. However, as revealed by CUT&RUN-seq, both troponin I and T are present in the nucleus of old, as well as young wildtype mice, and

their binding sites change with age. Further studies are needed to elucidate the effect that the troponins' age-related differential binding to DNA has on the gene transcription.

Materials and Methods

1. Animals

Wildtype C57BL/6 mice were bred and maintained at the Oklahoma Medical Research Foundation (OMRF) animal facility. Mice were fed a regular diet at libitum and had unrestricted access to water. Animal protocols were approved by OMRF's IACUC.

2. Western blotting

Western blots were done as previously described (Piekarz et al., 2020) and the following antibodies were used: anti-troponin I-FS (C-11), mouse monoclonal IgG1, Santa Cruz, sc-377451 1:200; anti-troponin C-FS (E-7), mouse monoclonal IgG1, Santa Cruz, sc-48347, 1:200; anti-troponin T-FS (H-8), mouse monoclonal IgG_{2a} κ, Santa Cruz, sc-166663, 1:200; goat anti-mouse IgG-HRP, Santa Cruz, sc-2005, 1:5,000. Shortly, isolated spinal cords were lysed in RIPA buffer, protein concentration was measured using Bradford assay, and 40 µg of total protein were loaded onto 10% acrylamide/bisacrylamide gel, ran at 120V until the proper separation was achieved. Membrane transfer was done overnight at 20V and 4°C, then the membranes were stained with Ponceau stain and cut accordingly, so they can be probed with multiple primary antibodies. Next, the membranes were washed for 30 min in TBS-T, blocked 1h RT in 5% milk in TBS-T and incubated with primaries overnight at 4°C. The following day the blots were washed 3x 10 min in TBS-T, probed with the secondary antibodies for 1h at RT, and washed again in TBS-T (3x 10 min), then incubated in ECL and imaged using G-box. The blots were quantified in ImageJ software and normalized to Ponceau.

3. Immunofluorescence

Spinal cord immunofluorescence was done as previously described on frozen cross-sections from the lumbar spinal cord (Piekarz et al., 2020). The following antibodies were used: anti-NeuN – Cell Signaling #D3S3I, rabbit mAB, 1:500; anti-troponin I-FS (C-11), mouse monoclonal IgG1, Santa Cruz, sc-377451 1:50; anti-troponin C-FS (E-7), mouse monoclonal IgG1, Santa Cruz, sc-48347, 1:50; anti-troponin T-FS (H-8), mouse monoclonal IgG_{2a} κ, Santa Cruz, sc-166663, 1:50; Alexa Fluor 488- conjugated AffiniPure donkey anti-rabbit IgG (H+L) – Jackson ImmunoResearch, #711-545-152, 1:200; CyTM3 AffiniPure donkey anti-mouse IgG (H+L) – Jackson ImmunoResearch, #715-165-150, 1:200. The slides were imaged on the Nikon C2 confocal microscope and the images were analyzed in ImageJ software.

4. Motor neuron isolation from adult mouse spinal cord

Motor neuron isolation was performed according to the protocol published by Beaudet et al. (2015) with modifications. After dissection, the isolated spinal cords were placed in 1 ml of ice-cold DMEM with 10% FBS in a 12-well plate, then the samples were transferred to another well with 1 ml of pre-warmed papain (2 mg/ml in 60% HBSS, 40% DMEM) and incubated for 10 min at RT. During the incubation, the spinal cords were cut into small pieces with a scalpel. The plate was then placed in an incubator for 30 min at 37°C. Every 5 min the plate was taken out and shaken for 10 s. During the last 5 min of the incubation, 3 mls of ice-cold DMEM (without FBS from now on) were passed through the 100-micron mesh strainer placed on 50-ml falcon kept on ice. Then 3 mls of DMEM were added to papain to stop digestion and the samples were pipetted up and down six times in a glass serological pipette to facilitate cell dislodging. Medium was collected and transferred onto the strainer, and 3 more mls of DMEM medium were added into the well, to the tissue pieces. The samples were again pipetted up and down and the medium was transferred to the strainer. Another 3 mls of DMEM were added to the

remaining spinal cord pieces, and all the process was repeated 3 times in total. After the last round of pipetting up and down, whole sample (both medium and remaining small pieces of the spinal cord) was transferred to the strainer. The empty wells were washed with another 3 mls of DMEM that was then transferred to the strainer as well. Any remaining tissue was ground in the strainer with the glass pipette and the strainer was washed with 3 mls of DMEM. The cells were then mixed in the falcon by inversion and centrifugated for 20 min 280 x g at 4°C. Resulting supernatant containing glia was only processed during optimization and validation steps, and wasn't used in the MN isolation experiment. The pellet containing motor neurons was resuspended in first 1 ml of DMEM by gentle pipetting, then another 5 mls of DMEM were added. The 6 mls of MN suspension were layered onto Nycoprep (9.84 mls of 1.077g/ml Nycoprep and 0.16 ml of DMEM) and spun 900 x g for 20 min at 4°C, without brake. MNs were then aspirated from the interface and transferred to a new 50-ml falcon, which was topped up with DMEM and centrifugated for 10 min at 425 x g, 4°C. The resulting pellet was then used in further experiments.

Successful motor neuron isolation was confirmed by immunoblotting (as described above, the pellet was resuspended in 30 µl of RIPA buffer with protease inhibitor 1:100; primary antibodies used: anti-NeuN – Cell Signaling #D3S3I, rabbit mAB, 1:1000; anti-GFAP – Abcam, ab7260, rabbit polyclonal, 1:10,000; anti-ChAT – Abcam, ab178850, rabbit monoclonal, 1:1000), and by culturing up to 5 DIV. For culturing, the cells were plated in 250 µl of motor neuron medium (Neurobasal A medium, 2% horse serum, 2% B-27 minus antioxidants, 0.5 mM Glutamax, 1% penicillin/streptomycin, 10 ng/ml of BDNF, GDNF, NT-3, CNTF from Peprotech) on round coverslips coated with poly-D-lysine (100 µg/ml) placed in 12-well plate, left to attach overnight, then 1 ml of motor neuron medium was added the next day. After few days, cells were stained in the following manner: the cells were fixed on coverslips in 4% PFA for 30 min, then rinsed

3x 5 min in HBSS at RT. The excess of HBSS was removed by gently placing the edge of the coverslip on a kimwipe, then the coverslips were blocked for 1h at RT by placing them cell-side down onto a droplet of blocking solution (8 μ l for a 12-mm coverslip, 20 μ l for 20-mm one) on parafilm-covered culture plate lid. Then the cells were incubated in a similar manner with the primary antibody (anti-ChAT – Abcam, ab178850, rabbit monoclonal, 1:2000) overnight at 4°C. Next day, the cells were rinsed 3x 10 min in HBSS, incubated with a secondary antibody for 75 min at RT, rinsed again 3x 10 min in HBSS, and mounted on a slide on a droplet of 5 μ l of mounting medium.

5. Nuclear isolation

Isolated motor neurons were used for the nuclear isolation, which was done as described by Nabbi and Riabowol (2015) using the sucrose method. The samples were then used for immunoblotting, as described above, and probed for troponins.

6. CUT&RUN-seq

Isolated motor neurons (isolation procedure described above) from the spinal cords of young and old wildtype mice served as the input for the CUT&RUN-seq. The motor neuron pellet was processed accordingly to the protocol posted on protocols.io, version 3 (CUT&RUN: Targeted in situ genome-wide profiling with high efficiency for low cell numbers V.3 by Janssens and Henikoff, <https://www.protocols.io/view/cut-amp-run-targeted-in-situ-genome-wide-profiling-zcpf2vn>; Skene and Henikoff, 2017; Meers et al., 2019) with some modifications. The cells were resuspended in 1.5 ml of wash buffer (RT) and centrifugated for 3 min at 600 x g at RT. The washing was repeated two more times, then the cells were resuspended in 1 ml of wash buffer, and pre-activated concavalin-A-coated magnetic beads slurry was added (10 μ l/sample; kind gift of Sansam lab), while gently vortexing. The samples were rotated for 10 min at RT, and after a

shortspin, the samples were placed in a magnetic stand for 1 min and the liquid was discarded. Next, the beads were gently resuspended in 150 µl of antibody buffer, then the antibodies were added (anti-troponin I-FS (C-11), mouse monoclonal IgG1, Santa Cruz, sc-377451 1:50 anti-troponin T-FS (H-8), mouse monoclonal IgG_{2a} κ, Santa Cruz, sc-166663, 1:50; during validation and optimization anti-H3K27ac antibody was used; CST #8173, rabbit mAB, 1:100). The samples were incubated overnight at 4°C, without agitation (placing the samples in the rotator at this step caused in our case the premature cell lysis and beads clumping which led to a poor yield). The next day the liquid was removed using the magnetic stand, then the beads were washed twice in dig-wash buffer (150 µl of 5% digitonin in 30 ml of wash buffer) and resuspended in pAG-MNase (CUTANA by EpiCypher accordingly to manufacturer's protocol: 2.5 µl of the enzyme in 50 µl of dig-wash buffer), and incubated for 1h at 4°C on a rotator. Next, chromatin digestion and release step was done using option 1 (standard CUT&RUN), and the only change was digesting for 1h instead of 30 min (at the step 34 of the original protocol). Then we proceeded with Phenol/Chloroform Extraction (step 57). At the last step (#67), the air-dried DNA pellet was resuspended in 5 µl of 1 mM Tris-HCL pH 8 with 0.1 mM EDTA. The samples were then processed by OMRF Clinical Genomics Center. The samples were PCR-amplified and run on TapeStation for quality control purposes, then the libraries were prepared and the samples were sequenced on NovaSeq (Illumina, 20M reads).

The results were analyzed in Galaxy hub (usegalaxy.org, Afgan et al., 2018) and the following tools were used: Map with BWA - map short reads (Galaxy Version 0.7.17.4) for mapping reads to mouse reference genome; Filter SAM or BAM, output SAM or BAM files on FLAG MAPQ RG LN or by region (Galaxy Version 1.8+galaxy1) for postprocessing and filtering mapped reads by quality (the reads with score below q20 were discarded); multiBamSummary (Galaxy Version 3.3.2.0.0), plotCorrelation

(Spearman correlation, Galaxy Version 3.3.2.0.0), plotFingerprint (Galaxy Version 3.3.2.0.0) for quality control and assessing the enrichment; MACS2 predictd (Galaxy Version 2.1.1.20160309.1) and MACS2 callpeak (Galaxy Version 2.1.1.20160309.6) for peak calling; Intersect the intervals of two datasets (Galaxy Version 1.0.0) for generating high confidence peak set for each animal and the list of common peaks in all young and old mice; Subtract the intervals of two datasets (Galaxy Version 1.0.0) to generate a list of peaks specific for old mice (peaks in old minus peaks in young); Extract Genomic DNA using coordinates from assembled/unassembled genomes (Galaxy Version 3.0.3+galaxy2) and MEME-ChIP - motif discovery, enrichment analysis and clustering on large nucleotide datasets (Galaxy Version 4.11.2+galaxy1) to determine common motifs across the binding sites. Lastly, Integrative Genomics Viewer (IGV) version 2.4.19 was used to visualize the localization of the peaks in mouse genome (Robinson et al., 2011).

7. Statistical analysis

The data were analyzed in Graphpad Prism 8 and 9. To compare young and old groups, unpaired t-test was used, after confirming normality.

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CHAPTER VII

DISCUSSION

In this work we describe changes occurring in the aging spinal cord, as well as propose interventions aiming at reversing those changes. The main age-related change in the spinal cord that we observe is a **decrease in number of α -MNs** that precedes changes at the neuromuscular junction and muscle degeneration (Chapter II). Our results show a 41% decrease in α -MN number in the lumbar part of the spinal cord in wildtype mice that corresponds well with an axonal loss of 40% by 25 months of age. Although the majority of the literature reports the age-related α -MN loss (Kawamura et al., 1977a; Kawamura et al., 1977b; Tomlinson and Irving, 1977; Zhou et al., 1996 – to name a few), some articles do not see any change in α -MN number with age (Fu et al., 2015; Maxwell et al., 2018). However, Fu et al. (2015) did not perform immunostaining in the tissue section, but homogenized the whole spinal cord and isolated nuclei which were subsequently stained with DAPI (to assess total cell number) and with anti-NeuN antibody (to assess the neuronal cell number). A disadvantage of this approach consists in the fact that some nuclei can be lost during the procedure at the rate that may differ with age, which would bias the results. One reason behind it may be that the nuclear envelope in old α -MNs is more vulnerable (Gillon et al., 2020; Martins et al. 2020). Furthermore, even though the authors are aware that NeuN does not stain all neurons, they only mention Purkinje cells and neuron subpopulation in olfactory bulb as cell types that are not visualized by NeuN staining, NeuN also fails to stain putative small gamma motor neurons, which is relevant in spinal cord and can alter the results (Friesse et al., 2009). Also, NeuN antigenicity can be temporarily altered in pathological states such as ischemia (Unal-Cevik et al., 2004; McPhail et al., 2004) or in response to ionizing

radiation (Wu et al., 2010). To be sure that age-related α -MN loss that we observe is not a result of any bias related to NeuN specificity, we also performed MN counts on Nissl- and nuclear fast red-stained samples (Chapter II) and we have consistently saw the age-related α -MN loss. Moreover, in another study that do not report age-related α -MN loss (Maxwell et al., 2018), the authors actually do not perform neuronal counts, but just assess instead the MN number based on changes of gene expression and abundance of NeuN protein and transcript (via Western blotting and qPCR), which is not a direct measure of MN number and should not be used to formulate conclusions about the number of α -MNs. Furthermore, α -MN count, which would be an objective and direct measure, can be easily performed using simple anti-NeuN staining, Nissl staining, etc. It is worth noting that NeuN is expressed not only by α -MNs, but by other neuronal populations as well, therefore the changes of NeuN protein abundance or transcript levels in spinal cord lysates cannot be used to draw conclusions about any specific neuronal population.

Another striking change associated with aging is an **increase in α -MN cell body area** (Chapter II). Even though the increase in cell body size in aging was noted as early as 1977 (Machado-Salas et al.), its meaning remains unknown. Machado-Salas et al. reported progressive age-related changes in MN cell body morphology such as increased size, dendritic swelling and nodularization, loss of spines, distortion of axons, and degeneration of dendritic distal parts. The increase in cell body size might be a first step in neuronal degeneration process. We suspect that the MN size might play a role in neuronal death, hypothesizing that there may be something intrinsic that makes bigger MNs more vulnerable (such as increased oxidative stress, etc., however at this point it is merely a hypothesis). Large MNs are more vulnerable in ALS patients (Conradi and Ronnevi, 1993) and one potential explanation given in the paper concerns the functional anatomy of these cells, possibly involving acetylcholine esterase turnover. Large MNs

are also more susceptible to polio virus (Hodes, 1949; Hodes et al., 1949). Age-related changes in the concentration of osmolytes (such as myo-inositol) can be potentially one of the reasons behind the increase in cell body area. Interestingly, papers that report no MN loss in aging usually do not report increased cell body size either (Liu et al., 1996; the number of 'large' neurons – α -MNs – did not change in aged cats, while the cell body area of that cell population decreased by 6%), suggesting that smaller size could be somehow protective against cellular degeneration.

Other changes occurring in the aging spinal cord that we uncovered and that promote neuronal loss are higher abundance of cleaved caspase-3 and metabolites such as glutamate, taurine or myo-inositol, as well as decrease in NAA levels, upregulation of transcripts involved in fibrotic pathway (e.g., MMPs and ECM components), as well as increased BSCB permeability and inflammation, in terms of both astrocyte and microglia activation (Chapter II). Especially the latter was linked to decreased α -MN counts in aging (Buchman et al., 2019; also Chapter III), as microglia activation was shown to be negatively correlated with α -MN number.

Despite the fact that the spinal cord aging is a universal problem, currently there are no cures or commercially available treatments aimed at reversing age-related α -MN loss. The only **potential interventions** reported in the literature are exercise and caloric restriction, however the results are inconsistent (see Waters et al., 2010 for a review).

Exercise was shown to ameliorate α -MN loss, decrease neuromuscular degeneration and maintain healthy nucleocytoplasmic transport in 24 months old wildtype mice that were allowed access to a running wheel for 4 months (Gillon et al., 2018). Exercise (12-week long exercise program on treadmill) also improved balance in rats, however without affecting oxidative stress markers nor blood glucose levels (Silveira et al., 2019). Both exercise and **caloric restriction** were reported to ameliorate age-related changes at the neuromuscular junction (Valdez et al., 2010). A 40% caloric restriction from 4 to 24

months of age preserved NMJ structure in 24 months old wildtype mice and the beneficial effect on NMJ was also observed in 22 months old mice after one month of exercise on a running wheel. Furthermore, caloric restriction, but not exercise decreased age-related α -MN loss. The authors do not investigate however, whether the effect of combining caloric restriction and exercise would increase the beneficial effect on the α -MNs. Chopek and Gardiner (2010) show that 40% caloric restriction in rats, starting at 14th week of age, decreased age-related caspase 3 activation in α -MN, but it had no effect on α -MN number. Interestingly, in this case the authors did not observe α -MN loss with age whatsoever, nor age-related increase in α -MN cell body area, which is consistent with our observation that the papers that do not report age-related α -MN loss, do not report α -MN soma swelling with age either, which suggests that there is a relationship between increase in the area of the soma and neuronal loss, as discussed above. Although caloric restriction was shown to be neuroprotective (e.g., Valdez et al., 2010), a study on rats showed that it does not normalize α -MN electrophysiological properties that alter with age, such as greater afterhyperpolarization potential, longer AHP half-decay time, higher input resistance, etc. (Kalmar et al., 2009). Also, gastrocnemius muscle overloading in rats did not prevent age-related α -MN loss (Kanda et al., 1996). However, the exercise was shown to modulate **microglia activation**, which had a neuroprotective effect on aging α -MNs (Giorgetti et al., 2019). The authors demonstrate as well that pharmacological microglia inhibition with BLZ945, a CSF1R inhibitor, has a similar beneficial effect on α -MNs in aging, suggesting that microglia play a crucial role in age-related α -MN loss. Our results support this hypothesis, as we observe increased microglia activation in the aging spinal cord (Chapter II), and decreased microglia activation correlates with higher α -MN counts on old mice (Chapter III). Treatment with OKN-007 also decreases microglia activation and attenuates α -MN loss in both old wildtype mice and ALS mouse model (Chapters IV and V). Another

intervention proposed for preventing age-related α -MN is **administration of exogenous testosterone** (in rats; Fargo et al., 2007). Also **PRMT8** was identified as a potential therapeutic target (Simadi et al., 2018). PRMT8 is a protein arginine methyltransferase that regulates the abundance of asymmetric dimethyl arginines in neurons, which increases α -MN resistance to cellular stress.

Despite all the ongoing research, there are still no definitive conclusions about the neuroprotective effect on aging α -MNs of the most commonly investigated interventions, such as exercise and caloric restriction. Other experimental approaches (exogenous testosterone, manipulating PRMT8 levels) may be difficult to translate into the clinic or may cause unwanted adverse effects. However, the interventions described in this work have better translational potential. OKN-007 has been successfully administered in clinical trials to human patients suffering from glioblastoma, without negative side effects even after multiple years of administration. We demonstrate that OKN-007 attenuates age-related α -MN loss (Chapter IV), as well as increases α -MN survival in ALS mouse model (Chapter V). OKN-007 delayed significantly disease progression, which makes it also a potential therapeutic in ALS. Furthermore, OKN-007 decreases microglia activation and improves BSCB integrity. We observe similar benefits in old MMP-9 and MMP-12 knockout mice (Chapter III). Specific pharmaceutical MMP inhibitors are available commercially, however more research is needed before they could be tested in a clinical trial. Furthermore, we identify troponins as potential targets for therapeutic intervention (Chapter VI).

We show for the first time that all three troponins (I, T, and C) are expressed in α -MNs of wildtype mice and that troponins I and T can translocate to the nucleus where they differentially bind DNA in an age-dependent manner. Further studies are needed to establish, if troponins' age-related increase in cytoplasmic abundance alone can lead to their translocation to the nucleus or if their nuclear presence is a sign of a more general

problems with nucleocytoplasmic transport. A recent study by Gillon et al. (2020) shows that **nuclear permeability** increases with age in murine α -MN due to a disruption of nuclear barrier and that change favors neuronal loss. Interestingly, exercise (4 months of training on running wheel, starting at 20 months of age) helped maintain proper nucleocytoplasmic transport. Age-dependent defects in nucleocytoplasmic transport are also preserved and recapitulated in neurons derived from human fibroblasts, and were attributed to a decrease in the levels of RanBP17 (Martens et al., 2015). A defect in nucleocytoplasmic transport and mislocalization of nuclear proteins, such as TDP-43, and their presence in cytoplasm was also reported by Hideyama et al. (2012) in α -MNs of old mice. TDP-43 toxic aggregates in the cytoplasm are also a hallmark of ALS (see Dewey et al., 2012; Prpar Mihevic et al., 2017; and Burk and Pasterkamp, 2019; Ding and Sepehrimanesh, 2021 for review) which points towards disruption of nucleocytoplasmic transport. Moreover, a GGGGCC repeat in *C9orf72* was identified as another factor leading to defects in nucleocytoplasmic transport seen in ALS, as well as in frontotemporal dementia (Freibaum et al., 2015; Zhang et al., 2015). Similarly, a poly-glutamine expansion in huntingtin was shown to disrupt nucleocytoplasmic transport in Huntington's disease in an age-dependent manner (Gasset-Rosa et al., 2017). Moreover, nuclear envelope disruption is a feature of progeria and nuclear lamin mutations are associated with accelerated aging (Li et al., 2016). Further studies are needed to determine, whether the interventions that we propose, such as treatment with OKN-007 or inhibiting MMPs, positively affect the integrity of nuclear envelope and maintain proper nucleocytoplasmic transport in aging α -MNs.

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SUMMARY

Aging is a universal problem concerning the whole society. Age-related loss of muscle mass and function leads to motor deficits and affects greatly the quality of life of an affected individual, but it also has a broader negative impact on economy, healthcare system, etc. Our laboratory demonstrated that the loss of innervation is a primary cause leading to sarcopenia, therefore it is important to address changes occurring in the aging spinal cord, especially those concerning alpha-motor neurons. Spinal cord aging, however, is an understudied subject, and this work fills the gap in the knowledge about this important aspect of the aging process. We uncovered a 41% decrease in α -MNs number in the lumbar spinal cord of old wildtype mice. Neuronal loss is accompanied by swelling of the soma, axonal loss and demyelination, as well as increased inflammation and BSCB permeability. The analysis of the transcriptome reveals an upregulation of fibrotic pathway, especially MMPs and ECM components, such as collagens.

Currently there are no treatments available aiming at preventing age-related α -MN loss, therefore in this study, we propose several interventions that could become potential therapeutics suitable for clinical use in humans. Firstly, we observed that MMP-12 was the most upregulated transcript in the aging spinal cord (172-fold increase), which led us to look closer into the role of MMPs in the spinal cord aging and α -MN loss. Using MMP-9 and MMP-12 knockout in mice, we uncovered that both molecules contribute to age-related α -MN loss and increased neuroinflammation, as MMP-9 and MMP-12 KO mice exhibited higher α -MN counts than wildtype age-matched controls. Knocking out MMP-12 also improved BSCB barrier integrity. Furthermore, we see similar improvements in old wildtype mice treated with OKN-007, a neuroprotective, anti-inflammatory and anti-oxidant compound. Moreover, OKN-007 has a neuroprotective and anti-inflammatory effect SOD1 G93A ALS mice model which exhibits an accelerated

α -MN loss. The treatment with OKN-007 also significantly slowed down the disease progression, however without delaying the disease onset. In this work we also discovered and described for the first time the presence of troponins in the spinal cord. Troponins I, C and T localize to α -MNs and their abundance changes with age. Also, troponins I and T can translocate to the nucleus and bind DNA and compete for binding sites with transcription factors such as FoxD3. Therefore, troponins can be another potential target of an intervention aiming at preventing age-related α -MN loss.

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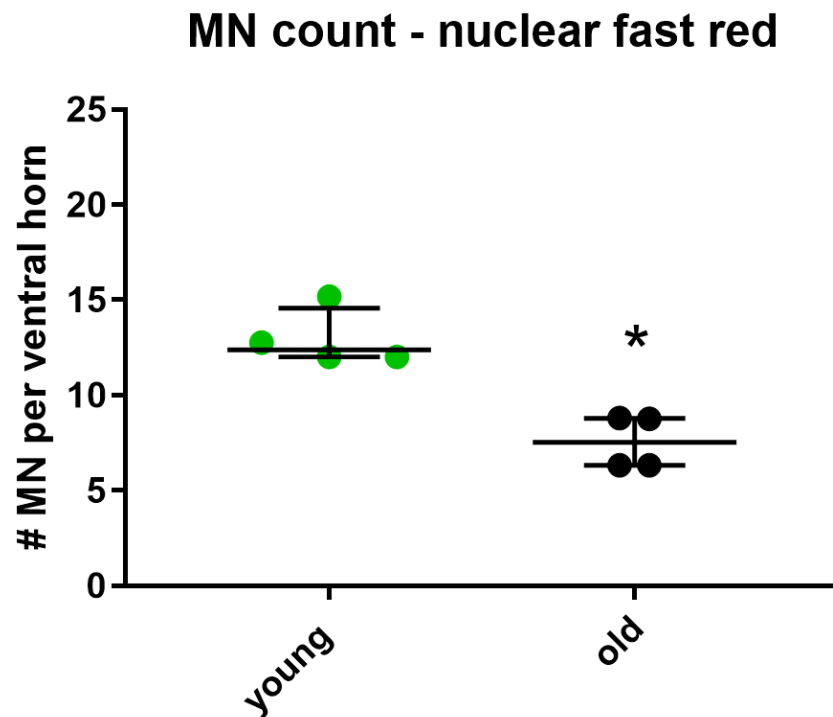
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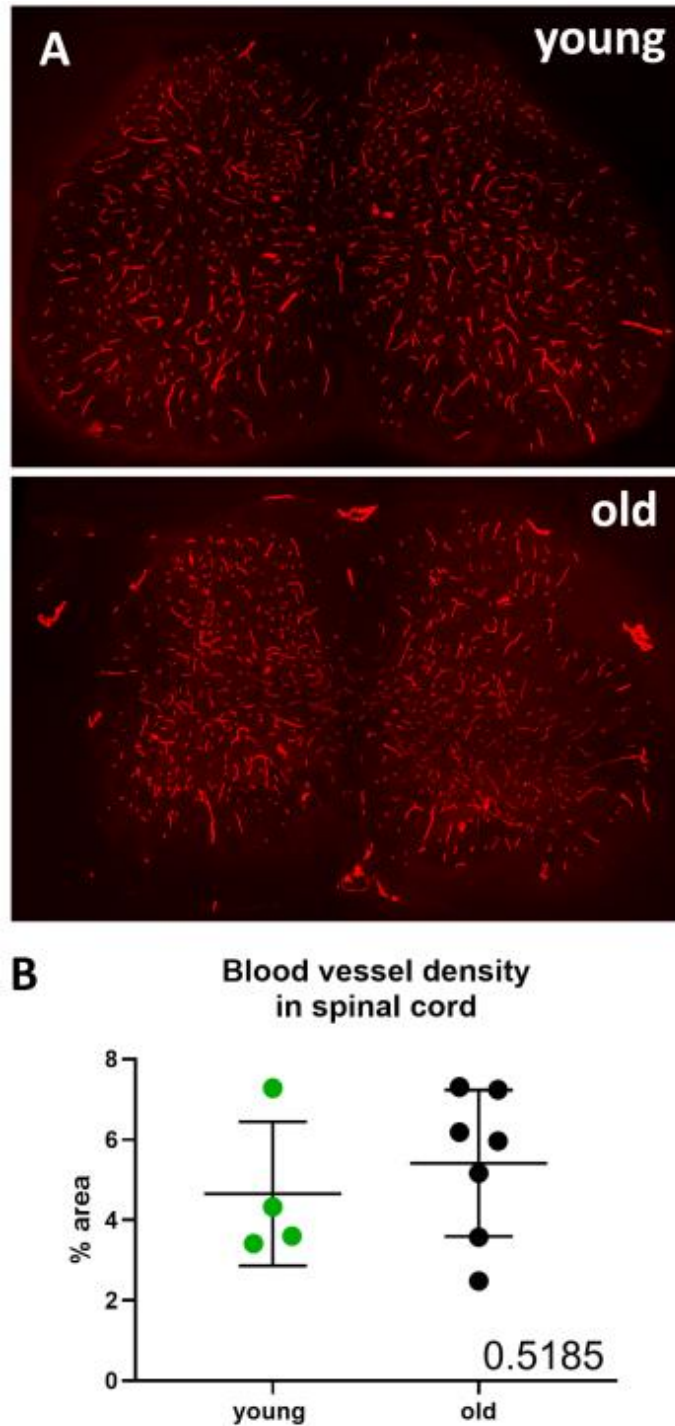
APPENDIX A

MOLECULAR CHANGES ASSOCIATED WITH SPINAL CORD AGING

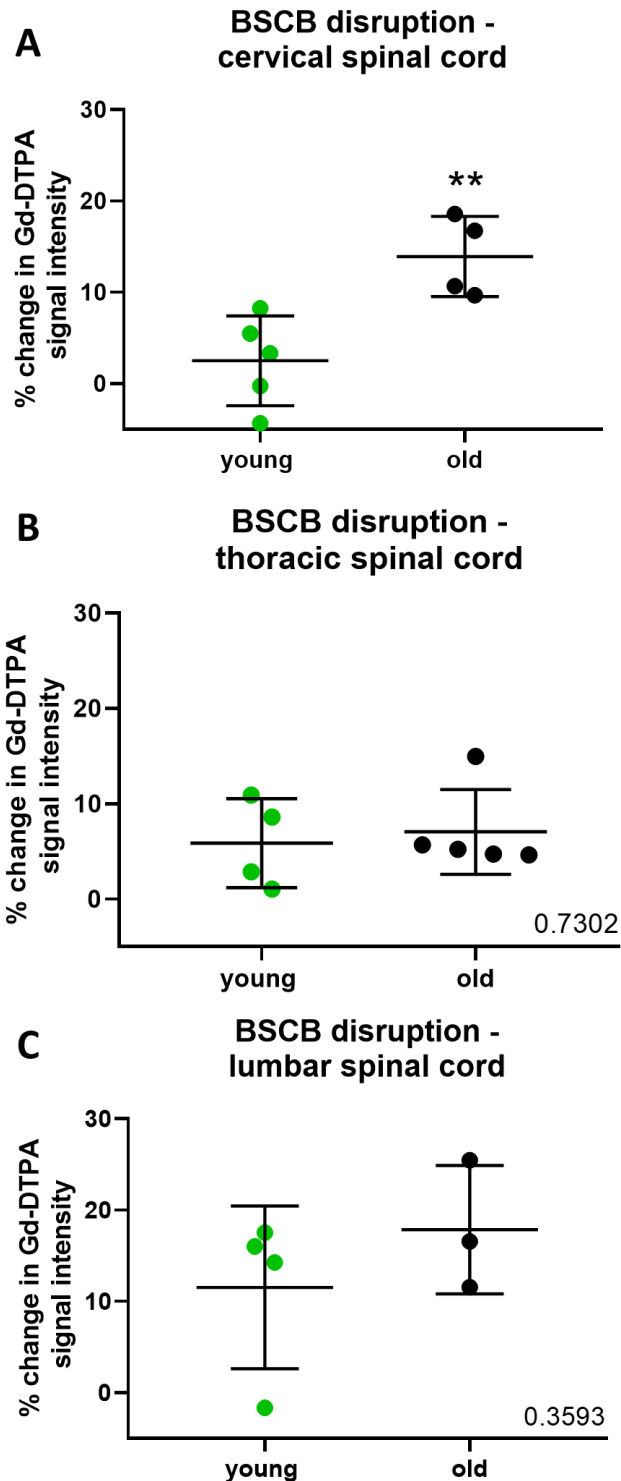
SUPPLEMENTARY MATERIAL



Supplemental Figure A.1. Age-related α -motor neuron loss. Nuclear Fast Red staining of lumbar spinal cord sections. There is approximately a 39% decrease in α -MN number in old mice (females, 27 mo, $n = 4$, median and IQR = 7.542 and 2.455) comparing to young (6-7 mo, $n = 4$, median and IQR = 12.38 and 2.56; Mann Whitney test, $p = 0.0286$).



Supplemental Figure A.2. Blood vessel density does not decrease with age. **A** – A representative image of a cross section of lumbar spinal cord of a young (3 mo) and old (29 mo) mice stained for CD31 to visualize blood vessels (tiling of four 10x images, z-stack of 5, maximum intensity projection); **B** – quantification of blood vessel density a percentage of total SC area. There is no significant difference between young ($n = 4$, mean $\pm$ SD = 4.66 ± 1.79) and old mice ($n = 7$, mean $\pm$ SD = 5.42 ± 1.82 ; two-tailed unpaired t-test, $t = 0.6720$, $p = 0.5185$).



Supplemental Figure A.3. BSCB permeability to Gd-DTPA. MRI assessment of BSCB permeability; percentage change in the signal intensity pre- and post-Gd-DTPA injection in young (3-4 mo) and old (26-27 mo) mice measured in cervical (A), thoracic (B), and lumbar (C) spinal cord.

Ingenuity Canonical Pathways	-log(B-H p-value)	Ratio	z-score
Hepatic Fibrosis / Hepatic Stellate Cell Activation	9.19	0.209	
Dendritic Cell Maturation	7.16	0.195	3.9
Complement System	6.79	0.6	1.633
Phagosome Formation	5.89	0.232	
T Helper Cell Differentiation	5.85	0.264	
Granulocyte Adhesion and Diapedesis	5.57	0.202	
LXR/RXR Activation	5.35	0.237	-2.111
Atherosclerosis Signaling	4.9	0.203	
Acute Phase Response Signaling	4.73	0.173	1.155
Coagulation System	4.73	0.5	-0.378
Th1 and Th2 Activation Pathway	4.73	0.156	
Altered T Cell and B Cell Signaling in Rheumatoid Arthritis	4.15	0.214	
Antigen Presentation Pathway	3.73	0.308	
Agranulocyte Adhesion and Diapedesis	3.62	0.176	
TREM1 Signaling	3.62	0.204	3.317

Ingenuity Canonical Pathways	$-\log(\text{B-H p-value})$	Ratio	z-score
Th2 Pathway	3.62	0.153	0.333
B Cell Development	3.62	0.429	
Communication between Innate and Adaptive Immune Cells	3.62	0.222	
Fcγ Receptor-mediated Phagocytosis in Macrophages and Monocytes	3.24	0.183	3.317
Leukocyte Extravasation Signaling	3.04	0.135	3.742

Supplemental Table A.1 IPA analysis results on RNA-seq data from young and old spinal cord. Top 20 canonical pathways

Symbol	Entrez Gene Name	Ensembl	Expr Log Ratio	Expr p-value
MMP12	matrix metalloproteinase 12	ENSMUSG00000049723	7.43	0.0003
SERPINA3	serpin family A member 3	ENSMUSG00000058207	-6.39	0.0001
COL28A1	collagen type XXVIII alpha 1 chain	ENSMUSG00000068794	5.62	0.00005
TNNI2	troponin I2, fast skeletal type	ENSMUSG00000031097	5.32	0.00005
SERPINA1	serpin family A member 1	ENSMUSG00000072849	-5.14	0.00035
ITGAX	integrin subunit alpha X	ENSMUSG00000030789	4.92	0.00005
CCL11	C-C motif chemokine ligand 11	ENSMUSG00000020676	4.87	0.00005
IL1RN	interleukin 1 receptor antagonist	ENSMUSG00000026981	4.65	0.00015
MSR1	macrophage scavenger receptor 1	ENSMUSG00000025044	4.32	0.0001
EGR2	early growth response 2	ENSMUSG00000037868	4.27	0.00005
LYZ	lysozyme	ENSMUSG00000069516	4.26	0.00005
CLEC4D	C-type lectin domain family 4 member D	ENSMUSG00000030144	4.2	0.00005
LGALS3	galectin 3	ENSMUSG00000050335	4.18	0.00005
GLDN	gliomedin	ENSMUSG00000046167	3.97	0.00005
APOA1	apolipoprotein A1	ENSMUSG00000032083	-3.92	0.00005
CLEC7A	C-type lectin domain containing 7A	ENSMUSG00000079293	3.87	0.00005
BCL2A1	BCL2 related protein A1	ENSMUSG00000089929	3.82	0.00005
CST7	cystatin F	ENSMUSG00000068129	3.74	0.00005
HPSE	heparanase	ENSMUSG00000035273	3.68	0.00005
CD22	CD22 molecule	ENSMUSG00000030577	3.59	0.00005

Supplemental Table A.2 IPA analysis results on RNA-seq data from young and old spinal cord. Top 20 molecules

Top Diseases and Functions - Categories	Focus Molecules
Cellular Movement, Hematological System Development and Function, Immune Cell Trafficking	35
Cellular Function and Maintenance, Cell-To-Cell Signaling and Interaction, Cell Death and Survival	25
Connective Tissue Disorders, Organ Morphology, Organismal Injury and Abnormalities	16
Cellular Movement, Immune Cell Trafficking, Inflammatory Response	21
Inflammatory Response, Organismal Injury and Abnormalities, Cellular Movement	20
Cellular Development, Cellular Growth and Proliferation, Hematological System Development and Function	20
Cell Death and Survival, Cell-To-Cell Signaling and Interaction, Hematological System Development and Function	19
Inflammatory Disease, Immunological Disease, Cellular Movement	18
Cell Cycle, Lymphoid Tissue Structure and Development, Tissue Morphology	15
Digestive System Development and Function, Embryonic Development, Endocrine System Development and Function	1
Infectious Diseases, Inflammatory Disease, Organismal Injury and Abnormalities	1
Cellular Compromise, Cardiovascular Disease, Developmental Disorder	1
Ophthalmic Disease, Organismal Injury and Abnormalities, Neurological Disease	1

Supplemental Table A.3 IPA analysis results on RNA-seq data from young and old spinal cord. Diseases – main categories

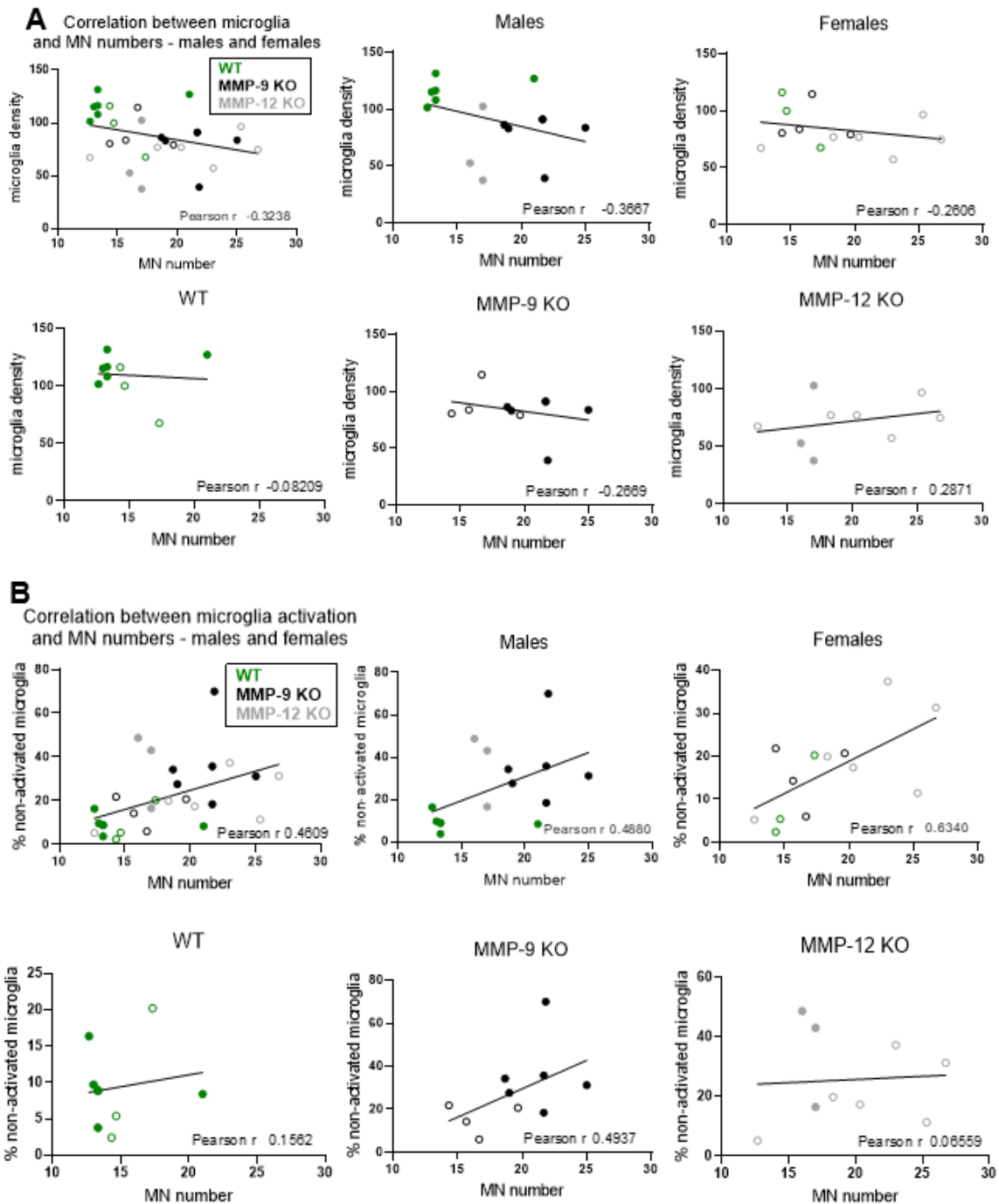
Upstream Regulator	Expr Log Ratio	Molecule Type	Predicted Activation State	Activation z-score
IFNG		cytokine	Activated	3.321
IL10RA		transmembrane receptor		-0.131
PTGER4		G-protein coupled receptor		-1.414
MYD88		other	Activated	3.373
TGFB1	1.2	growth factor		0.277
STAT1		transcription regulator		1.732
PPARD		ligand-dependent nuclear receptor	Activated	2.415
TNF		cytokine	Activated	2.312
PPARG		ligand-dependent nuclear receptor		0.886
TYROBP	2.05	transmembrane receptor		
CX3CR1		G-protein coupled receptor		1
CEBPB		transcription regulator		0.707
LDLR		transporter		
TICAM1		other	Activated	2.813
PPARA		ligand-dependent nuclear receptor		
SMAD4		transcription regulator		
IL10		cytokine		0.626
G6PC3		phosphatase		
IL6R	1.16	transmembrane receptor		
TLR2	1.33	transmembrane receptor		0.128

Supplemental Table A.4 IPA analysis results on RNA-seq data from young and old spinal cord. Top 20 upstream regulators

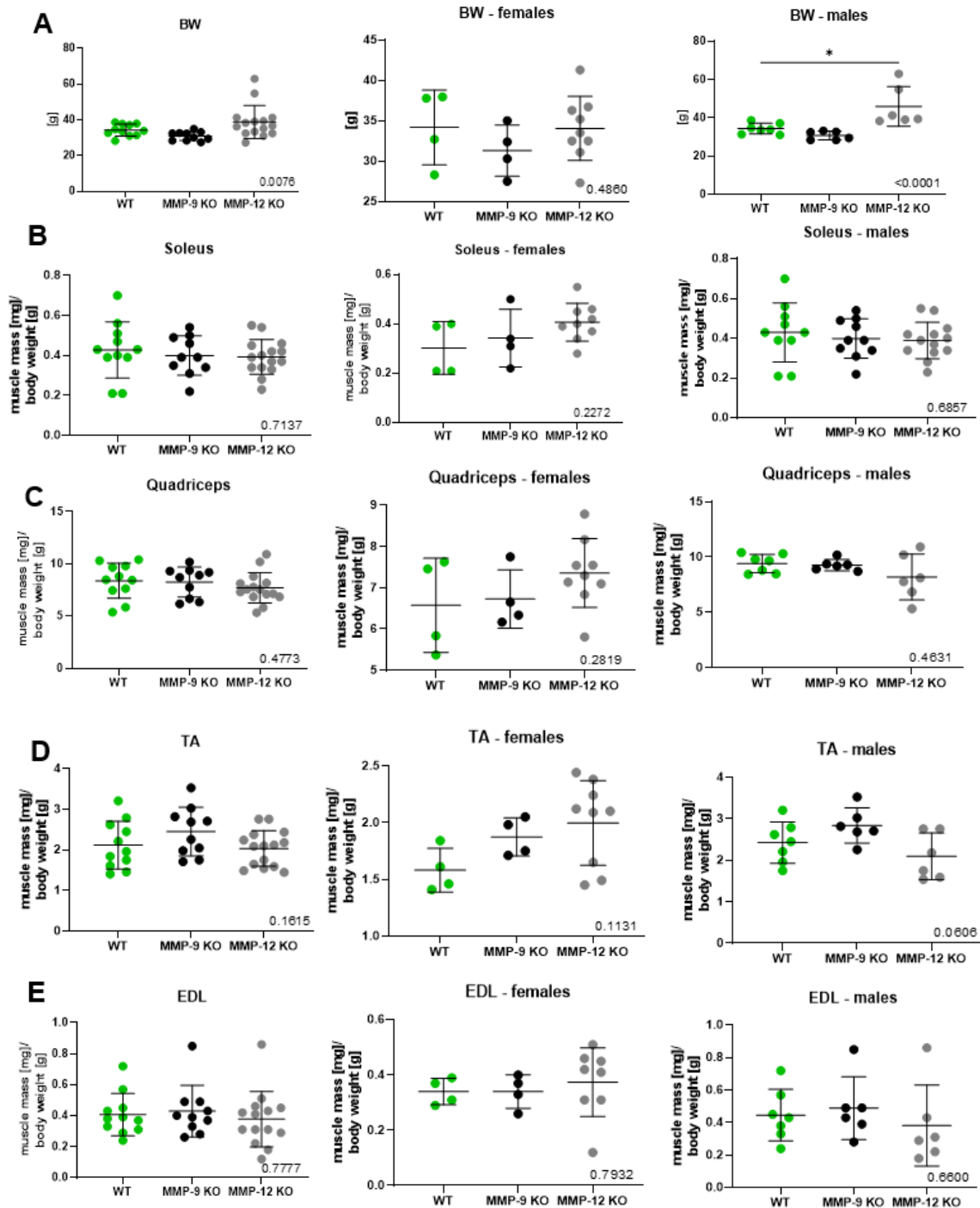
APPENDIX B

KNOCKING OUT MMP-9 AND MMP-12 HAS A DIFFERENTIAL AND SEX-SPECIFIC EFFECT ON AGE-RELATED A-MOTOR NEURON LOSS AND DECLINE IN MUSCLE

MASS



Supplemental Figure B.1. Correlation between MN number and microglia density and activation. **A** – overall, there is a negative correlation between α -MN number and microglia density (number of Iba1+ cells per mm²; open circles indicate females; Pearson $r = -0.3238$; males: $r = -0.3667$, females: $r = -0.2606$). However, in MMP-12 KO mice the correlation is reversed ($r = 0.2871$, MMP-9 KO: $r = -0.2669$, WT: $r = -0.08209$); **B** – there is an overall positive correlation between α -MN number and microglia activation (the percentage of non-activated microglia per ventral horn; Pearson $r = 0.4609$; males: $r = 0.4880$, females: $r = 0.6340$). In MMP-12 KO mice the correlation is basically abolished ($r = 0.06559$, MMP-9 KO: $r = 0.4937$, WT: $r = 0.1562$).

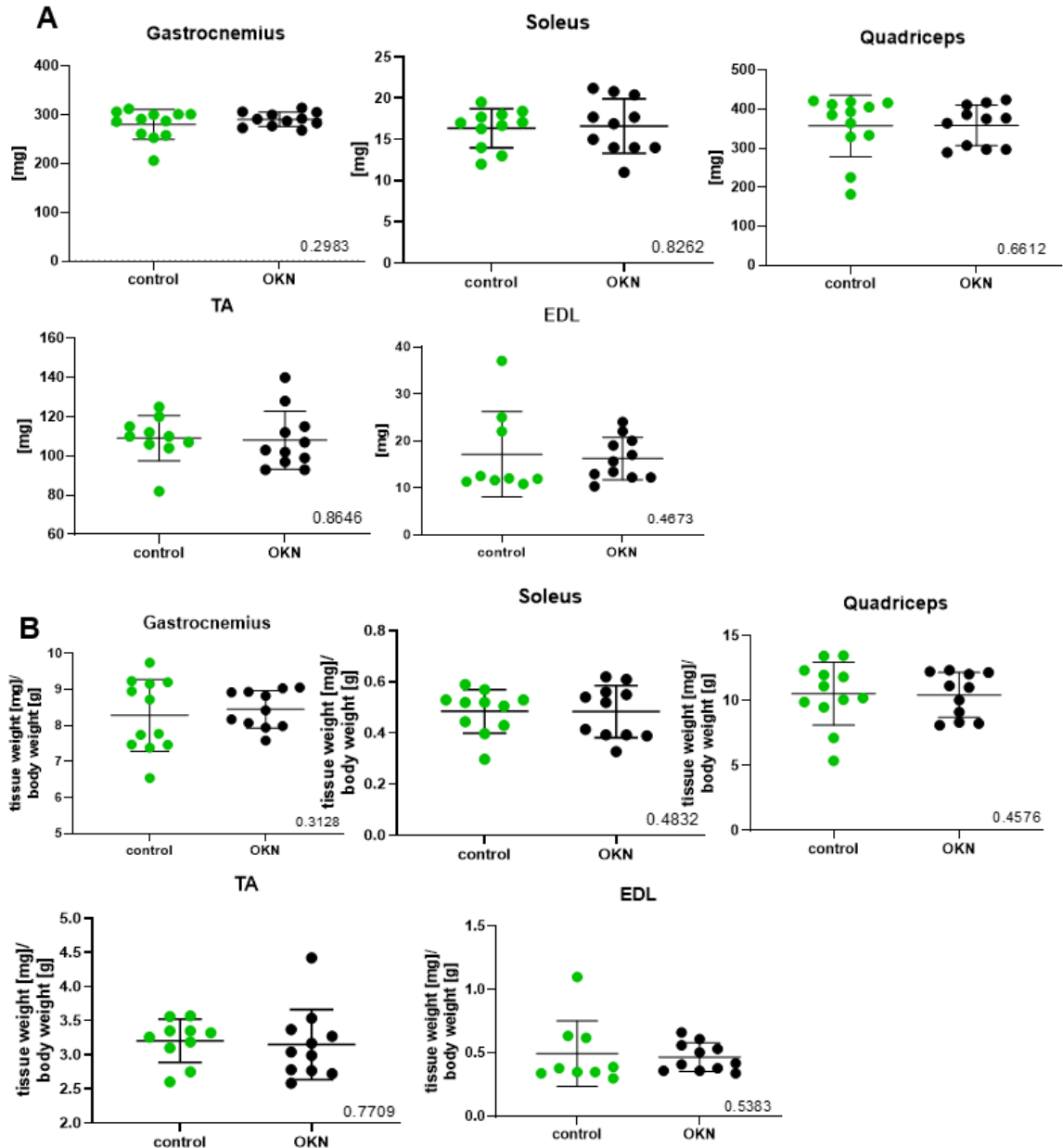


Supplemental Figure B.2. Body weight and hindlimb muscle weights in 25 mo MMP-9 and MMP-12 KO mice. A – MMP-9 KO and MMP-12 KO mice's body weight is not significantly different from WT controls, except for MMP-12 KO males (both males and females: Kruskal-Wallis $p = 0.0076$, Dunn's post hoc WT vs MMP-9 KO $p = 0.1170$, WT vs MMP-12 KO $p = 0.5193$; females: ordinary one-way ANOVA $p = 0.4860$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.4866$, WT vs MMP-12 KO $p = 0.9971$;

males: Kruskal-Wallis $p < 0.0001$, Dunn's post hoc WT vs MMP-9 KO $p = 0.2524$, WT vs MMP-12 KO $p = 0.0486$; **B** – soleus muscle weight normalized to body weight. Both males and females: ordinary one-way ANOVA $p = 0.7137$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.7759$, WT vs MMP-12 KO $p = 0.6385$; females: Kruskal-Wallis $p = 0.2272$, Dunn's post hoc WT vs MMP-9 KO $p > 0.9999$, WT vs MMP-12 KO $p = 0.2048$; males: ordinary one-way ANOVA $p = 0.6857$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.7639$, WT vs MMP-12 KO $p = 0.6011$; **C** – quadriceps muscle weight normalized to body weight. Both males and females: ordinary one-way ANOVA $p = 0.4773$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.9753$, WT vs MMP-12 KO $p = 0.4251$; females: ordinary one-way ANOVA $p = 0.2819$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.9548$, WT vs MMP-12 KO $p = 0.2625$; males: Welch's ANOVA $p = 0.4631$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.9243$, WT vs MMP-12 KO $p = 0.3871$; **D** – TA muscle weight normalized to body weight. Both males and females: ordinary one-way ANOVA $p = 0.1615$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.2701$, WT vs MMP-12 KO $p = 0.8930$; females: ordinary one-way ANOVA $p = 0.1131$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.3153$, WT vs MMP-12 KO $p = 0.0697$; males: ordinary one-way ANOVA $p = 0.0606$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.2646$, WT vs MMP-12 KO $p = 0.4077$; **E** – EDL muscle weight normalized to body weight. Both males and females: Kruskal-Wallis $p = 0.7777$, Dunnett's post hoc WT vs MMP-9 KO $p > 0.9999$, WT vs MMP-12 KO $p > 0.9999$; females: ordinary one-way ANOVA $p = 0.7932$, Dunnett's post hoc WT vs MMP-9 KO $p > 0.9999$, WT vs MMP-12 KO $p = 0.7926$; males: ordinary one-way ANOVA $p = 0.6600$, Dunnett's post hoc WT vs MMP-9 KO $p = 0.9030$, WT vs MMP-12 KO $p = 0.7977$.

APPENDIX C

OKN-007 DECREASES AGE-RELATED ALPHA-MOTOR NEURON LOSS AND IMPROVES MUSCLE FUNCTION IN WILDTYPE MICE



Supplemental Figure C.1 OKN-007 does not change hindlimb muscle mass in 25 mo mice. A – absolute muscle masses; the treatment does not affect hindlimb muscle mass (gastrocnemius: Mann Whitney test $p = 0.2983$; soleus: two-tailed unpaired t-test $p = 0.8262$; quadriceps: Mann Whitney test $p = 0.6612$; TA: two-tailed unpaired t-test $p = 0.8646$; EDL: Mann Whitney test $p = 0.4673$); **B** – after normalizing to body weight, there is still no difference between the groups; (gastrocnemius: two-tailed unpaired t-test $p =$

0.3128; soleus: two-tailed unpaired t-test $p = 0.4832$; quadriceps: two-tailed unpaired t-test $p = 0.4576$; TA: two-tailed unpaired t-test $p = 0.7709$; EDL: Mann Whitney test $p = 0.5383$).

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
-0.47	-4.71	-4.24	7.80E-05	Ighg2b	immunoglobulin heavy constant gamma 2B
-5.89	-3	2.88	3.60E-05	Slc22a1	solute carrier family 22 (organic cation transporter), member 1
-5.38	-2.52	2.86	0.00014	Gm48478	predicted gene, 48478
-1.18	1.66	2.83	8.80E-06	Gm38372	predicted gene, 38372
-5.89	-3.21	2.68	0.00011	Gm10581	predicted gene 10581
-5.53	-2.88	2.65	3.80E-05	Gm16202	predicted gene 16202
-5.89	-3.26	2.62	4.30E-06	Gm3448	predicted gene 3448
-2.12	-4.63	-2.51	0.00012	Slc5a12	solute carrier family 5 (sodium/glucose cotransporter), member 12
-3.38	-5.89	-2.5	6.90E-05	Dcpp3	demilune cell and parotid protein 3
-5.89	-3.45	2.44	0.00014	Gm34176	predicted gene, 34176
-3.5	-5.89	-2.38	5.70E-05	Gm48773	predicted gene, 48773
-3.53	-5.89	-2.36	5.50E-06	Wnt3a	wingless-type MMTV integration site family, member 3A
-3.73	-5.89	-2.16	0.00011	Gm9451	predicted gene 9451
-3.96	-5.89	-1.93	4.30E-05	4930583P06Rik	RIKEN cDNA 4930583P06 gene
1.27	3.14	1.87	1.90E-05	Gucy1a2	guanylate cyclase 1, soluble, alpha 2
-0.08	-1.93	-1.85	0.00013	Cxcl13	chemokine (C-X-C motif) ligand 13
-1.5	0.32	1.82	9.20E-05	Gm28322	predicted gene 28322
-4.09	-5.89	-1.79	0.00013	AI646519	expressed sequence AI646519
-4.09	-5.89	-1.79	0.00013	Gm48407	predicted gene, 48407
0.84	2.5	1.66	8.00E-05	Gm26945	predicted gene, 26945
-0.18	1.41	1.59	2.50E-05	Far1os	fatty acyl CoA reductase 1, opposite strand
-5.89	-4.3	1.58	0.00011	Gm5979	predicted gene 5979
3.83	5.41	1.57	6.10E-06	Lnpep	leucyl/cystinyl aminopeptidase

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
-1.68	-0.12	1.56	3.70E-05	Gm43665	predicted gene 43665
-2.31	-0.81	1.5	7.20E-05	Gm21671	predicted gene, 21671
-0.68	0.65	1.33	4.60E-05	Tekt4	tektin 4
-0.39	0.85	1.25	6.00E-05	6430511E19Rik	RIKEN cDNA 6430511E19 gene
3	1.93	-1.07	7.70E-05	Cartpt	CART prepropeptide
2.08	3	0.92	8.60E-05	Gm38020	predicted gene, 38020
4.08	4.98	0.9	8.10E-05	Zfp871	zinc finger protein 871
4.47	5.33	0.87	1.50E-06	Dio2	deiodinase, iodothyronine, type II
3.75	2.93	-0.82	2.50E-05	Mafa	v-maf musculoaponeurotic fibrosarcoma oncogene family, protein A (avian)
1.85	2.66	0.81	6.90E-05	Mgp	matrix Gla protein
3.64	4.45	0.8	5.00E-05	Slc10a4	solute carrier family 10 (sodium/bile acid cotransporter family), member 4
2.62	3.42	0.8	9.80E-05	Kcnh5	potassium voltage-gated channel, subfamily H (eag-related), member 5
4.41	5.16	0.75	9.30E-05	Slc18a3	solute carrier family 18 (vesicular monoamine), member 3
2.36	3.08	0.72	9.60E-05	Kcng1	potassium voltage-gated channel, subfamily G, member 1
2.29	3	0.71	2.50E-05	Smyd1	SET and MYND domain containing 1
4.29	4.97	0.68	8.50E-05	Kcnj14	potassium inwardly-rectifying channel, subfamily J, member 14
1.11	1.75	0.65	7.40E-05	Gm9856	predicted gene 9856
5.95	6.6	0.65	0.00011	Gm45884	predicted gene 45884
6.46	5.81	-0.65	7.70E-06	Ebf1	early B cell factor 1
4.85	4.21	-0.65	9.10E-05	A230103L15Rik	RIKEN cDNA A230103L15 gene
2.48	3.11	0.62	2.10E-05	Onecut2	one cut domain, family member 2
3.1	2.5	-0.6	4.80E-05	Gbx1	gastrulation brain homeobox 1
4.74	4.15	-0.59	0.00014	Tshz3	teashirt zinc finger family member 3

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
4.36	3.79	-0.56	1.50E-05	Cpne4	copine IV
3.25	2.7	-0.55	4.10E-05	Pax8	paired box 8
4.73	4.2	-0.53	3.80E-05	Col5a2	collagen, type V, alpha 2
3.29	2.77	-0.53	7.00E-05	Kcnp2	Kv channel-interacting protein 2
4.35	3.82	-0.53	7.50E-05	Tac1	tachykinin 1
4.99	4.46	-0.53	0.00014	Syt5	synaptotagmin V
4.63	4.11	-0.52	4.40E-06	Unc13b	unc-13 homolog B
4.86	4.34	-0.52	0.00012	Adcyap1	adenylate cyclase activating polypeptide 1
4.88	5.39	0.51	0.00012	Chat	choline acetyltransferase
5.29	4.81	-0.48	1.00E-04	Cdh8	cadherin 8
7.04	7.51	0.47	7.00E-05	Sacs	sacsin
4.43	3.96	-0.47	0.00012	Klhl1	kelch-like 1
4.63	5.09	0.46	0.00013	Atm	ataxia telangiectasia mutated
4.69	4.23	-0.46	3.10E-05	Plch2	phospholipase C, eta 2
3.25	3.69	0.45	0.00015	Wnk4	WNK lysine deficient protein kinase 4
4.26	3.81	-0.45	2.00E-05	Plxnc1	plexin C1
6.97	6.52	-0.45	8.40E-05	Cacna2d1	calcium channel, voltage-dependent, alpha2/delta subunit 1
6.12	5.68	-0.44	2.70E-05	Wdr6	WD repeat domain 6
4.21	3.77	-0.44	3.00E-05	Lbx1	ladybird homeobox 1
4.73	4.29	-0.44	1.00E-04	Scn3b	sodium channel, voltage-gated, type III, beta
4.1	3.67	-0.43	0.00014	AI480526	expressed sequence AI480526
3.87	3.45	-0.42	0.00014	Pou4f1	POU domain, class 4, transcription factor 1
5.92	5.51	-0.41	0.00012	Sst	somatostatin

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
7.27	6.87	-0.4	8.30E-05	Cacna1e	calcium channel, voltage-dependent, R type, alpha 1E subunit
7.19	6.8	-0.39	0.00013	Col11a1	collagen, type XI, alpha 1
4.75	4.36	-0.39	0.00014	Cacna1h	calcium channel, voltage-dependent, T type, alpha 1H subunit
6.22	5.84	-0.39	0.00016	B3gat1	beta-1,3-glucuronyltransferase 1 (glucuronosyltransferase P)
5.98	6.36	0.38	0.00013	Ptpn4	protein tyrosine phosphatase, non-receptor type 4
5.18	4.81	-0.38	0.00014	Col27a1	collagen, type XXVII, alpha 1
6.93	7.3	0.37	5.80E-05	Smg1	SMG1 homolog, phosphatidylinositol 3-kinase-related kinase (C. elegans)
7.83	7.46	-0.37	6.40E-05	Syt1	synaptotagmin I
7.19	7.56	0.36	3.70E-05	Hook3	hook microtubule tethering protein 3
5.79	5.43	-0.36	6.30E-05	Sez6	seizure related gene 6
7.77	7.43	-0.34	4.10E-05	Trank1	tetratricopeptide repeat and ankyrin repeat containing 1
5.15	4.81	-0.34	0.00012	Cadps2	Ca2+-dependent activator protein for secretion 2
6.96	6.65	-0.31	4.90E-05	Col16a1	collagen, type XVI, alpha 1
5.71	6	0.3	3.40E-05	Chka	choline kinase alpha
6.44	6.74	0.3	6.40E-05	Rgs4	regulator of G-protein signaling 4
6.22	5.92	-0.3	9.70E-05	Pnmal1	PNMA-like 1
7.31	7.6	0.28	1.80E-05	Ranbp2	RAN binding protein 2
6.34	6.61	0.28	6.80E-05	Gpd1	glycerol-3-phosphate dehydrogenase 1 (soluble)
6.24	6.51	0.28	0.00011	Bicd1	BICD cargo adaptor 1
5.96	5.69	-0.28	1.10E-05	Fgf13	fibroblast growth factor 13
5.39	5.11	-0.28	1.00E-04	Bahcc1	BAH domain and coiled-coil containing 1
4.73	4.44	-0.28	0.00014	Necab2	N-terminal EF-hand calcium binding protein 2
5.69	5.42	-0.27	5.60E-05	Tmem94	transmembrane protein 94

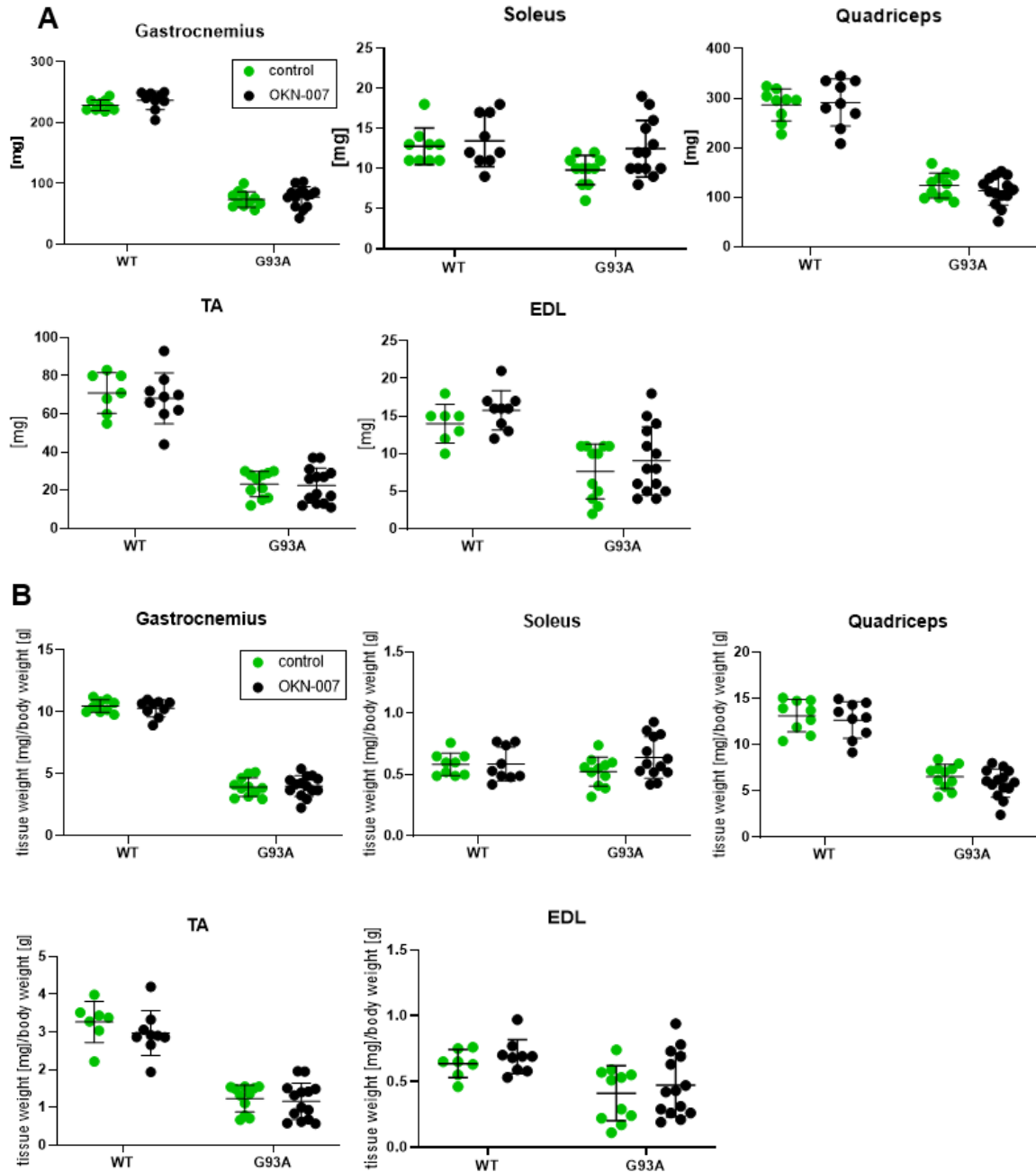
OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
5.16	4.89	-0.27	8.10E-05	Pnma3	paraneoplastic antigen MA3
6.74	6.47	-0.27	0.00015	Pkia	protein kinase inhibitor, alpha
7.86	7.6	-0.26	9.70E-05	Tspoap1	TSPO associated protein 1
9.77	9.51	-0.26	0.00013	Srrm2	serine/arginine repetitive matrix 2
7.62	7.37	-0.26	0.00015	Adcy1	adenylate cyclase 1
6.72	6.97	0.25	6.40E-05	Mtmr2	myotubularin related protein 2
4.23	4.48	0.25	0.00012	Ttl12	tubulin tyrosine ligase-like family, member 12
5.53	5.29	-0.25	9.60E-05	Fbrsl1	fibrosin-like 1
11.7	11.9	0.24	8.60E-05	mt-Nd2	mitochondrially encoded NADH dehydrogenase 2
8.84	8.6	-0.24	4.40E-05	Gpm6a	glycoprotein m6a
6.66	6.42	-0.24	0.00015	Usp11	ubiquitin specific peptidase 11
6.38	6.61	0.23	1.90E-05	Pmp22	peripheral myelin protein 22
7.59	7.36	-0.23	0.00013	Pam	peptidylglycine alpha-amidating monooxygenase
5.91	6.13	0.22	6.10E-05	Paqr4	progesterin and adipoQ receptor family member IV
8.29	8.51	0.22	0.00016	Fgf1	fibroblast growth factor 1
7.98	8.19	0.21	3.40E-05	Cdc37l1	cell division cycle 37-like 1
8.95	8.74	-0.21	0.00014	Chd3	chromodomain helicase DNA binding protein 3
10.2	10	-0.2	1.60E-05	Gprasp1	G protein-coupled receptor associated sorting protein 1
8.56	8.75	0.19	0.00015	Endod1	endonuclease domain containing 1
6.83	7.01	0.18	6.80E-05	Lgi3	leucine-rich repeat LGI family, member 3
7.96	7.78	-0.18	0.00013	Slc17a6	solute carrier family 17 (sodium-dependent inorganic phosphate cotransporter), member 6
9.38	9.21	-0.17	2.00E-05	Pnmal2	PNMA-like 2
7.01	6.84	-0.17	5.20E-05	Ppfia4	protein tyrosine phosphatase, receptor type, f polypeptide (PTPRF), interacting protein (liprin), alpha 4

OKN value	ctrl value	lgFold	p_value	SYMBOL	GENENAME
8.29	8.13	-0.17	5.30E-05	Srrm1	serine/arginine repetitive matrix 1
8.88	8.71	-0.17	6.80E-05	Timp2	tissue inhibitor of metalloproteinase 2
8.77	8.93	0.16	7.90E-05	Hspa4l	heat shock protein 4 like
8.15	8.05	-0.1	6.40E-05	Eif4b	eukaryotic translation initiation factor 4B

Supplemental Table C.1 The list of differentially expressed genes from RNA-seq analysis (old treated vs old control).

APPENDIX D

OKN-007 TREATMENT SLOWS DOWN DISEASE PROGRESSION IN SOD1 G93A
ALS MOUSE MODEL BY DECREASING ALPHA-MOTOR NEURON LOSS AND
MICROGLIA ACTIVATION, WITHOUT AFFECTING MUSCLE MASS AT P145



Supplemental Figure D.1 The treatment with OKN-007 doesn't affect hindlimb muscle mass in G93A ALS mouse model. **A** – absolute muscle masses; there is the genotype effect on all dissected muscles: gastrocnemius ($p < 0.0001$), soleus ($p = 0.0323$), quadriceps ($p < 0.0001$), TA ($p < 0.0001$), EDL ($p < 0.0001$); **B** – muscle masses normalized to body weight; the genotype effect remains significant in all muscles except soleus: gastrocnemius ($p < 0.0001$), soleus ($p = 0.9544$), quadriceps ($p < 0.0001$), TA ($p < 0.0001$), EDL ($p = 0.0011$).

APPENDIX E:

TROPONINS ARE EXPRESSED IN ALPHA-MOTOR NEURONS OF WILDTYPE
MOUSE AND CAN TRANSLOCATE TO THE NUCLEUS TO POTENTIALLY ACT AS
TRANSCRIPTION FACTORS

peaks specific to old (old-young)						
chromosome	start	stop	gene	upstream of	downstream of	location
chr3	5860425	5860754		1700008P02Rik	Pex2	intergenic
chr9	123371012	123371252	Lars2			exon
chr13	3372266	3372484		2810429I04Rik	Speer6-ps1	intergenic
chr16	57391513	57391781	Cmss1, Filip1l			intron
chrX	109484401	109484742	Apool	Satl1, Zfp711		intergenic
peaks in young						
chromosome	start	stop	gene	upstream of	downstream of	location
chr2	98502482	98502997		B230118H07Rik, Rag2	Lrrc4c	intergenic
chr2	98506542	98507523		B230118H07Rik, Rag2	Lrrc4c	intergenic
chr6	103598974	103599374	Chl1			intron
chr9	3000272	3000728		mir101c		intergenic
chr9	3001218	3001520		mir101c		intergenic
chr9	3006620	3006995		mir101c		intergenic
chr9	3023367	3023732		mir101c		intergenic
chr9	3023979	3024664		mir101c		intergenic
chr9	35112776	35113253		Cdon	4933422A05Rik	intergenic
chr12	3109789	3110175		Rab10os		intergenic
chr15	74916811	74917373		Ly6c2	Ly6c1	intergenic
Chr16	11144083	11144340	Zc3h7a			intron

peaks in old						
chromosome	start	stop	gene	upstream of	downstream of	location
chr2	98502411	98502996		B230118H07Rik, Rag2	Lrrc4c	intergenic
chr2	98506516	98507513		B230118H07Rik, Rag2	Lrrc4c	intergenic
chr3	5860425	5860754		1700008P02Rik	Pex2	intergenic
chr6	103598984	103599363	Chl1			intron
chr9	3000286	3000701		mir101c		intergenic
chr9	3001233	3001561		mir101c		intergenic
chr9	3006638	3006982		mir101c		intergenic
chr9	3023378	3023721		mir101c		intergenic
chr9	3023982	3024650		mir101c		intergenic
chr9	35112784	35113245		Cdon	4933422A05Rik	intergenic
chr9	123371012	123371252	Lars2			exon
chr12	3109797	3110162		Rab10os		intergenic
chr13	3372266	3372484		2810429I04Rik	Speer6-ps1	intergenic
chr15	74916761	74917440		Ly6c2	Ly6c1	intergenic
chr16	11144114	11144360	Zc3h7a			intron
chr16	57391513	57391781	Cmss1, Filip1l			intron
chrX	109484401	109484742	Apool	Satl1, Zfp711		intergenic

Supplemental Table E.1. CUT&RUN peaks for troponin T.

peaks specific to old (old-young)						
chromosome	start	stop	gene	upstream of	downstream of	location
chr1	183299200	183299209	Aida			intron
chr2	98666409	98666417		Gm33780	Lrcc4c	intergenic
chr15	75086410	75086419		Ly6c2	Ly6c1	intergenic
peaks in young						
chromosome	start	stop	gene	upstream of	downstream of	location
chr1	183299361	183299362	Aida			intron
chr2	98662683	98662684		Gm33780	Lrcc4c	intergenic
chr2	98666775	98666776		Gm33780	Lrcc4c	intergenic
chr3	5860628	5860629		XR_001783789.1	4930555M17Rik	intergenic
chr6	103649147	103649148	Chl1			intron
chr9	3000506	3000507		mir101c		intergenic
chr9	3006820	3006821		mir101c		intergenic
chr9	3023535	3023536		mir101c		intergenic
chr9	3024466	3024467		mir101c		intergenic
chr9	35305444	35305445	Gm33838			intron
chr12	3109993	3109994		Rab10os		intergenic
chr14	19417261	19417262		Gm41101	Gm33022	intergenic
chr15	75086731	75086732		Ly6c2	Ly6c1	intergenic
peaks in old						
chromosome	start	stop	gene	upstream of	downstream of	location
chr1	183299200	183299535	Aida			intron
chr2	98662324	98662840		Gm33780	Lrcc4c	intergenic
chr2	98666409	98667368		Gm33780	Lrcc4c	intergenic
chr3	5860407	5860689		XR_001783789.1	4930555M17Rik	intergenic
chr6	103648981	103649381	Chl1			intron
chr9	3000297	3000715		mir101c		intergenic
chr9	3001251	3001563		mir101c		intergenic

chr9	3006621	3006989		mir101c		intergenic
chr9	3023448	3023714		mir101c		intergenic
chr9	3023968	3024663		mir101c		intergenic
chr9	35305195	35305668	Gm33838			intron
chr12	3109788	3110169		Rab10os		intergenic
chr14	19417141	19417494		Gm41101	Gm33022	intergenic
chr15	75086410	75086970		Ly6c2	Ly6c1	intergenic

Supplemental Table E.2. CUT&RUN peaks for troponin I.